



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

SECP2523: DATABASE

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DATABASE DESIGN DESCRIPTION: LOGICAL DATABASE DESIGN FOR Hasta Travel Vehicle Rental Management System

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1.0 Introduction

1.1 Overview of the Company

Hasta Travel is a private limited company (Sdn. Bhd.) operating within Universiti Teknologi Malaysia (UTM). Established in 2017, the company provides low-cost car rental services primarily to UTM students and staff. Its main operations are based at the UTM Student Mall. In 2020, Hasta Travel officially obtained its licence as a travel and tour operator.

Currently, the company serves more than 200 customers per month and manages a growing volume of booking, rental and deposit transactions. As a sales-oriented service provider, Hasta Travel relies heavily on efficient handling of customer, vehicle and payment information to support its daily operations and service delivery.

1.2 The Existing System

The existing car rental management system at Hasta Travel operates using a semi-manual approach with minimal system integration. The overall process still depends heavily on manual coordination by staff and adherence to a Standard Operating Procedure (SOP) although some digital tools are used. Core business activities such as booking management, customer eligibility verification, payment confirmation, vehicle pickup, vehicle return, and record keeping are largely handled manually.

Customers initiate the booking process by contacting Hasta Travel through WhatsApp and completing a pre-set WhatsApp form. Staff members then manually check vehicle availability using physical booking schedules and internal systems. After that, staff would verify manually the customer eligibility by viewing submitted documents to identify whether the customers are UTM students and staff. Proof of deposit and rental payments is also sent by customers through WhatsApp. Then, the staff must download, verify and organize the receipts into monthly folders for record keeping.

Vehicle pickup and return processes are also conducted manually. Rental documents are generated, printed, signed and stored physically. Vehicle condition inspections are recorded through photographs sent via WhatsApp. On the other hand, the return details such as fuel level, mileage, damages and late return charges are manually assessed and recorded using spreadsheets. Deposit refunds and penalty deductions are calculated manually and communicated to customers through WhatsApp messages.

From an operational and financial perspective, the system lacks integration between booking, payment, fleet management and accounting processes. Data is scattered across multiple platforms including WhatsApp, Excel files, Google Drive, GPS tracking software, and paper documents. This fragmented structure increases the risk of data inconsistency, duplicate records, delayed updates and human error. Other than that, financial reconciliation of deposits, rentals, penalties and refunds is time-consuming and error-prone. This makes the process difficult to ensure accurate reporting and regulatory compliance.

Furthermore, there is no centralized platform. As a result, the customers cannot manage their own bookings, payments and rental history. This increases reliance on staff for routine tasks. As transaction volumes grow, this manual approach becomes more and more inefficient and challenging to scale. With the upcoming enforcement of LHDN e-Invoicing requirements in June 2025, the current system is no longer adequate to support accurate data management, compliance needs and efficient service delivery. Consequently, it highlights the necessity for a fully integrated car rental management system.

1.3 The Proposed System

1.3.1 Mission Statement

The proposed Hasta Travel Vehicle Rental Management System aims to digitalize and streamline the vehicle rental process for UTM students and staff. The system will ensure efficient user verification, secure payment handling, transparent booking management and a reward mechanism in order to encourage customer loyalty. This truly enhances the convenience, trust and also operational efficiency.

1.3.2 Overview

The to-be system is a web-based vehicle rental management platform designed for Universiti Teknologi Malaysia (UTM) users. It will handle every stage of the vehicle rental process, which is from user registration and verification to vehicle booking, payment and loyalty rewards. The system will also allow the administrators to manage user data, vehicle information, bookings, payments and blacklisted accounts. Other than that, customers can easily make bookings and view vehicle availability in real-time via this system.

The main objectives of this proposed system are to simplify the vehicle rental and verification process for UTM students and staff. Moreover, it will ensure only verified and eligible users which are non-blacklisted can access the system. Most importantly, this system plans to provide users real-time vehicle availability using the booking calendar so the users need not to ask the staff through Whatsapp about the availability of the vehicles. With vouchers rewarded to loyal customers, this system aims to encourage repeated use from customers. Lastly, it truly helps staff to efficiently verify documents, approve bookings, manage blacklists and monitor transactions.

1.3.3 Objectives

The objectives of this project have been clearly defined to ensure that the proposed database system effectively solves the problems faced in the current vehicle rental booking process and meet the requirements of clients.

The objectives of the proposed system are as follows:

1. To create a digital platform that allows customers to review vehicle details, prices, and available time, and complete the entire booking process without manual booking through WhatsApp.
2. To reduce manual work by enabling staff to create, edit, delete, and retrieve customer, vehicle, booking, and payment data efficiently.

3. To improve the accuracy of data such as booking details, payment records, and vehicle availability by applying the use of a centralized database.
4. To help admin manage rental operations more effectively, including booking confirmation, document verification, payment tracking, and vehicle status updates.
5. To provide automatic generated reports that allow management to view and analyse rental activities and track financial flows clearly in purpose to support decision-making.

1.3.4 Scope

This section describes the scope of the proposed vehicle rental booking system. This defines the features included in the vehicle rental booking system and the features that are not included in this project.

1.3.4.1 In Scope

1.3.4.1.1 User Management

While registering the account, users need to fill in their email and have to create a username and password which are used for login purposes. Users can login into the system and have a look at the vehicle models and respective details. However, before proceeding to book a vehicle, it is mandatory for the users to fill in all the information required in the profile section. For example, users must enter their name, matric number, faculty, address and other personal data. Also, users need to upload their identity card or passport, matric card and also driving license for verification. Besides, emergency contact details are required. Manual verification will be done by staff before they can book the vehicle.

If a user is blacklisted because of outstanding payments, they will still be able to log in and view the vehicle models and details of the vehicle. However, they would be restricted from accessing other system features. Once the blacklisted user wants to rent a vehicle, they will be redirected to a payment page to clear their outstanding payments.

1.3.4.1.2 Vehicle Management

The system will display vehicle models handled under the company and their respective details such as price, colour, engine, power and number of seats. Furthermore, there will be real-time availability shown via an interactive calendar. The calendar highlights when the vehicle is booked and enforces a 1-hour cooling period between bookings.

1.3.4.1.3 Booking Management

The system will allow customers to view the availability of vehicles and make bookings easily. When filling the booking form, the customer can directly choose their rental time on the calendar which could reduce the redundancy as the customer needs not to refer back to another page for the availability. Other than that, if a customer chooses other locations instead of Student Mall for pick up location and return location, extra fee would be charged. The booking fee is automatically calculated based on rental period, pick up point, deposit of RM50 per booking and voucher used. After the staff has verified the payment, customers will receive a confirmation notification through their email and also on the website.

Before picking up the vehicle, customers would have to take photos of the vehicle before rental, fill in the current fuel level and upload in the system besides the agreement form. Then, the system would automatically start the countdown after customers finish uploading the documents. On the other hand, for returning, the customers are required to upload the photos of the vehicle and the inspection form. They can even give feedback and ratings on this rental whether about the service or the vehicle.

On the other hand, if the customer wishes to cancel the booking, they have to make it 24 hours before picking up the vehicle. Then, they would get the refund including the deposit. Nonetheless, if the customer only manages to cancel the booking within 24 hours before the pick up, they would not get their refund back.

1.3.4.1.4 Payment and Penalty Handling

The system supports a payment method which is QR payment. Payment receipts must be uploaded for verification. The system will also automatically calculate and apply penalties for late return. For this reason, there is a feature that can track and start the time countdown once the customer gets the vehicle key. Users have a 30-minute grace period after return time, after this period, regardless of whether the vehicle is a car or a motorcycle, an additional fine of RM50 will be charged for each hour of overdue return.

1.3.4.1.5 Loyalty Rewards

The user will get a loyalty stamp when they make a booking that is more than 9 hours. When the user gets 3 stamps, the user would be allowed to redeem discount vouchers such as 12-hour free booking that can be used on a random day. Vouchers are viewable and redeemable through the user's account.

1.3.4.1.6 Admin Dashboard

The admin dashboard would display the important activity and information needed to be reviewed by the admin such as total revenue, active rental, pending task, quick links and action center. While the manager dashboard contains more visualized graphs that show performance trends, top rented models, customer analytics, fleet and booking status. On the other hand, both manager and admin can verify user documents, approve or reject the users bookings besides updating vehicle availability by having the overview of all vehicles. Moreover, they can manage the blacklists and track the payments.

1.3.4.2 Out of Scope

1.3.4.2.1 Vehicle Insurance Processing

The system does not handle vehicle insurance claims or policy renewals. All insurance-related matters would remain under Hasta Travels external insurance provider to avoid integration with third-party insurance APIs and confidential legal data.

1.3.4.2.2 Integration with External Banking APIs

Although the system accepts payment receipts, it does not connect directly to bank systems. Nevertheless, all payment verifications are done manually by staff to avoid the complexity of financial API compliance.

1.3.4.2.3 Automated Staff Assignment

If a customer requests delivery, manual staff assignment will be used. The system will not automatically schedule or dispatch drivers based on the route optimization.

1.3.4.2.4 Real-time vehicle tracking using GPS

The system does not include real-time vehicle tracking using GPS. Vehicle location monitoring and live tracking are not implemented to avoid additional hardware requirements and system complexity.

1.3.5 System Boundaries

The system serves as the main platform which connects customers which are UTM students or staff with Hasta Travel & Tours admin and manager. It interacts with external entities only through limited manual data exchange.

Table 1.3.5.1 Interaction between entities and system

Entities	Description
Customer (UTM student or staff)	Registers, logins, update personal details, uploads verification documents, views vehicle models, prices and details, checks availability, books vehicles, makes payments, fills pickup and return form, earns loyalty stamp, redeem vouchers, cancels or modifies bookings
Admin	Verifies users documents, approves or rejects bookings, updates vehicle availability and status, handles payment verification, adds or removes users from blacklist, manages vouchers
Manager	Views faculty and residential college booking statistics, reviews financial reports, manages admin

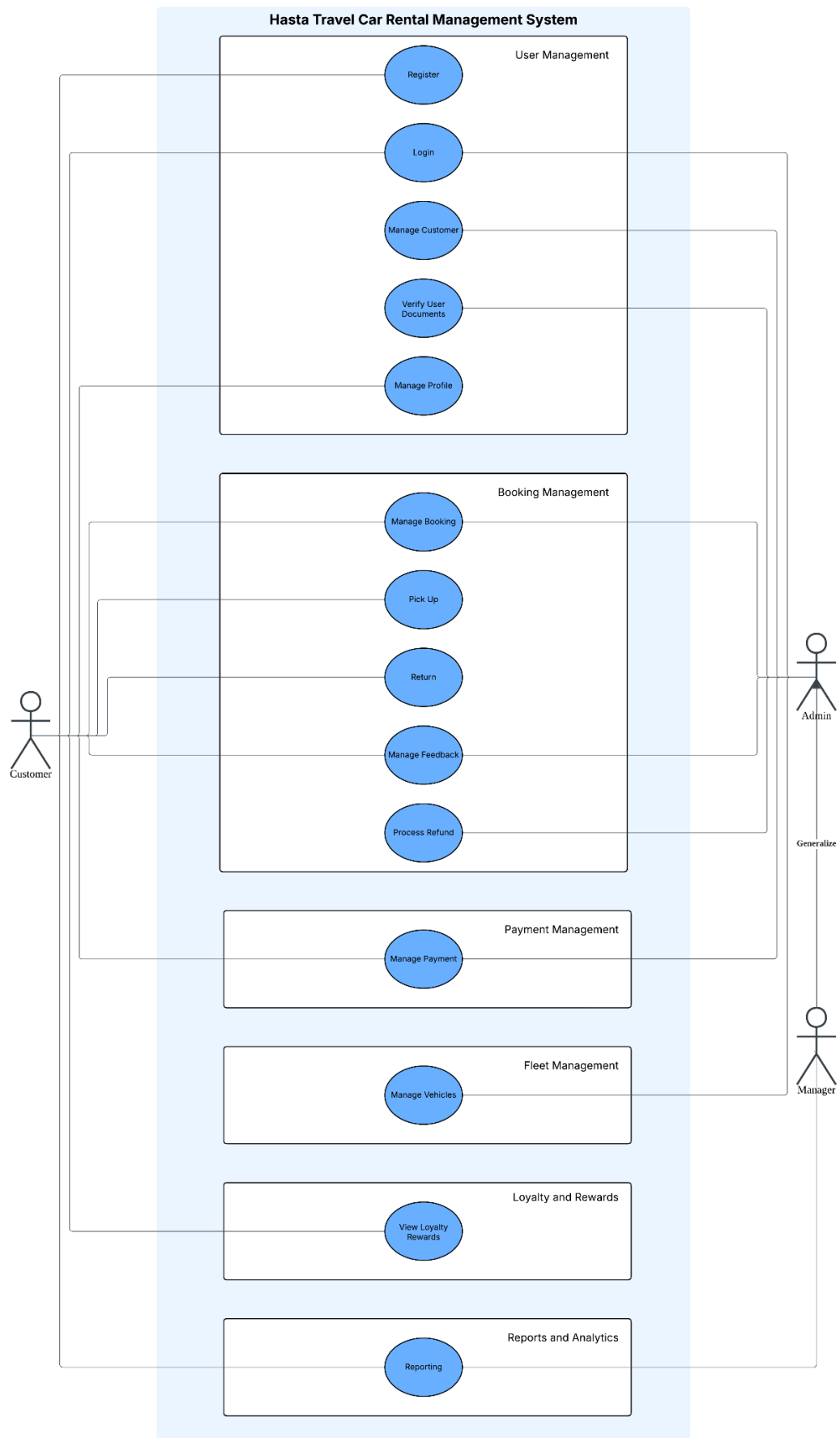


Figure 1.3.5.1 Use Case Diagram

1.3.6 Proposed Technologies

Table 1.3.6.1 Technology used for Proposed System

Component	Proposed Technology
Frontend	HTML, CSS and Javascript
Backend	PHP
Database	MySQL
Hosting	Laragon
Security	User authentication, blacklist access control and input validation

2.0 Overview of Project

This project focuses on the logical design of a database system to support the operations of the vehicle rental management system for Hasta Travel and Tours. The previous phases of the project have determined the business requirements for the system and the system requirements. Therefore, this phase will transform those business and system requirements into a logical relational database design. The design emphasizes proper data organization, integrity and consistency. This is to ensure that operational data can be stored, retrieved and managed efficiently.

The project involves identifying major system entities, defining their attributes, establishing relationships among entities and applying normalization techniques to minimize data redundancy. Logical design artifacts such as the Entity Relationship Diagram (ERD), Enhanced ERD (EERD), relational schema and data dictionary are developed to represent the structure of the database clearly. These components provide a blueprint for implementing the database in a relational database management system.

By following the Logical Design Report structure, this project ensures that the proposed database design aligns with the data and transactional requirements of the system. It also supports future scalability and complies with data management best practices. The outcome of this project serves as a foundation for physical database implementation and system development in subsequent phases.

3.0 Database conceptual design

3.1 Updated Business Rule

3.1.1 User Management Module

1. All new public users are registered with the customer role by default
2. Customers must complete their specific profile data which are driving license, Identification Card (IC) or passport and matric card before they are permitted to book a vehicle
3. Uploaded documents require manual approval by an admin or manager
4. Admins or managers can approve or reject these documents and a reason is needed when rejection occurs
5. Booking privileges are blocked until documents are approved
6. Customers with outstanding penalties are marked as blacklisted and cannot make new bookings until they upload a payment receipt which is then verified by an admin

3.1.2 Fleet Management Module

1. The system designates vehicles in two primary states which are 'active' for those currently in the operational fleet and 'retired' for those permanently inactive or the owner does not want to rent the vehicle anymore
2. Vehicle availability is tracked dynamically which 'available' indicates readiness for booking while 'unavailable' indicates the vehicle is currently rented or disabled
3. A maintenance status overrides standard availability which prevents bookings when a vehicle is undergoing repairs
4. Rental prices are defined individually for each vehicle across standard durations which are one hour, three hours, five hours, twelve hours and one day. The system also applies different pricing tiers for weekends versus weekdays to accommodate demand fluctuations
5. The system records detailed owner information for each vehicle
6. Critical documents such as the vehicle grant, road tax and insurance cover note are uploaded and tracked digitally
7. The system actively tracks expiry dates for road tax, insurance and other recurring obligations. Admin can set due date and receive alerts for upcoming renewals
8. Service records log details of maintenance work, vendor name, cost and date while meter updates records tracks the odometer readings periodically to monitor vehicle usage

3.1.3 Booking Management Module

1. Customers must have a complete profile with all approved uploaded documents before they can initiate a booking
2. The system enforces a conflict check to strictly prevent any booking that might overlap with an existing pickup or return for the selected vehicle

3. The system applies additional fees for pickups or returns that occur at designated surcharge locations such as outside campus or during specific surcharge times. Surcharge rules are configurable via the admin and manager panel
4. Customers are only permitted to cancel a confirmed booking if the scheduled pickup time is more than 24 hours away.
5. Cancellations are eligible for a refund which is processed manually within 3 to 5 business days
6. If a booking using a voucher is cancelled, the voucher will be automatically restored for future use
7. Both pickup and return processes require a digital handover where the customer must upload four angle photos of the vehicle, a dashboard photo and note any damage
8. Fuel levels are recorded at both ends of the transaction to calculate penalties
9. Customers must digitally acknowledge the rental agreement at pickup
10. Customers are required to fill the ratings and feedback at return
11. Late return is calculated only after a 30-minute grace period from the scheduled return time and RM50 would be charged per hour
12. Fuel shortage penalties are applied based on the number of fuel bars missing as recorded during inspection which are RM10 per missing fuel bar
13. The penalties are subtracted from the deposit. If the penalty is less than deposit, the remaining deposit is refundable

3.1.4 Payment Management Module

1. A deposit is mandatory and collected for every single booking iteration
2. Admin must visually verify the uploaded receipt image before a booking can be confirmed
3. Rejected receipts will trigger a re-upload request or booking cancellation. Only “incorrect amount” triggers a refund workflow while other reasons typically require re-upload
4. The system records the specific admin ID who verified each payment for accountability

3.1.5 Loyalty and Rewards Module

1. Customers earn 1 loyalty stamp for every completed rental transaction with a duration of 9 hours or more
2. Customers can redeem collected stamps for vouchers based on defined tiers. Redemption immediately deducts the stamp count from the customer’s balance
3. Vouchers can optionally grant a percentage discount or specific free hours
4. A voucher can only be used once
5. If the booking is cancelled or rejected, the voucher will be automatically restored for future use
6. Admin and manager can assign voucher to the customers

3.1.6 Report and Analytics Module

1. Admin dashboard focuses on operational or daily tasks while manager dashboard focuses on strategic performance
2. Revenue is aggregated into Daily (Revenue vs Expenses) and Monthly (Total Sales Count and Amount) views
3. The dashboard has fleet status overview which shows whether the fleet are available, rented or under maintenance
4. Feedback Scores view the vehicle rating especially those ratings below 3 stars to alert admin and manager

3.2 Conceptual ERD

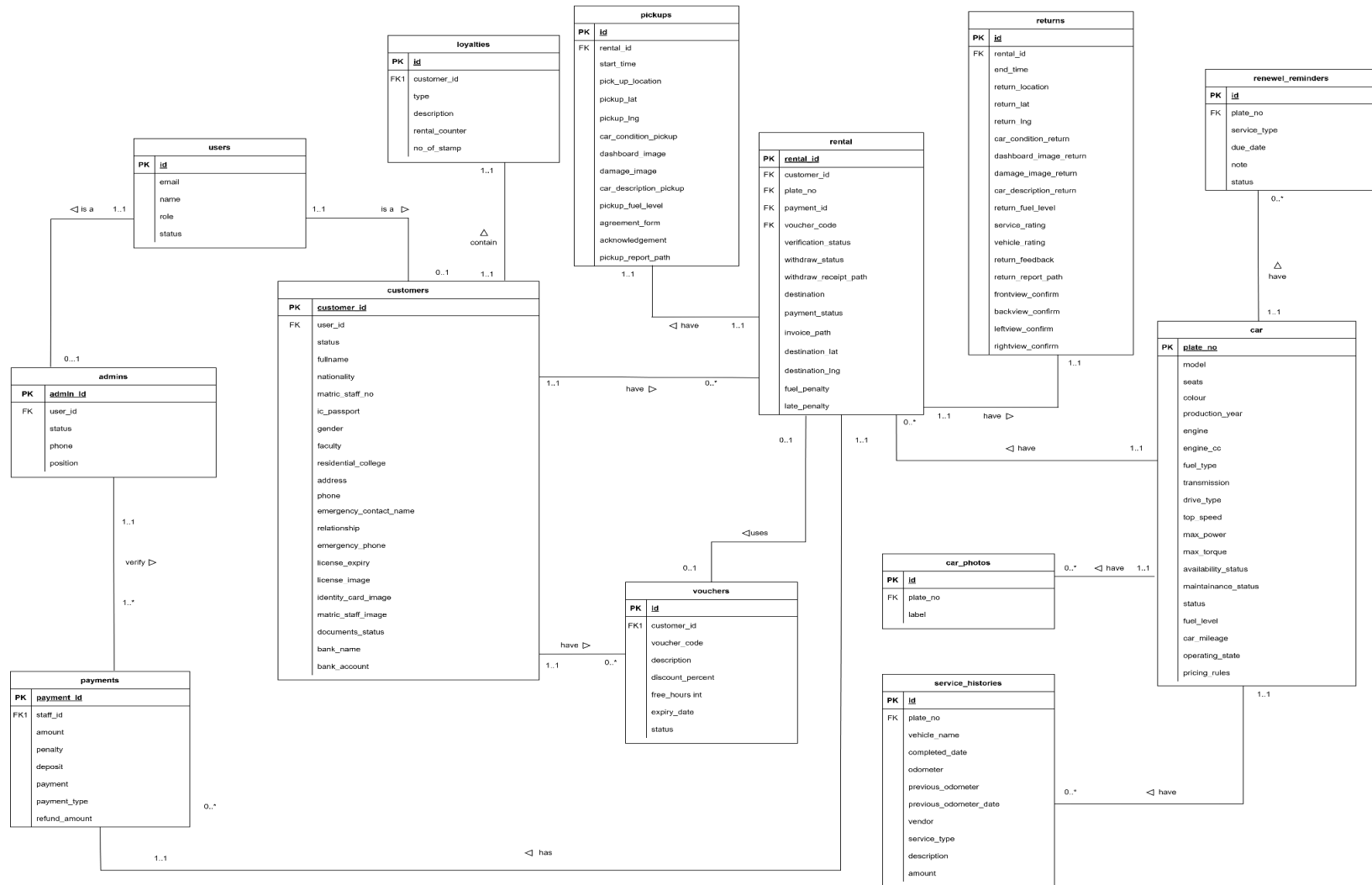


Figure 3.2.1 Conceptual ERD

4.0 DB Logical Design

4.1 Logical ERD

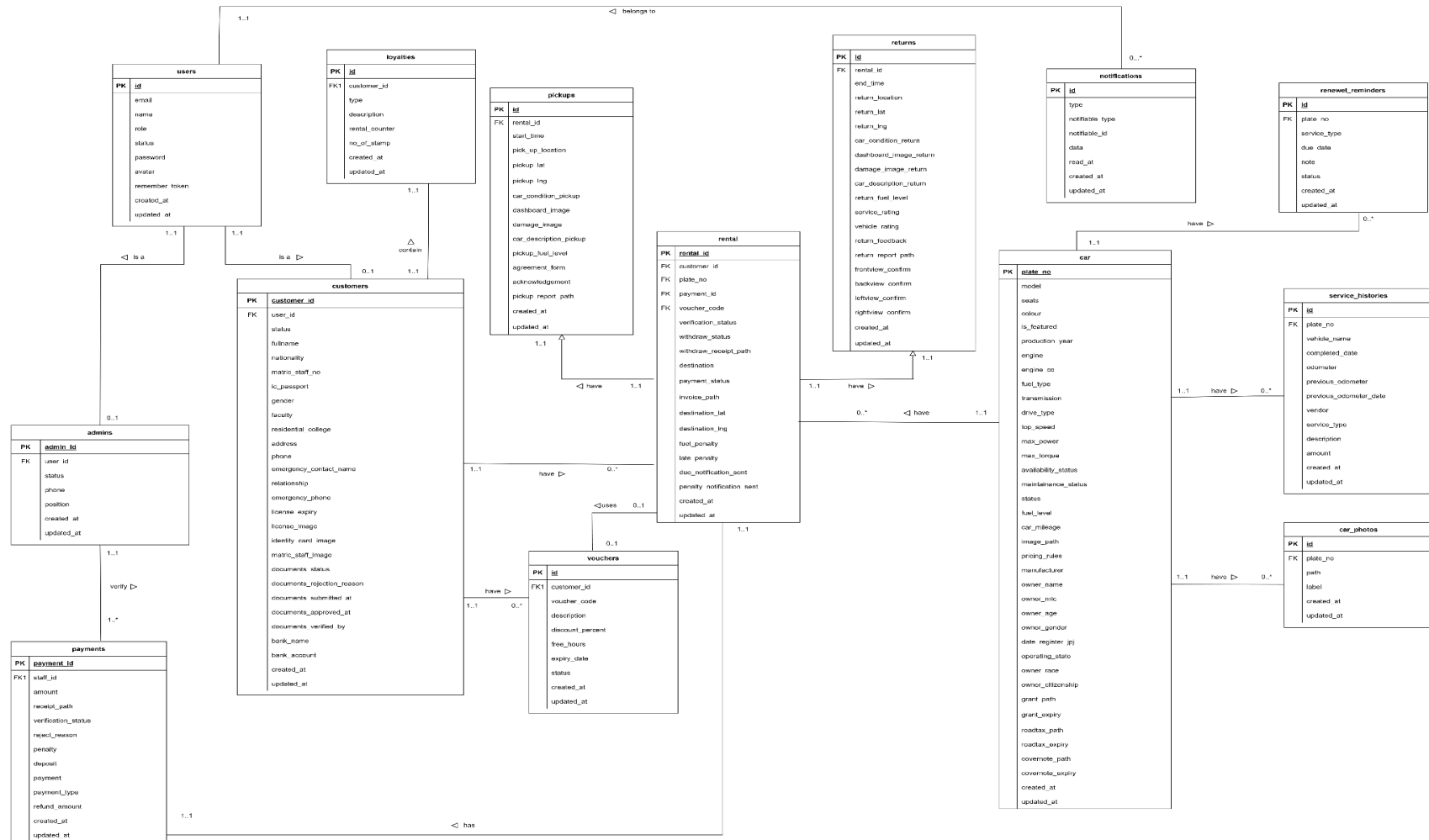


Figure 4.1.1 Logical ERD

4.2 Updated Data Dictionary

1. admins

Column	Data Type	Length	Description	Allowed Values	Relationship	Required
admin_id	varchar	50	Unique admin identifier	-	Primary Key	Yes
user_id	bigint	-	Unique user identifier	-	Foreign Key => users.id	Yes
status	varchar	20	Current status of the admin account	active/deactivated	-	Yes
phone	varchar	20	Admin's contact phone number	-	-	No
position	varchar	20	Admin's job position	-	-	No
created_at	timestamp	n/a	Records when a row was first added	Any valid timestamp	-	No
updated_at	timestamp	n/a	Records the last time the row was modified	Any valid timestamp	-	No

2. car

Column	Data Type	Length	Description	Allowed Values	Relationship	Required
plate_no	varchar	20	Vehicle's plate number	Alphanumeric	Primary Key	Yes
model	varchar	150	Vehicle's model	-	-	Yes
seats	tinyint	-	Number of seats	1 – 9	-	No
colour	varchar	30	Vehicle colour	-	-	No
is_featured	tinyint	1	Featured vehicle flag	0 = No, 1 = Yes	-	Yes
production_year	year	-	Year of manufacture	YYYY	-	No
availability_status	tinyint	1	Availability of vehicle	available / unavailable	-	Yes
status	varchar	20	Operational status of car	active / retired	-	Yes
fuel_level	integer	-	The fuel level for the vehicle	-	-	Yes
car_mileage	decimal	10, 2	Vehicle's mileage	≥ 0	-	Yes
image_path	varchar	255	Vehicle image	File path	-	No
engine	varchar	100	Engine description	-	-	No

engine_cc	integer	-	Engine capacity(cc)	-	-	No
fuel_type	varchar	20	Type of fuel	Petrol/Diesel/E V	-	No
transmission	varchar	20	Transmission type	Automatic / Manual	-	No
drive_type	varchar	20	Drive type	FWD/RWD/A WD	-	No
top_speed	integer	-	Maximum speed (km/h)	-	-	No
max_power	decimal	8, 2	Maximum power (HP)	-	-	No
max_torque	decimal	8, 2	Maximum torque (Nm)	-	-	No
maintenance_status	tinyint	1	Maintenance state	0 = OK, 1 = Under Maintenance	-	Yes
pricing_rules	json	-	Hourly pricing rates of the vehicle	-	-	Yes
owner_name	varchar	100	Registered owner name	-	-	No
manufacturer	varchar	50	Vehicle manufacturer	-	-	No
owner_nric	varchar	20	Owner's identification card number	-	-	No

owner_age	integer	-	Owner's age	-	-	No
owner_gender	varchar	20	Owner's gender	Male/Female	-	No
date_register_jpj	date	-	JPJ registration date	-	-	No
operating_state	varchar	50	State where vehicle operates	-	-	No
owner_race	varchar	30	Owner's race	-	-	No
owner_citizenship	varchar	30	Owner's citizenship	-	-	No
grant_path	varchar	255	Vehicle grant document path	File path	-	No
grant_expiry	date	-	Grant expiry date	-	-	No
roadtax_path	varchar	255	Road tax document path	File path	-	No
roadtax_expiry	date	-	Road tax expiry date	-	-	No
covernote_path	varchar	255	Insurance cover note path	File path	-	No
covernote_expiry	date	-	Insurance expiry date	-	-	No
created_at	timestamp	n/a	Records when a row was first added	Any valid timestamp	-	No

updated_at	timestamp	n/a	Records the last time the row was modified	Any valid timestamp	-	No
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3. car_photos

Column	Data Type	Length	Description	Allowed Values	Relationship	Required
id	bigint (UNSIGNED)	-	Unique identifier for each vehicle photo record	Auto-increment integer	Primary Key	Yes
plate_no	varchar	20	Vehicle's plate number	Alphanumeric	Foreign key => car.plate_no	Yes
path	varchar	255	Vehicle's image path	File path	-	Yes
label	varchar	50	Label of the image	Front, Back, Left, Right, Interior 1, Interior 2	-	No
created_at	timestamp	n/a	Records when a row was first added	Any valid timestamp	-	No
updated_at	timestamp	n/a	Records the last time the row was modified	Any valid timestamp	-	No

4. customers

Column	Data Type	Length	Description	Allowed Values	Relationship	Required
customer_id	varchar	50	Unique customer identifier	-	Primary key	Yes
user_id	bigint	-	Unique user identifier	-	Foreign Key => users.id	Yes
status	varchar	20	Current status of the customer account	active/ blacklisted	-	Yes
fullname	varchar	100	Customer's full name as per IC/passport	-	-	No
nationality	varchar	100	Customer's nationality	-	-	No
matric_staff_no	varchar	20	Student or staff matric number	-	Unique key	No
ic_passport	varchar	15	IC number (local) or passport number (foreign)	-	-	No
gender	varchar	50	Customer's gender	"Male", "Female", "Prefer not to say"	-	No
faculty	varchar	100	Customer's faculty	-	-	No

residential_coll ege	varchar	100	Customer's residential college	-	-	No
address	text	-	Customer's home address	-	-	No
phone	varchar	20	Customer's contact phone number	Digits, '+' allowed	Unique key	No
emergency_con tact_name	varchar	100	Name of emergency contact person	-	-	No
relationship	varchar	50	Relationship between customer and emergency contact person	-	-	No
emergency_pho ne	varchar	20	Emergency contact phone number	Digits '+' allowed	-	No
license_expiry	date	-	Driving license expiry date	-	-	No
license_image	varchar	255	File path to driving license image	-	-	No
identity_card_i mage	varchar	255	File path to identity card image	-	-	No
matric_staff_im age	varchar	255	File path to matric card image	-	-	No

documents_status	enum	-	Verification status of submitted documents	'pending', 'approved', 'rejected'	-	No
documents_rejection_reason	text	-	Reasons if the document is rejected	-	-	No
documents_submitted_at	timestamp	-	Records the date and time when the customer submitted documents	Any valid timestamp	-	No
documents_approved_at	timestamp	-	Records the date and time when the customer's documents were approved	Any valid timestamp	-	No
documents_verified_by	varchar	255	Identifier of the admin or manager who verified the documents	admin/manager	-	No
bank_name	varchar	100	Customer's bank name	-	-	No
bank_account	varchar	30	Customer's bank account number	-	-	No
created_at	timestamp	n/a	Records when a row was first added	Any valid timestamp	-	No
updated_at	timestamp	n/a	Records the last time the row was modified	Any valid timestamp	-	No

5. loyalties

Column	Data Type	Length	Description	Allowed Values	Relationship	Required
id	bigint	-	Unique loyalty identifier	-	Primary Key	Yes
customer_id	varchar	50	ID of the customer who owns this loyalty record	-	Foreign Key -> customers.customer_id	Yes
type	varchar	50	Indicates if stamp was earned or redeemed	'earned', 'redeemed'	-	Yes
description	varchar	255	Description of the loyalty action	-	-	Yes
rental_counter	integer	-	Current redeemable stamps	0 or above	-	Yes
no_of_stamp	integer	-	Lifetime total stamps	0 or above	-	Yes
created_at	timestamp	n/a	Records when a row was first added	Any valid timestamp	-	No
updated_at	timestamp	n/a	Records the last time the row was modified	Any valid timestamp	-	No

6. notifications

Column	Data Type	Length	Description	Allowed Values	Relationship	Required
id	char	36	Unique loyalty identifier	Auto-increment integer	Primary Key	Yes
type	varchar	100	Types of notifications the customer will receive.	'Rejected', 'Approved'	-	Yes
notifiable_type	varchar	100	Stores the model class name	-	-	Yes
notifiable_id	varchar	50	Stores Id that the notification will be sent to	Specified integer for specific customer	-	Yes
data	text	-	Record the status,Unique rental IDs and notification messages for vehicle rental transactions.	-	-	No
read_at	timestamp	n/a	Records when the customer reads the notification	Any valid timestamp	-	No
created_at	timestamp	n/a	Records when a row was first added	Any valid timestamp	-	No

updated_at	timestamp	n/a	Records the last time the row was modified	Any valid timestamp	-	Optional
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7. payments

Column	Data Type	Length	Description	Allowed Values	Relationship	Required
payment_id	varchar	20	Unique payment identifier	Auto-increment positive integer	Primary Key	Yes
staff_id	varchar	50	Unique staff identifier	From existing Staff table	Foreign Key -> admins.admin_id	No
amount	decimal	10, 2	Amount need to pay	In Malaysian Ringgit currency decimal value	-	Yes
receipt_path	varchar	255	File path to receipt	Valid file path	-	No
verification_status	varchar	50	Status of verification	pending / approved / rejected	-	No

reject_reason	text	-	Reasons for payment rejection	Incorrect amount/Invalid account/others	-	No
penalty	decimal	10, 2	Penalty needs to pay	In Malaysian Ringgit currency decimal value	-	Yes
deposit	decimal	10, 2	Deposit needs to pay	In Malaysian Ringgit currency decimal value	-	Yes
payment	varchar	20	Payment method to choose	Valid bank QR	-	Yes
payment_type	varchar	20	Reason of payment whether for rental or penalty	booking / penalty	-	Yes
refund_amount	decimal	10, 2	The amount that needed to be refunded to the customer	In Malaysian Ringgit currency decimal value	-	Yes
created_at	timestamp	n/a	Records when a row was first added	Any valid timestamp	-	Optional
updated_at	timestamp	n/a	Records the last time the row was modified	Any valid timestamp	-	Optional

8. renewal_reminders

Column	Data Type	Length	Description	Allowed Values	Relationship	Required
id	bigint (UNSIGNED)	-	Unique identifier for each renewal reminder record	Auto-increment integer	Primary Key	Yes
plate_no	varchar	20	Vehicle plate number associated with the reminder	Alphanumeric	Foreign key => car.plate_no	Yes
service_type	enum	-	Type of renewal or service reminder	Insurance, Investor Commission, Monthly Installment, Roadtax, Others	-	Yes
due_date	date	-	Due date for the renewal or service	Valid date	-	Yes
note	text	-	Additional remarks or notes for the reminder	Free text	-	
status	enum	-	Current reminder status	upcoming, notified,	-	Yes

				completed		
created_at	timestamp	n/a	Records when a row was first added	Any valid timestamp	-	Optional
updated_at	timestamp	n/a	Records the last time the row was modified	Any valid timestamp	-	Optional

9. rental

Column	Data Type	Length	Description	Allowed Values	Relationship	Required
rental_id	bigint	-	Unique identifier for each rental transaction	Auto-increment positive integer	Primary Key	Yes
customer_id	varchar	50	Identifier of the customer who made the rental	Existing customer IDs	Foreign Key -> customers.customer_id	Yes
plate_no	varchar	10	Vehicle plate number rented	Registered vehicle plate numbers	Foreign Key -> car.plate_no	Yes
payment_id	varchar	50	Unique payment id	System-generated payment IDs	Foreign Key -> payments.payment_id	Yes

voucher_code	varchar	50	Voucher code applied to rental	Existing voucher codes	Foreign Key -> vouchers.voucher_code	No
verification_status	varchar	50	Status of verification	pending/ verified/ done/ renting/ Cancelled/ failed	-	Yes
withdraw_status	varchar	20	Status of deposit withdrawal/refund	pending/ done	-	Yes
withdraw_receipt_path	varchar	255	The receipt for withdraw deposit uploaded by admin	Valid file path	-	No
destination	varchar	255	Destination entered by customer	Free text location	-	Yes
payment_status	varchar	50	The status to record the payment of user	pending/ paid/ failed/ cancelled	-	No
invoice_path	varchar	255	The autogenerated invoice of this rental	Valid file path	-	No

destination_lat	decimal	10,7	Latitude coordinate of destination location	Valid latitude coordinate	-	Yes
destination_lng	decimal	10,7	Longitude coordinate of destination location	Valid longitude coordinate	-	Yes
fuel_penalty	decimal	8, 2	Charges incurred when fuel level during return is less than pickup	-	-	
late_penalty	decimal	10,2	Charges incurred for late return		-	Yes
due_notification_sent	timestamp	n/a	Time when due reminder notification was sent	Any valid timestamp	-	No
penalty_notification_sent	timestamp	n/a	Time when penalty notification was sent	Any valid timestamp	-	No
created_at	timestamp	n/a	Record when a row was first added	Any valid timestamp	-	Yes
updated_at	timestamp	n/a	Record the last time the row updated	Any valid timestamp	-	Yes

10. pickups

Column	Data Type	Length	Description	Allowed Values	Relationship	Required
id	bigint	-	Unique booking identifier	Auto-increment	Primary Key	Yes
rental_id	bigint	-	References the rental associated with this pickup	Existing rental.rental_id	Foreign Key-> rental.rental_id	Yes
start_time	datetime	-	Scheduled pickup start time and date	Valid datetime	-	Yes
pick_up_location	varchar	255	The pickup location	Text	-	Yes
pickup_lat	decimal	10,7	Latitude coordinate of pickup location	Any valid latitude	-	No
pickup_lng	decimal	10,7	Longitude coordinate of pickup location	Any valid longitude	-	No
car_condition_pickup	text	-	Text describing vehicle condition at pickup	Text	-	No
dashboard_image	varchar	255	Image path of vehicle dashboard at pickup	File path	-	No
damage_image	varchar	255	Image path showing any vehicle damage at pickup	File path	-	No

car_description_pickup	text	-	Additional notes describing vehicle condition at pickup	Text	-	No
pickup_fuel_level	integer	1	Fuel level at pickup	1 to 8	-	No
agreement_form	varchar	255	Signed rental agreement file path	File path	-	No
acknowledgement	tinyint	1	Customer acknowledgement of pickup	0,1	-	Yes
pickup_report_path	varchar	255	Generated pickup inspection report file path	File path	-	No
created_at	timestamp	n/a	Records when a row was first added	Any valid timestamp	-	No
updated_at	timestamp	n/a	Records the last time the row was modified	Any valid timestamp	-	No

11. returns

Column	Data Type	Length	Description	Allowed Values	Relationship	Required
id	bigint	-	Unique identifier for return record	Auto-increment	Primary Key	Yes

rental_id	bigint	-	References the rental associated with this return	Existing rental.rental_id	Foreign Key -> rental.rental_id	Yes
end_time	datetime	-	Actual return date and time of the vehicle	Valid datetime	-	Yes
return_location	varchar	255	Vehicle return location name or address	Text	-	Yes
return_lat	decimal	10,7	Latitude of return location	Any valid latitude	-	No
return_lng	decimal	10,7	Longitude of return location	Any valid longitude	-	No
car_condition_return	text	-	Text describing vehicle condition at return	Text	-	No
dashboard_image_return	varchar	255	Image path of dashboard at return	File path	-	No
damage_image_return	varchar	255	Image path showing vehicle damage at return	File path	-	No
car_description_return	text	-	Additional notes on vehicle condition during return	text	-	No
return_fuel_level	int	-	Fuel level at time of return	0-8	-	No

service_rating	int	-	Customer rating for rental service	1-5	-	No
vehicle_rating	int	-	Customer rating for vehicle condition	1-5	-	No
return_feedback	text	-	Customer feedback or comments after return	text	-	No
return_report_path	varchar	255	Generated return inspection report file path	File path	-	No
frontview_confirm	tinyint	1	Confirmation that front view inspection is completed	0/1	-	Yes
backview_confirm	tinyint	1	Confirmation that back view inspection is completed	0/1	-	Yes
leftview_confirm	tinyint	1	Confirmation that left view inspection is completed	0/1	-	Yes
rightview_confirm	tinyint	1	Confirmation that right view inspection is completed	0/1	-	Yes
created_at	timestamp	n/a	Record creation timestamp	Any valid timestamp	-	Yes
updated_at	timestamp	n/a	Last update timestamp	Any valid timestamp	-	Yes

12. service_histories

Column	Data Type	Length	Description	Allowed Values	Relationship	Required
id	bigint (UNSIGNED)	-	Unique identifier for service record	Auto-increment integer	Primary Key	Yes
plate_no	varchar	20	Vehicle plate number associated with the service	Alphanumeric	Foreign key => car.plate_no	Yes
vehicle_name	varchar	100	Vehicle model/name	Text	-	Yes
completed_date	date	-	Date when the service was completed	Valid date	-	Yes
odometer	integer	-	Vehicle mileage at service time (km)	≥ 0	-	No
previous_odometer	integer	-	Previous recorded mileage	≥ 0	-	No
previous_odometer_date	date	-	Date of previous mileage record	Valid date	-	No
vendor	varchar	100	Service provider / workshop	Text	-	No
service_type	varchar	50	Type of service	Oil Change, Meter Update,	-	No

				Repair, Inspection		
description	text	-	Additional service details	Free text	-	No
amount	decimal	10,2	Service cost (MYR)	≥ 0.00	-	Yes
created_at	timestamp	n/a	Records when a row was first added	Any valid timestamp	-	Optional
updated_at	timestamp	n/a	Records the last time the row was modified	Any valid timestamp	-	Optional

13. users

Column	Data Type	Length	Description	Allowed Values	Relationship	Required
id	bigint (UNSIGNED)	-	Unique user identifier used for authentication and internal references	-	Primary key	yes
email	varchar	100	User's login email address	Valid email format	Unique key	yes
role	varchar	50	Defines the role of the user in the system	customer/ admin/ manager	-	yes

name	varchar	50	Username use in system	-	-	Yes
status	varchar	20	Account status	active / deactivated	-	Yes
password	varchar	255	Encrypted password used for authentication	Hashed value	-	Yes
avatar	varchar	255	Stores the path to the user's profile image	-	-	No
remember_token	varchar	100	Token used by Laravel to remember logged-in sessions	System-generated	-	No
created_at	timestamp	n/a	Records when a row was first added	Any valid timestamp	-	No
updated_at	timestamp	n/a	Records the last time the row was modified	Any valid timestamp	-	No

14. vouchers

Column	Data Type	Length	Description	Allowed Values	Relationship	Required
id	bigint	-	Unique identifier for the voucher	-	Primary key	Yes

customer_id	varchar	50	ID of the customer who owns the voucher	-	Foreign Key -> customers.customer_id	Yes
voucher_code	varchar	50	Unique code for each voucher	-	Unique Key	Yes
description	string	255	Detail about the voucher	-	-	No
discount_percent	integer	-	Percentage discount amount	-	-	No
free_hours	integer	-	Free rental hours granted	-	-	No
expiry_date	date	-	The date when the voucher expires	-	-	Yes
status	varchar	50	Current state of the voucher	'active', 'used', 'expired'	-	Yes
created_at	timestamp	n/a	Records when a row was first added	Any valid timestamp	-	No
updated_at	timestamp	n/a	Records the last time the row was modified	Any valid timestamp	-	No

4.3 Normalization

4.3.1 Unnormalized Form (UNF)

Before normalization, the rental system data contained repeating groups and multi-valued attributes. Here's what the unnormalized data structure looked like:

UNF Relation Schema

1. users (user_id, user_name, user_email, user_role, user_status, password)
2. customers (customer_id, customer_status, nationality, matric_staff_no, ic_passport, gender, faculty, residential_college, address, phone, emergency_contact_name, emergency_phone, license_image, identity_card_image, documents_status, bank_name, bank_account)
3. admins (admin_id, admin_name, admin_phone, admin_position)
4. car (plate_no, model, manufacturer, seats, colour, availability_status, fuel_level, car_mileage, production_year, engine, transmission, owner_name, owner_nric, grant_path, grant_expiry, roadtax_path, roadtax_expiry, status)
5. car_photos (path, label)
6. service_records (date, type, vendor, amount, odometer)
7. renewal_reminders (type, due_date, status)
8. payments (payment_id, amount, receipt_path, verification_status, penalty, deposit, refund_amount)
9. rental (rental_id, destination, destination_lat, destination_lng, voucher_code, fuel_penalty, late_penalty)
10. pickups (pickup_id, start_time, pick_up_location, pickup_lat, pickup_lng, car_condition_pickup, dashboard_image, pickup_fuel_level, acknowledgement)
11. returns (return_id, end_time, return_location, return_lat, return_lng, car_condition_return, return_fuel_level, service_rating, vehicle_rating, return_feedback)
12. vouchers (code, discount, expiry, status)
13. layalties (type, description, stamps)
14. notifications (type, data, read_at)

UNF Sample Data Examples

1. Example: car_photos

rental_id	plate_no	path	label
ROO1	UTM6522	Fleet/UTM6522/front.png	Front
		Fleet/UTM6522/back.png	Back
		Fleet/UTM6522/left.png	Left
		Fleet/UTM6522/right.png	Right

		Fleet/UTM6522/interior.png	Interior
R002	JQU1957	Fleet/JQU1957/front.png	Front

2. Example: service_records

plate_no	date	type	vendor	amount
UTM6522	2025-01-05	Oil Change	Proton Service	170.00
	2025-06-10	Brake Service	Brake Shop	250.00
	2025-12-15	Tire Rotation	NULL	80.00
UTM137	2026-01-05	Oil Change	Proton Service	170.00

3. Example: vouchers

customer_id	code	amount	expiry	status
C000001	REWARD-ABC123	50	2026-03-01	Active
	SUMMER-2026	20	2026-06-30	used
C000007	LOYALTY-15	15	2026-04-01	Active
	HAPPY-2026	20	2026-07-30	used

4. Example: notifications

user_id	type	data	read_at
2	Pickup Success	Pickup for Honda...	2026-01-31 02:01:35
	Payment Verified	Payment approved	2026-01-31 02:01:35
5	Document Approved	Documents OK	NULL

Problem:

1. Repeating group exist, one column have multiple value
2. Data Redundancy
 - a. Car information (model, manufacturer, owner) duplicated for every rental
 - b. Customer information (name, phone, address) duplicated for every rental
3. Update Anomalies
 - a. Changing car manufacturer requires updating multiple rental records
 - b. Updating customer phone number requires changing all their rental records
 - c. Risk of inconsistent data if updates are partial

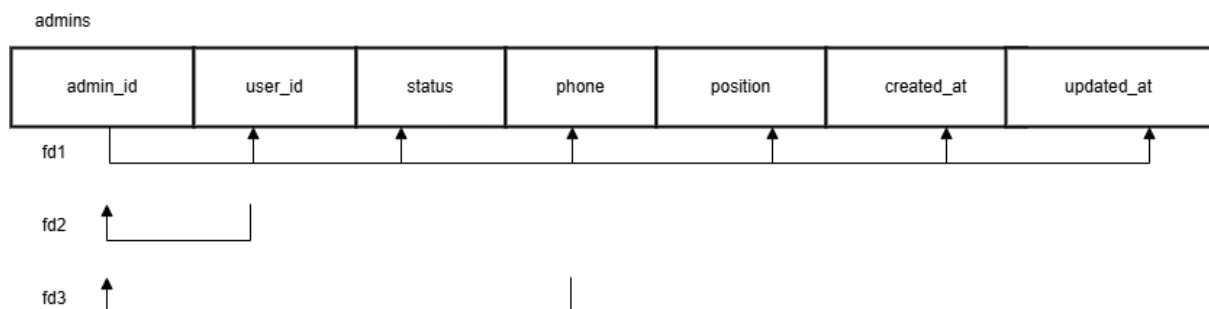
4.3.2 First Normal Form (1NF)

First Normal Form (1NF) is a format that is concerned with the organization of data to exclude repetitions and ambiguity. To fulfill the 1NF, one should eliminate all the repeating groups to assure that the fields can have a single value only. A table should not have multiple or composite values but individual (single, indivisible) values in every attribute. The tables should also have a well-known primary key to distinguish each record. Moreover, the data that is related to various themes or is concerned with various entities should be differentiated into different tables to preserve clarity and coherence.

After Applying 1NF:

1. admins

admins (admin_id, user_id, status, phone, created_at, updated_at)



Functional Dependencies:

fd1: $\text{admin_id} \rightarrow \text{user_id}, \text{status}, \text{phone}, \text{created_at}, \text{updated_at}$

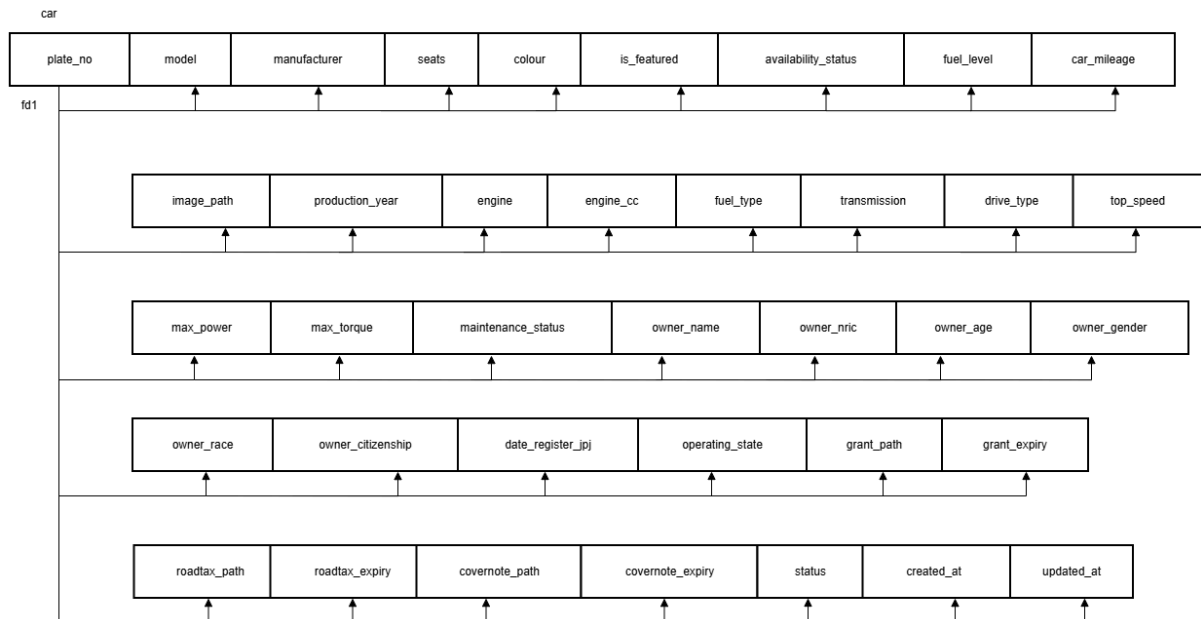
fd2: $\text{user_id} \rightarrow \text{admin_id}$

fd3: $\text{phone} \rightarrow \text{admin_id}$

1NF: All attributes contain atomic values. Each admin has single-valued attributes like phone, position, and status with no multi-valued or composite attributes.

2. car

car (plate_no, model, manufacturer, seats, colour, is_featured, availability_status, fuel_level, car_mileage, image_path, production_year, status, engine, engine_cc, fuel_type, transmission, drive_type, top_speed, max_power, max_torque, maintenance_status, owner_name, owner_nric, owner_age, owner_gender, owner_race, owner_citizenship, date_register_jpj, operating_state, grant_path, grant_expiry, roadtax_path, roadtax_expiry, covernote_path, covernote_expiry, created_at, updated_at)



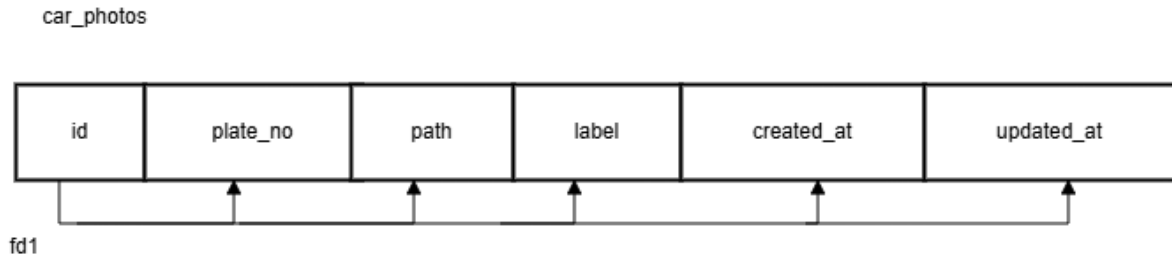
Functional Dependencies:

fd1: plate_no → model, manufacturer, seats, colour, is_featured, availability_status, fuel_level, car_mileage, image_path, production_year, status, engine, engine_cc, fuel_type, transmission, drive_type, top_speed, max_power, max_torque, maintenance_status, owner_name, owner_nric, owner_age, owner_gender, owner_race, owner_citizenship, date_register_jpj, operating_state, grant_path, grant_expiry, roadtax_path, roadtax_expiry, covernote_path, covernote_expiry, created_at, updated_at

1NF: All attributes are atomic. Vehicle specifications like engine_cc, seats, and fuel_level are stored as single values without nested structures or arrays.

3. car_photos

car_photos (id, plate_no, path, label, created_at, updated_at)



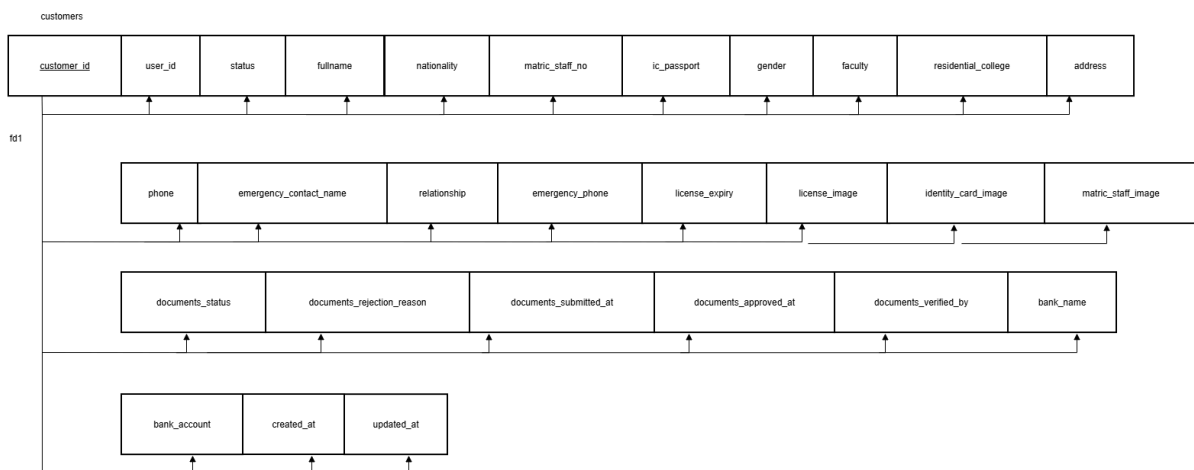
Functional Dependencies:

fd1: id $\rightarrow$ plate_no, path, label, created_at, updated_at

1NF: All attributes contain atomic values. Each photo has single-valued attributes like path and label (Front, Back, Left, Right, Interior) without multi-valued fields.

4. customers

customers (customer_id, status, user_id, fullname, nationality, matric_staff_no, ic_passport, gender, faculty, residential_college, address, phone, emergency_contact_name, relationship, emergency_phone, license_no, license_expiry, license_image, identity_card_image, matric_staff_image, documents_status, documents_rejection_reason, documents_submitted_at, documents_approved_at, documents_verified_by, bank_name, bank_account, created_at, updated_at)



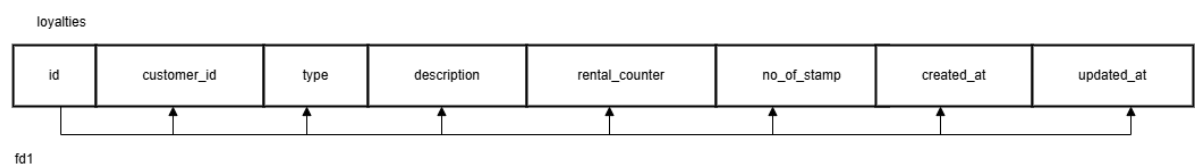
Functional Dependencies:

fd1: customer_id $\rightarrow$ status, user_id, fullname, nationality, matric_staff_no, ic_passport, gender, faculty, residential_college, address, phone, emergency_contact_name, relationship, emergency_phone, license_no, license_expiry, license_image, identity_card_image, matric_staff_image, documents_status, documents_rejection_reason, documents_submitted_at, documents_approved_at, documents_verified_by, bank_name, bank_account, created_at, updated_at

1NF: All attributes are atomic with single values. Customer information like fullname, address, and phone are stored as individual scalar values without repeating groups.

5. loyalties

loyalties (id, customer_id, type, description, rental_counter, no_of_stamp, created_at, updated_at)



Functional Dependencies:

fd1: id → customer_id, type, description, rental_counter, no_of_stamp, created_at, updated_at

1NF: All attributes are atomic. Loyalty details like type (earned/redeemed), rental_counter, and no_of_stamp are stored as single values without repeating groups.

6. notifications

notifications (id, user_id, type, notifiable_type, notifiable_id, data, read_at, created_at, updated_at)



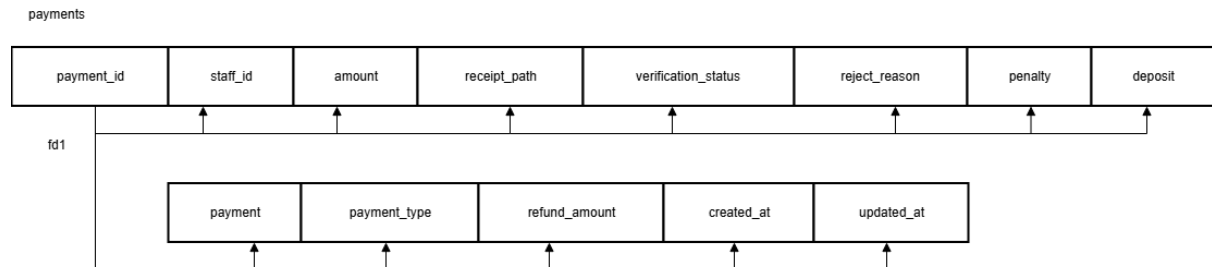
Functional Dependencies:

fd1: id → user_id, type, notifiable_type, notifiable_id, data, read_at, created_at, updated_at

1NF: All attributes contain atomic values. Notification details like type, notifiable_type, notifiable_id, and read_at are stored as single values. The data attribute stores JSON but is treated as a single atomic text value.

7. payments

payments (payment_id, staff_id, amount, receipt_path, verification_status, reject_reason, penalty, deposit, payment, payment_type, refund_amount, created_at, updated_at)



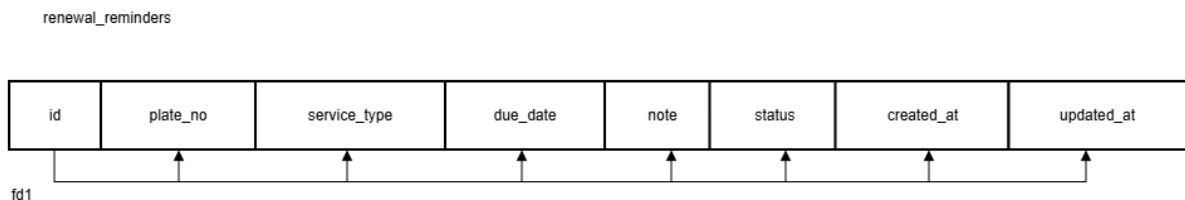
Functional Dependencies:

fd1: payment_id → staff_id, amount, receipt_path, verification_status, reject_reason, penalty, deposit, payment, payment_type, refund_amount, created_at, updated_at

1NF: All attributes are atomic. Payment details like amount, penalty, deposit, and refund_amount are stored as single decimal values without repeating groups.

8. renewal_reminders

renewal_reminders (id, plate_no, service_type, due_date, note, status, created_at, updated_at)



Functional Dependencies:

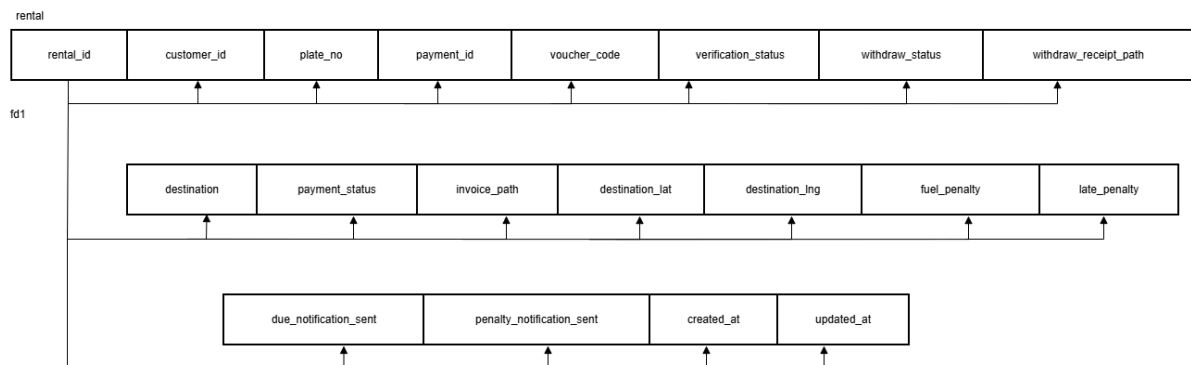
fd1: id → plate_no, service_type, due_date, note, status, created_at, updated_at

1NF: All attributes contain atomic values. Reminder details like service_type (Insurance, Roadtax, etc.), due_date, and status are stored as single values without multi-valued attributes.

9. rental

rental (rental_id, customer_id, plate_no, payment_id, , voucher_code, verification_status, withdraw_status, withdraw_receipt_path, destination, payment_status, invoice_path,

destination_lat, destination_lng, fuel_penalty, late_penalty, due_notification_sent, penalty_notification_sent, created_at, updated_at)



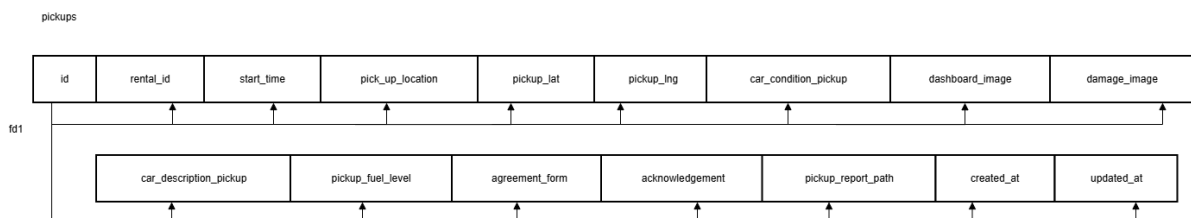
Functional Dependencies:

fd1: rental_id → customer_id, plate_no, payment_id, voucher_code, verification_status, withdraw_status, withdraw_receipt_path, destination, payment_status, invoice_path, created_at, updated_at, destination_lat, destination_lng, fuel_penalty, late_penalty, due_notification_sent, penalty_notification_sent

1NF: All attributes contain atomic values. Rental details like destination_lat, destination_lng, deposit_deduction, and late_penalty are stored as single scalar values.

10. pickups

pickups (id, rental_id, start_time, pick_up_location, pickup_lat, pickup_lng, car_condition_pickup, dashboard_image, damage_image, car_description_pickup, pickup_fuel_level, agreement_form, acknowledgement, pickup_report_path, created_at, updated_at)



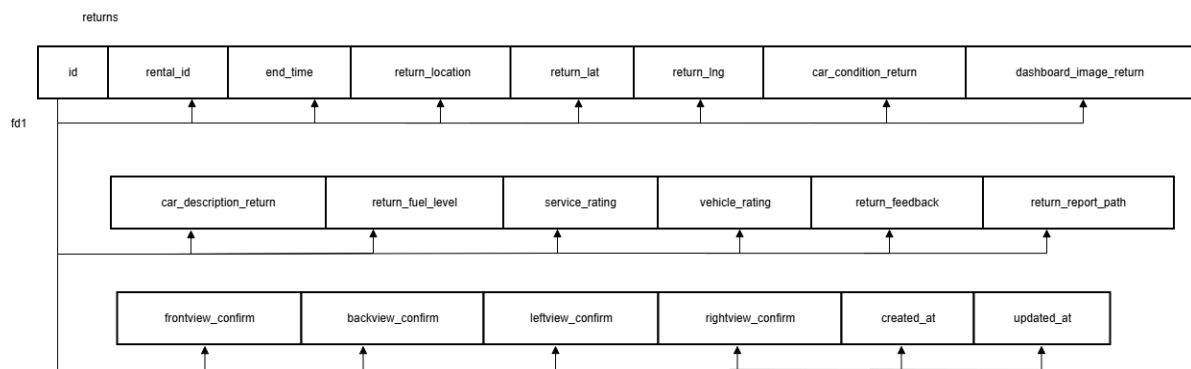
Functional Dependencies:

fd1: id → rental_id, start_time, pick_up_location, pickup_lat, pickup_lng, car_condition_pickup, dashboard_image, damage_image, car_description_pickup, pickup_fuel_level, agreement_form, acknowledgement, pickup_report_path, created_at, updated_at

1NF: All attributes are atomic. Pickup details like pickup_lat, pickup_lng, pickup_fuel_level, and acknowledgement are stored as single values without arrays or nested structures.

11. returns

returns (id, rental_id, end_time, return_location, return_lat, return_lng, car_condition_return, dashboard_image_return, damage_image_return, car_description_return, return_fuel_level, service_rating, vehicle_rating, return_feedback, return_report_path, frontview_confirm, backview_confirm, leftview_confirm, leftview_confirm, rightview_confirm, return_report_path, created_at, updated_at)



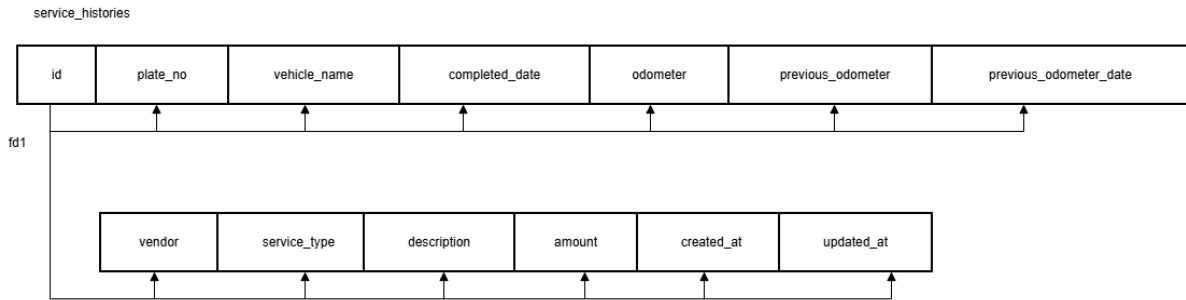
Functional Dependencies:

fd1: id→rental_id, end_time, return_location, return_lat, return_lng, car_condition_return, dashboard_image_return, damage_image_return, car_description_return, return_fuel_level, service_rating, vehicle_rating, return_feedback, return_report_path, frontview_confirm, backview_confirm, leftview_confirm, leftview_confirm, rightview_confirm, return_report_path, created_at, updated_at

1NF: All attributes contain atomic values. Return details like return_lat, return_lng, return_fuel_level, service_rating, and vehicle_rating are single-valued without multi-valued or composite attributes.

12. service_histories

service_histories (id, plate_no, vehicle_name, completed_date, odometer, previous_odometer, previous_odometer_date, vendor, service_type, description, amount, created_at, updated_at)



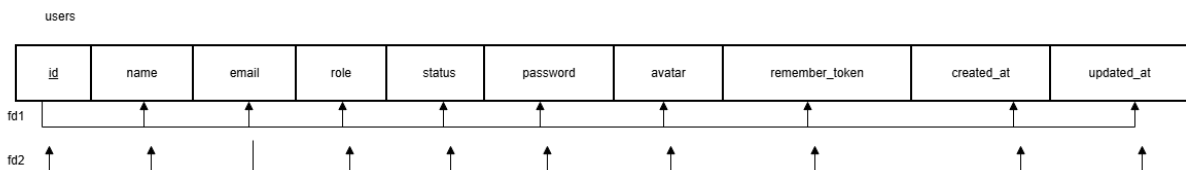
Functional Dependencies:

fd1: $id \rightarrow plate_no, vehicle_name, completed_date, odometer, previous_odometer, previous_odometer_date, vendor, service_type, description, amount, created_at, updated_at$

1NF: All attributes are atomic. Service record details like odometer, previous_odometer, amount, and completed_date are stored as single values without repeating groups.

13. users

users (id, name, email, role, status, email_verified_at, password, avatar, remember_token, created_at, updated_at)



Functional Dependencies:

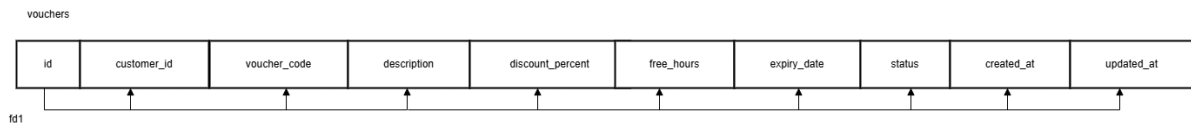
fd1: $id \rightarrow name, email, role, status, email_verified_at, password, avatar, remember_token, created_at, updated_at$

fd2: $email \rightarrow id, name, role, status, password, avatar, remember_token, created_at, updated_at$

1NF: All attributes contain atomic values (no repeating groups or multi-valued attributes). Each column contains only single values like email, role, and status.

14. vouchers

vouchers (id, customer_id, voucher_code, description, discount_percent, free_hours, expiry_date, status, created_at, updated_at)



Functional Dependencies:

fd1: $\text{id} \rightarrow \text{customer_id}, \text{voucher_code}, \text{description}, \text{discount_percent}, \text{free_hours}, \text{expiry_date}, \text{status}, \text{created_at}, \text{updated_at}$

1NF: All attributes are atomic. Voucher details like discount_percent, free_hours, expiry_date, and status are stored as single values without arrays or nested structures.

4.3.3 Second Normal Form (2NF)

A database table is in Second Normal Form (2NF) when it initially fulfills all the criteria of First Normal Form (1NF). Moreover every non-key attribute should be totally functionally reliant on the whole primary key. It implies that the non-key attributes must not rely on a subset of the primary key. Depending on the type of primary key used in a table, partial dependencies can always exist only where a composite primary key is present and therefore must be removed to reach 2NF.

2NF Analysis

The database has all tables that satisfy the Second Normal Form. The tables have a single attribute key as a primary key instead of composite key. It is impossible to have a partial dependence in this database design since the only form of dependence possible is partial dependency which occurs in the presence of a composite primary key. Also, the non-key attributes of all the tables are all functionally dependent as far as their primary keys. Thus, the database design is relevant to the needs of Second Normal Form.

Example of 2NF Compliance

1. rental table

Primary Key: rental_id (single attribute)

Functional Dependency:

$\text{rental_id} \rightarrow \text{customer_id}, \text{plate_no}, \text{payment_id}, \text{verification_status}, \text{withdraw_status}, \text{withdraw_receipt_path}, \text{destination}, \text{payment_status}, \text{invoice_path}, \text{destination_lat}, \text{destination_lng}, \text{voucher_code}, \text{fuel_penalty}, \text{late_penalty}, \text{due_notification_sent}, \text{penalty_notification_sent}, \text{created_at}, \text{updated_at}$

2. car_photos table

Primary Key: id (single attribute)

Functional Dependency:

$id \rightarrow \text{plate_no}, \text{path}, \text{label}, \text{created_at}, \text{updated_at}$

3. customers table

Primary Key: customer_id (single attribute)

Functional Dependency:

customer_id $\rightarrow$ status, user_id, fullname, nationality, matric_staff_no, ic_passport, gender, faculty, residential_college, address, phone, emergency_contact_name, relationship, emergency_phone, license_no, license_expiry, license_image, identity_card_image, matric_staff_image, documents_status, documents_rejection_reason, documents_submitted_at, documents_approved_at, documents_verified_by, bank_name, bank_account, created_at, updated_at

4.3.4 Third Normal Form (3NF)

A table in Third Normal Form (3NF) satisfies the criteria of Second Normal Form (2NF) and has no transitive dependencies. It is so because any other non-key attributes should not be reliant on other non-key attributes but rather be reliant on the primary key. Eradicating transitive dependencies will facilitate less redundancy and will exclude anomaly in updates in the database.

3NF Analysis

Every table in the database is in Third Normal form. No transitive dependencies occur since none of the non-key attributes depends upon anything other than the primary key. The non key attribute is not functionally dependent on another non key attribute. So, the design of the database is used in accordance with the demands of the Third Normal Form.

Example of 3NF Compliance

1. customers table

customer_id $\rightarrow$ bank_name (direct dependency)

customer_id $\rightarrow$ bank_account (direct dependency)

2. car table

plate_no $\rightarrow$ roadtax_path (direct dependency)

plate_no $\rightarrow$ roadtax_expiry (direct dependency)

3. rental table:

rental_id → late_penalty (direct dependency)

rental_id → verification_status (direct dependency)

4. payments table

payment_id → amount (direct dependency)

payment_id → deposit (direct dependency)

payment_id → refund_amount (direct dependency)

4.3.5 Boyce-Codd Normal Form (BCNF)

A table in the Boyce-Codd Normal Form (BCNF) is one that satisfies all the conditions of the Third Normal Form (3NF) to begin with. Also, the determinant X should be a superkey of every functional dependency -. This implies that all the determinants in the table should be candidate keys that can identify a unique tuple. This stricter rule further eliminates redundancy and also removes anomalies that can still occur despite a table being in 3NF.

Functional Dependencies Analysis

1. users table

Primary Key: id

Candidate Keys: id, email (unique constraint)

Functional Dependencies:

fd1: id → name, email, role, status, password, avatar, remember_token, created_at, updated_at

fd2: email → id, name, role, status, password, avatar, remember_token, created_at, updated_at

Analysis:

- fd1: id is the primary key (superkey)
- fd2: email is a candidate key (superkey)
- All determinants are superkeys → BCNF satisfied

2. admins table

Primary Key: admin_id

Candidate Keys: admin_id, user_id, phone (unique constraints)

Functional Dependencies:

fd1: admin_id → user_id, status, phone, position, created_at, updated_at

fd2: $\text{user_id} \rightarrow \text{admin_id}$ (one-to-one with users)

fd3: $\text{phone} \rightarrow \text{admin_id}$ (unique identifier)

Analysis:

- All determinants are superkeys $\rightarrow$ BCNF satisfied

5.0 Relational DB Schemas (After Normalization)

users (id, name, email, role, status, email_verified_at, password, avatar, remember_token, created_at, updated_at)

PK: id

UNIQUE: email

customers (customer_id, status, user_id, fullname, nationality, matric_staff_no, ic_passport, gender, faculty, residential_college, address, phone, emergency_contact_name, relationship, emergency_phone, license_no, license_expiry, license_image, identity_card_image, matric_staff_image, documents_status, documents_rejection_reason, documents_submitted_at, documents_approved_at, documents_verified_by, bank_name, bank_account, created_at, updated_at)

PK: customer_id

FK: user_id REFERENCES User (id)

admins (admin_id, user_id, status, phone, created_at, updated_at)

PK: admin_id

FK: user_id REFERENCES User (id)

car (plate_no, model, manufacturer, seats, colour, is_featured, availability_status, fuel_level, car_mileage, image_path, production_year, status, engine, engine_cc, fuel_type, transmission, drive_type, top_speed, max_power, max_torque, maintenance_status, owner_name, owner_nric, owner_age, owner_gender, owner_race, owner_citizenship, date_register_jpj, operating_state, grant_path, grant_expiry, roadtax_path, roadtax_expiry, covernote_path, covernote_expiry, created_at, updated_at)

PK: plate_no

car_photos (id, plate_no, path, label, created_at, updated_at)

PK: id

FK: plate_no REFERENCES Car (plate_no)

rental (rental_id, customer_id, plate_no, payment_id, voucher_code, verification_status, withdraw_status, withdraw_receipt_path, destination, payment_status, invoice_path, created_at, updated_at, destination_lat, destination_lng, fuel_penalty, late_penalty, due_notification_sent, penalty_notification_sent)

PK:rental_id

FK1: customer_id REFERENCES Customers (customer_id)

FK2: plate_no REFERENCES Car (plate_no)

FK3: payment_id REFERENCES Payments (payment_id)

payments (payment_id, staff_id, amount, receipt_path, verification_status, reject_reason, penalty, deposit, payment, payment_type, refund_amount, created_at, updated_at)

PK: payment_id

FK1: staff_id REFERENCES Admins (admin_id)

loyalties (id, customer_id, type, description, rental_counter, no_of_stamp, created_at, updated_at)

PK: id

FK: customer_id REFERENCES Customers (customer_id)

vouchers (id, customer_id, voucher_code, description, discount_percent, free_hours, expiry_date, status, created_at, updated_at)

PK: id

FK: customer_id REFERENCES Customers (customer_id)

UNIQUE: voucher_code

notifications (id, user_id, type, notifiable_type, notifiable_id, data, read_at, created_at, updated_at)

PK: id

FK: user_id REFERENCES Users(id)

pickups (id, rental_id, start_time, pick_up_location, pickup_lat, pickup_lng, car_condition_pickup, dashboard_image, damage_image, car_description_pickup, pickup_fuel_level, agreement_form, acknowledgement, pickup_report_path, created_at, updated_at)

PK: id

FK: rental_id REFERENCES rental(rental_id)

returns (id, rental_id, end_time, return_location, return_lat, return_lng, car_condition_return, dashboard_image_return, damage_image_return, car_description_return, return_fuel_level, service_rating, vehicle_rating, return_feedback, return_report_path, frontview_confirm, backview_confirm, leftview_confirm, leftview_confirm, rightview_confirm, return_report_path, created_at, updated_at)

PK: id

FK: rental_id REFERENCES rental (rental_id)

service_histories (id, plate_no, vehicle_name, completed_date, odometer, previous_odometer, previous_odometer_date, vendor, service_type, description, amount, created_at, updated_at)

PK: id

FK: plate_no REFERENCES car (plate_no)

renewal_reminders (id, plate_no, service_type, due_date, note, status, created_at, updated_at)

PK: id

FK: plate_no REFERENCES car (plate_no)

6.0 SQL Statements (DDL & DML)

6.1 admins Table

DDL:

```
CREATE TABLE admins (  
    admin_id VARCHAR(50) NOT NULL,  
    user_id BIGINT UNSIGNED NOT NULL,  
    status VARCHAR(20) NOT NULL DEFAULT 'active',  
    phone VARCHAR(20),  
    position VARCHAR(20),  
    created_at TIMESTAMP NULL,  
    updated_at TIMESTAMP NULL,  
  
    CONSTRAINT pk_admins PRIMARY KEY (admin_id),  
    CONSTRAINT uq_admins_admin_id UNIQUE (admin_id),  
    CONSTRAINT uq_admins_phone UNIQUE (phone),  
    CONSTRAINT fk_admins_user  
        FOREIGN KEY (user_id)  
        REFERENCES users(id)  
        ON DELETE CASCADE  
);
```

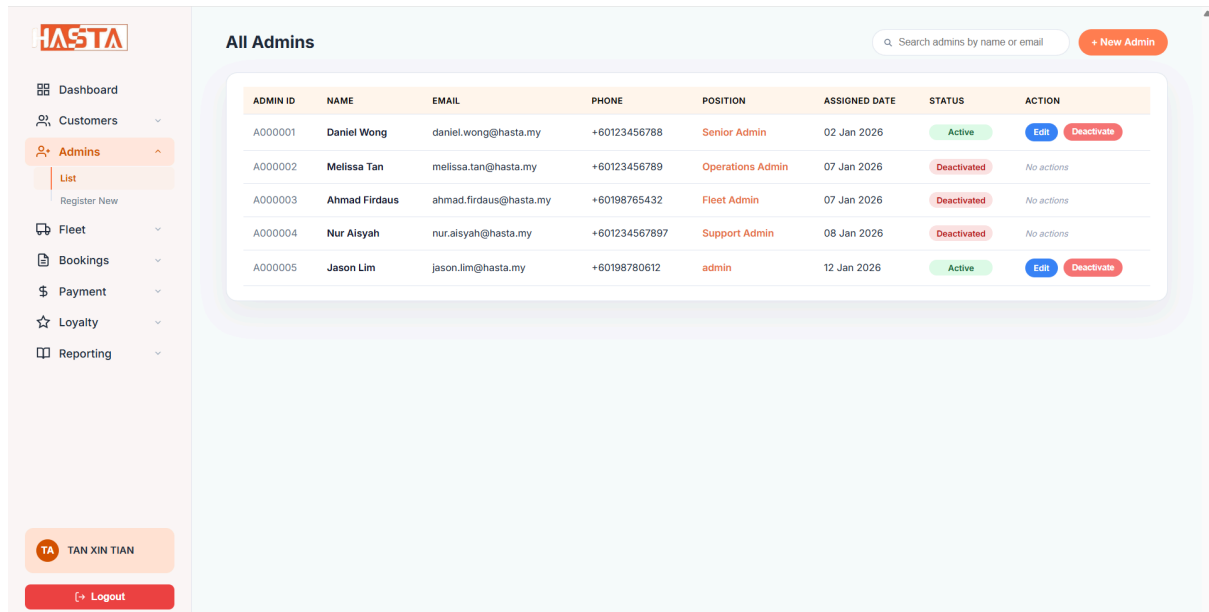


Figure 6.1.1 Admin List Page

DML#1: Select, Join

SELECT

a.admin_id,

u.name,

u.email,

a.phone,

a.position,

a.status,

a.created_at

FROM admins a

JOIN users u ON a.user_id = u.id

ORDER BY a.created_at DESC;

Figure 6.1.2 Create Admin Account Page

DML#2: Insert

```

INSERT INTO admins (
    admin_id,
    user_id,
    status,
    phone,
    position,
    created_at,
    updated_at
)
VALUES (
    'A000006',
    65,
    'active',
    '60123456789',
    'Operations Manager',
    CURRENT_TIMESTAMP,
    CURRENT_TIMESTAMP
);

```


Edit Admin
Update admin details and credentials.

← Back to List

Account Details

Full Name *
Daniel Wong

Email Address *
daniel.wong@hasta.my

Phone Number
+60199887766

Position *
Senior Admin

Change Password (Optional)
Leave blank if you don't want to change the password.

New Password
Min. 8 characters

Confirm New Password
Re-enter password

Cancel Update Admin

TA TAN XIN TIAN

Logout

Figure 6.1.3 Update Admin Account Page (Phone)

DML#3: Update

UPDATE admins

SET phone = '60199887766',

updated_at = CURRENT_TIMESTAMP

WHERE admin_id = 'A000001';

All Admins

Admin details updated successfully for Daniel Wong

dataate.test says
Are you sure you want to deactivate this admin?

OK Cancel

ADMIN ID	NAME	EMAIL	PHONE	POSITION	ASSIGNED DATE	STATUS	ACTION
A000001	Daniel Wong	daniel.wong@hasta.my	+60199887766	Senior Admin	02 Jan 2026	Active	Edit Deactivate
A000002	Melissa Tan	melissa.tan@hasta.my	+60123456789	Operations Admin	07 Jan 2026	Deactivated	No actions
A000003	Ahmad Firdaus	ahmad.firdaus@hasta.my	+60198765432	Fleet Admin	07 Jan 2026	Deactivated	No actions
A000004	Nur Aisyah	nur.aisyah@hasta.my	+601234567897	Support Admin	08 Jan 2026	Deactivated	No actions
A000005	Jason Lim	jason.lim@hasta.my	+60198780612	admin	12 Jan 2026	Active	Edit Deactivate

TA TAN XIN TIAN

Logout

Figure 6.1.4 Update Admin Account Page (Deactivated)

DML#4: Update (Deactivated)

UPDATE admins

SET status = 'deactivated',

updated_at = CURRENT_TIMESTAMP

WHERE admin_id = 'A000001';

ADMIN ID	NAME	EMAIL	PHONE	POSITION	ASSIGNED DATE	STATUS	ACTION
A000001	Daniel Wong	daniel.wong@hasta.my	+60123456788	Senior Admin	02 Jan 2026	Active	Edit Deactivate
A000002	Melissa Tan	melissa.tan@hasta.my	+60123456789	Operations Admin	07 Jan 2026	Deactivated	No actions
A000003	Ahmad Firdaus	ahmad.firdaus@hasta.my	+60198765432	Fleet Admin	07 Jan 2026	Deactivated	No actions
A000004	Nur Aisyah	nur.aisyah@hasta.my	+601234567897	Support Admin	08 Jan 2026	Deactivated	No actions
A000005	Jason Lim	jason.lim@hasta.my	+60198780612	admin	12 Jan 2026	Active	Edit Deactivate

Figure 6.1.5 Dummy Data of Admins

DML#5: Insert Dummy Data (more in SQL file)

INSERT INTO admins (

created_at,

updated_at,

admin_id,

user_id,

status,

phone,

position

) VALUES

('2026-01-02 01:17:07', '2026-01-11 09:26:56', 'A000001', 3, 'active', '+60123456788', 'Senior Admin'),

('2026-01-07 08:12:50', '2026-01-07 08:14:01', 'A000002', 8, 'deactivated', '+60123456789', 'Operations Admin'),

('2026-01-07 09:00:22', '2026-01-07 09:00:29', 'A000003', 9, 'deactivated', '+60198765432', 'Fleet Admin'),

('2026-01-07 19:06:55', '2026-01-07 19:07:18', 'A000004', 11, 'deactivated', '+601234567897', 'Support Admin'),

('2026-01-12 01:25:20', '2026-01-12 01:25:20', 'A000005', 63, 'active', '+60198780612', 'admin');

ADMIN ID	NAME	EMAIL	PHONE	POSITION	ASSIGNED DATE	STATUS	ACTION
A000001	Daniel Wong	daniel.wong@hasta.my	+60199887766	Senior Admin	02 Jan 2026	Active	Edit Deactivate
A000002	Melissa Tan	melissa.tan@hasta.my	+60111222333	Operations Manager	07 Jan 2026	Deactivated	No actions
A000003	Ahmad Firdaus	ahmad.firdaus@hasta.my	+60199887767	System Admin	07 Jan 2026	Deactivated	No actions
A000004	Nur Aisyah	nur.aisyah@hasta.my	+601234567897	Support Admin	08 Jan 2026	Deactivated	No actions
A000005	Jason Lim	jason.lim@hasta.my	+60198780612	admin	12 Jan 2026	Active	Edit Deactivate
A000006	Adam Johnson	adam.johnson@hasta.my	+60170000067	System Admin	01 Feb 2026	Active	Edit Deactivate
A000007	Benjamin Lee	benjamin.lee@hasta.my	+60170000068	Support Admin	01 Feb 2026	Active	Edit Deactivate
A000008	Charles Wong	charles.wong@hasta.my	+60170000069	Operations Admin	01 Feb 2026	Active	Edit Deactivate
A000009	Daniel Tan	daniel.tan@hasta.my	+60170000070	Finance Admin	01 Feb 2026	Active	Edit Deactivate
A000010	Ethan Lim	ethan.lim@hasta.my	+60170000071	HR Admin	01 Feb 2026	Active	Edit Deactivate
A000011	Felix Chan	felix.chan@hasta.my	+60170000072	System Admin	01 Feb 2026	Active	Edit Deactivate
A000012	George Ng	george.ng@hasta.my	+60170000073	Support Admin	01 Feb 2026	Active	Edit Deactivate

Figure 6.1.6 Updated Dummy Data of Admins

DML#6: Updated Dummy Data (more in SQL file)

UPDATE admins

SET

phone = CASE admin_id

WHEN 'A000002' THEN '+60111222333'

WHEN 'A000003' THEN '+60199887767'

END,

position = CASE admin_id

WHEN 'A000002' THEN 'Operations Manager'

WHEN 'A000003' THEN 'System Admin'

END,

updated_at = CURRENT_TIMESTAMP

WHERE admin_id IN ('A000002', 'A000003');

6.2 car Table

DDL:

```
CREATE TABLE `car` (  
    plate_no VARCHAR(20) NOT NULL,  
    model VARCHAR(150) NOT NULL,  
    seats TINYINT UNSIGNED DEFAULT NULL,  
    colour VARCHAR(30) DEFAULT NULL,  
    is_featured TINYINT(1) NOT NULL DEFAULT 0,  
    availability_status TINYINT(1) NOT NULL DEFAULT 1,  
    status VARCHAR(20) NOT NULL DEFAULT 'active',  
    maintenance_status TINYINT(1) NOT NULL DEFAULT 1,  
    fuel_level INT NOT NULL DEFAULT 100,  
    car_mileage DECIMAL(10,2) NOT NULL DEFAULT 0.00,  
    image_path VARCHAR(255) DEFAULT NULL,  
    pricing_rules JSON DEFAULT NULL,  
    production_year YEAR DEFAULT NULL,  
    engine VARCHAR(100) DEFAULT NULL,  
    engine_cc INT DEFAULT NULL,  
    fuel_type VARCHAR(20) DEFAULT NULL,  
    transmission VARCHAR(20) DEFAULT NULL,  
    drive_type VARCHAR(20) DEFAULT NULL,  
    top_speed INT DEFAULT NULL,  
    max_power DECIMAL(8,2) DEFAULT NULL,  
    max_torque DECIMAL(8,2) DEFAULT NULL,  
    owner_name VARCHAR(100) DEFAULT NULL,  
    manufacturer VARCHAR(50) DEFAULT NULL,  
    owner_nric VARCHAR(20) DEFAULT NULL,  
    owner_age TINYINT UNSIGNED DEFAULT NULL,  
    owner_gender VARCHAR(20) DEFAULT NULL,  
    date_register_jpj DATE DEFAULT NULL,  
    operating_state VARCHAR(50) DEFAULT NULL,
```

```

owner_race VARCHAR(30) DEFAULT NULL,
owner_citizenship VARCHAR(30) DEFAULT NULL,
grant_path VARCHAR(255) DEFAULT NULL,
grant_expiry DATE DEFAULT NULL,
roadtax_path VARCHAR(255) DEFAULT NULL,
roadtax_expiry DATE DEFAULT NULL,
covernote_path VARCHAR(255) DEFAULT NULL,
covernote_expiry DATE DEFAULT NULL,
created_at TIMESTAMP NULL,
updated_at TIMESTAMP NULL,

CONSTRAINT pk_car PRIMARY KEY (plate_no),
CONSTRAINT chk_car_status CHECK (status)
);

```

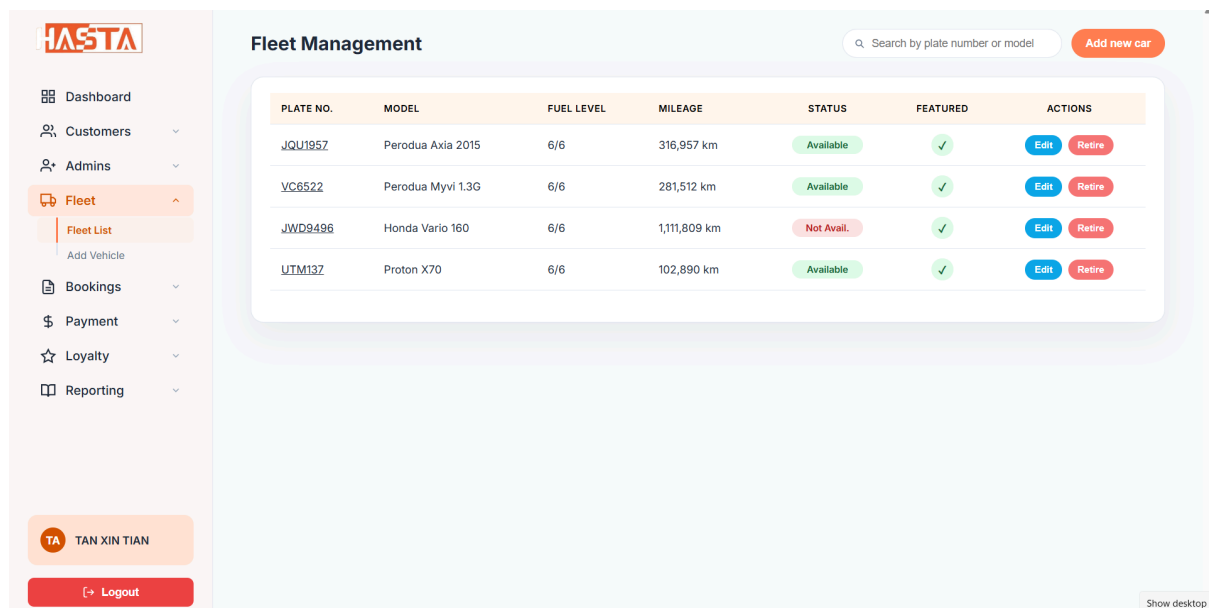


Figure 6.2.1 Fleet Management Page

DML#1: Select

```

SELECT
    plate_no,
    model,
    fuel_level,

```

car_mileage,
 availability_status,
 is_featured,
 status
 FROM car
 ORDER BY created_at DESC;

HASTA

Dashboard
Customers
Admins
Fleet
Fleet List
Add Vehicle
Bookings
Payment
Loyalty
Reporting

TAN XIN TIAN
Logout

Add New Car
Create a new car in your fleet.

Basic info

Plate number *
Model *
Production year
Engine
e.g. 1.3L Dual VVT-i
Text description, e.g. "1.3L Dual VVT-i" or "1.5L Turbo".
Colour
Seats
-- Select seats --

Specs

Engine CC
e.g. 1496
Numeric displacement in cc, e.g. "1496".
Fuel type
-- Select fuel type --
Transmission
-- Select transmission --
Drive type
-- Select drive type --
Top speed (km/h)
Max power (hp)
Max torque (Nm)

Status

Fuel level *
6/6
Mileage (km) *
Availability *
Available
Maintenance *
No (available)
Featured *
Yes (show in hero)

Image

Car image
Choose File No file chosen
Optional. JPG/PNG, up to 2MB.

Cancel Next: Set Pricing

Figure 6.2.2 Add Car Page

DML#2: Insert

INSERT INTO car (
 plate_no,
 model,
 seats,

```

colour,
is_featured,
production_year,
engine,
engine_cc,
fuel_type,
transmission,
drive_type,
top_speed,
max_power,
max_torque,
availability_status,
maintenance_status,
fuel_level,
car_mileage,
image_path,
created_at,
updated_at
) VALUES (
'UTM8888',
'Perodua Bezza 1.3X',
5,
'Blue',
0,
'2024',
'1.3L DOHC Dual VVT-i',
1329,
'Petrol',
'Automatic',
'FWD',
180,

```

```

94.00,
121.00,
1,
0,
6,
12000.00,
'Fleet/UTM8888/UTM8888_main.png',
CURRENT_TIMESTAMP,
CURRENT_TIMESTAMP
);

```

HASTA

- Dashboard
- Customers
- Admins
- Fleet**
 - Fleet List
 - Add Vehicle
- Bookings
- Payment
- Loyalty
- Reporting

TA TAN XIN TIAN

[Logout](#)

Set Pricing Rules

Configure hourly rates for weekday and weekend bookings.

Car Information

Plate: VC6522 | Model: Perodua Myvi 1.3G | Color: Silver

Weekday Pricing (Monday - Friday)

Duration	Rate
1 Hour	50
3 Hours	70
5 Hours	85
7 Hours	95
9 Hours	105
12 Hours	130
24 Hours	155

Weekend Pricing (Saturday - Sunday)

Duration	Rate
1 Hour	60
3 Hours	80
5 Hours	95
7 Hours	105
9 Hours	115
12 Hours	140
24 Hours	170

Figure 6.2.3 Update Pricing Rules Page

DML#3: Update

UPDATE car

SET pricing_rules = '{


```

"weekday": {
  "1": 50,
  "3": 70,
  "5": 85,
  "7": 95,
  "9": 105,
  "12": 130,
  "24": 155
},
"weekend": {
  "1": 60,
  "3": 80,
  "5": 95,
  "7": 105,
  "9": 115,
  "12": 140,
  "24": 170
}
}',
updated_at = CURRENT_TIMESTAMP
WHERE plate_no = 'VC6522';

```

The screenshot shows a web application interface for 'HASTA Fleet Management'. On the left is a sidebar with navigation links: Dashboard, Customers, Admins, Fleet (selected), Fleet List, Add Vehicle, Bookings, Payment, Loyalty, and Reporting. Below the sidebar is a user profile for 'TA TAN XIN TIAN' and a 'Logout' button. The main content area is titled 'Fleet Management' and includes a search bar and an 'Add new car' button. It displays a table with 12 rows of dummy car data. The table has columns for Plate No., Model, Fuel Level, Mileage, Status, Featured, and Actions. The 'Status' column uses color-coded labels: 'Maintenance' (yellow) and 'Available' (green). The 'Featured' column shows a green checkmark for all cars. The 'Actions' column contains 'Edit' and 'Retire' buttons for each car.

PLATE NO.	MODEL	FUEL LEVEL	MILEAGE	STATUS	FEATURED	ACTIONS
UUM1095	Perodua Alza (2nd Gen)	5/6	100,020 km	Maintenance	✓	Edit Retire
UTM9473	Perodua Axia (2nd Gen)	6/6	20,469 km	Available	✓	Edit Retire
UTM305Z	Perodua Bezza 1.3	6/6	20,504 km	Available	✓	Edit Retire
UTM3365	Perodua Axia (2nd Gen)	6/6	34,738 km	Available	✓	Edit Retire
UTM3655	Perodua Bezza 1.3	6/6	64,447 km	Available	✓	Edit Retire
CEX5224	Perodua Axia (2nd Gen)	7/6	31,484 km	Available	✓	Edit Retire
MCP6113	Perodua Axia (1st Gen)	5/6	83,060 km	Maintenance	✓	Edit Retire
JPN1416	Perodua Myvi (2nd Gen)	6/6	143,464 km	Available	✓	Edit Retire
NDD7803	Perodua Axia (1st Gen)	6/6	94,095 km	Available	✓	Edit Retire
JQU195Z	Perodua Axia 2015	6/6	316,957 km	Available	✓	Edit Retire

Figure 6.2.4 Dummy Data of Car

DML#4: Insert Dummy Data (more in SQL file)

INSERT INTO car (

plate_no, model, seats, colour, is_featured,
 production_year, engine, engine_cc, fuel_type,
 transmission, drive_type, top_speed,
 max_power, max_torque,
 availability_status, status, maintenance_status,
 fuel_level, car_mileage, image_path,
 pricing_rules, created_at, updated_at,
 owner_name, manufacturer, operating_state, owner_citizenship

) VALUES

(

'JQU1957', 'Perodua Axia 2015', 5, 'Green', 1,
 '2013', '3-Cylinder DOHC', 991, 'Petrol',
 'Automatic', 'FWD', 155,
 66.00, 90.00,
 1, 'active', 0,
 6, 316957.00, 'Fleet/JQU1957/JQU1957_main.png',
 '{"weekday":{"1":45,"3":65,"5":80,"7":90,"9":100,"12":120,"24":145},
 "weekend":{"1":55,"3":75,"5":90,"7":100,"9":110,"12":135,"24":165}}',

'2026-01-02 10:19:08', '2026-01-07 06:42:52',
 'Fleet Department', 'Perodua', 'Johor', 'Malaysia'
),
 (
 'VC6522', 'Perodua Myvi 1.3G', 5, 'Silver', 1,
 '2016', '1.3L DOHC VVT-i', 1298, 'Petrol',
 'Automatic', 'FWD', 164,
 90.00, 117.00,
 1, 'active', 0,
 6, 281512.00, 'Fleet/VC6522/VC6522_main.png',
 '{"weekday":{"1":50,"3":70,"5":85,"7":95,"9":105,"12":130,"24":155},
 "weekend":{"1":60,"3":80,"5":95,"7":105,"9":115,"12":140,"24":170}}',
 '2026-01-02 10:09:15', '2026-01-07 06:44:01',
 'Fleet Department', 'Perodua', 'Johor', 'Malaysia'
),
 (
 'JWD9496', 'Honda Vario 160', 1, 'Yellow', 1,
 '2023', 'Single Cylinder SOHC', 160, 'Petrol',
 'Automatic', 'RWD', 100,
 15.30, 13.80,
 0, 'active', 0,
 6, 1111809.00, 'Fleet/JWD9496/JWD9496_main.png',
 '{"weekday":{"1":20,"3":45,"5":65,"7":75,"9":85,"12":100,"24":120},
 "weekend":{"1":25,"3":55,"5":75,"7":85,"9":95,"12":110,"24":130}}',
 '2026-01-02 09:12:48', '2026-01-07 07:00:05',
 'Fleet Department', 'Honda', 'Johor', 'Malaysia'
),
 (
 'UTM137', 'Proton X70', 5, 'White', 1,
 '2022', '3 Cyl DOHC', 1477, 'Petrol',
 'Automatic', 'RWD', 195,

```
177.00, 255.00,  
1, 'active', 0,  
6, 102890.00, 'Fleet/UTM137/UTM137_main.png',  
'{"weekday":{"1":60,"3":80,"5":90,"7":95,"9":105,"12":135,"24":159.99},  
  "weekend":{"1":70,"3":90,"5":100,"7":105,"9":115,"12":145,"24":170}}',  
'2026-01-02 08:55:13', '2026-01-12 10:05:38',  
'Eleanor Ting Pik En', 'Proton', 'Sarawak', 'Malaysia'  
);
```

6.3 car_photos Table

DDL:

```
CREATE TABLE car_photos (  
    id BIGINT UNSIGNED NOT NULL AUTO_INCREMENT,  
    plate_no VARCHAR(20) NOT NULL,  
    path VARCHAR(255) NOT NULL,  
    label VARCHAR(50),  
    created_at TIMESTAMP NULL,  
    updated_at TIMESTAMP NULL,  
  
    CONSTRAINT pk_car_photos PRIMARY KEY (id),  
    CONSTRAINT fk_car_photos_car  
        FOREIGN KEY (plate_no)  
        REFERENCES car(plate_no)  
        ON DELETE CASCADE  
);
```

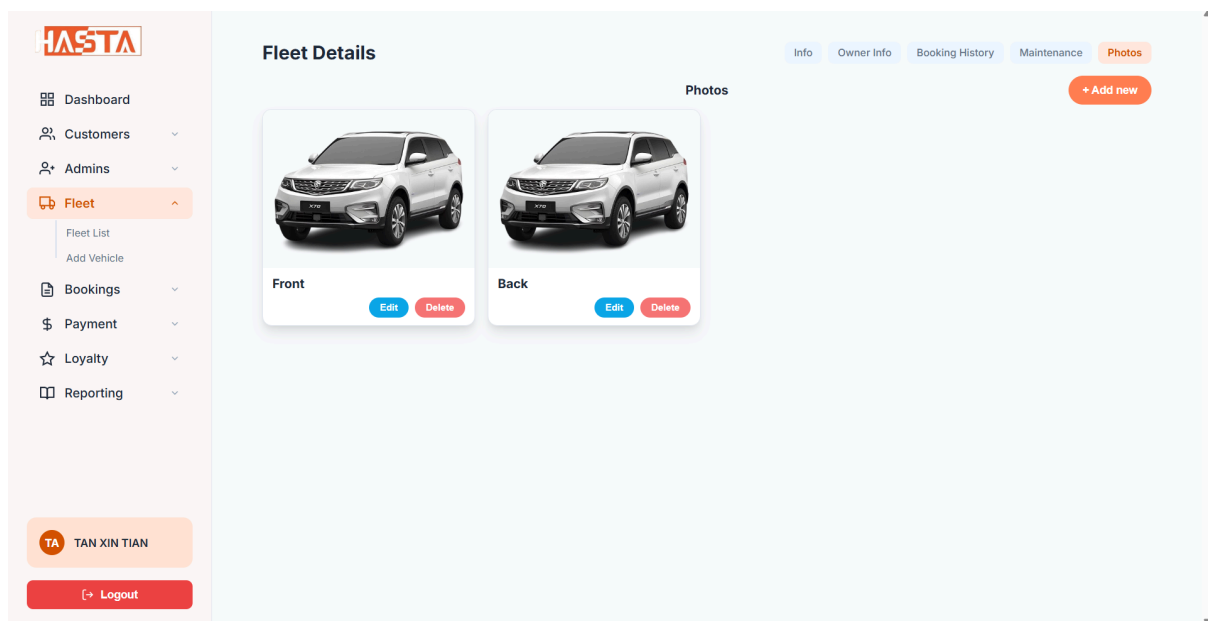


Figure 6.3.1 Fleet Details (Photos)

DML#1: Select

SELECT

id,

```

    plate_no,
    path,
    label,
    created_at
FROM car_photos
WHERE plate_no = 'UTM137'
ORDER BY created_at ASC;

```

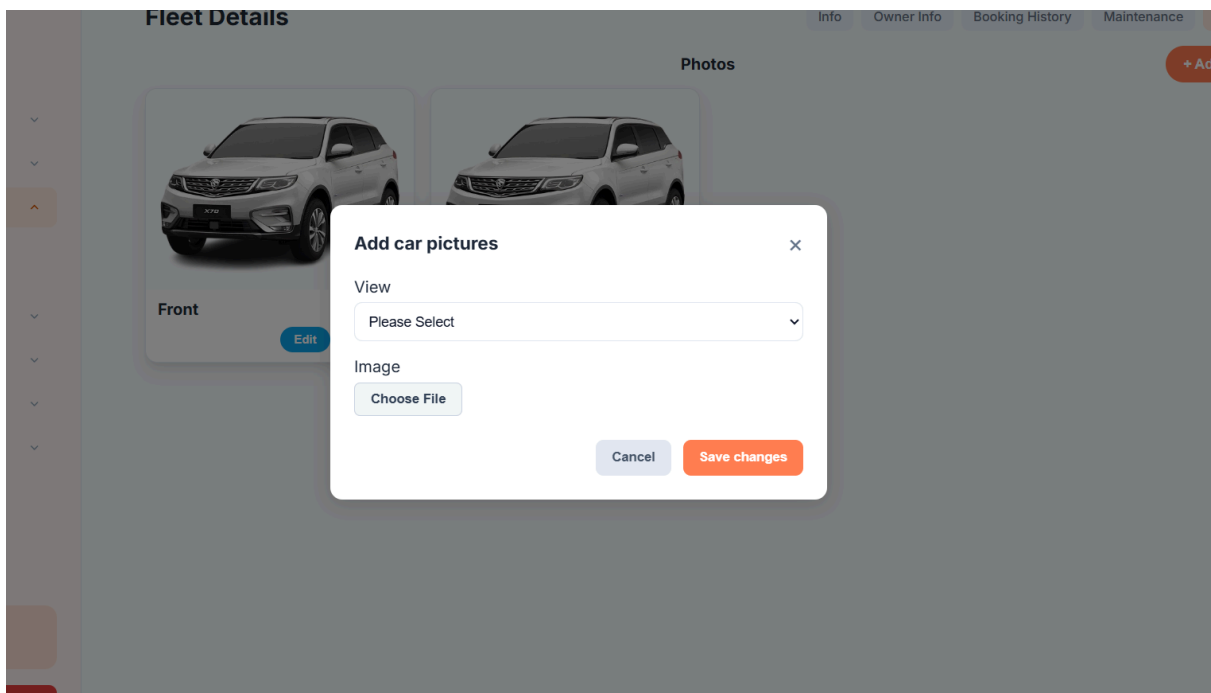


Figure 6.3.2 Insert Car Photos

DML#2: Insert

```

INSERT INTO car_photos (
    plate_no,
    path,
    label,
    created_at,
    updated_at
)
VALUES (
    'UTM137',

```

```
'Fleet/UTM137/UTM137_front_1767798564.png',  
'Front',  
CURRENT_TIMESTAMP,  
CURRENT_TIMESTAMP  
);
```

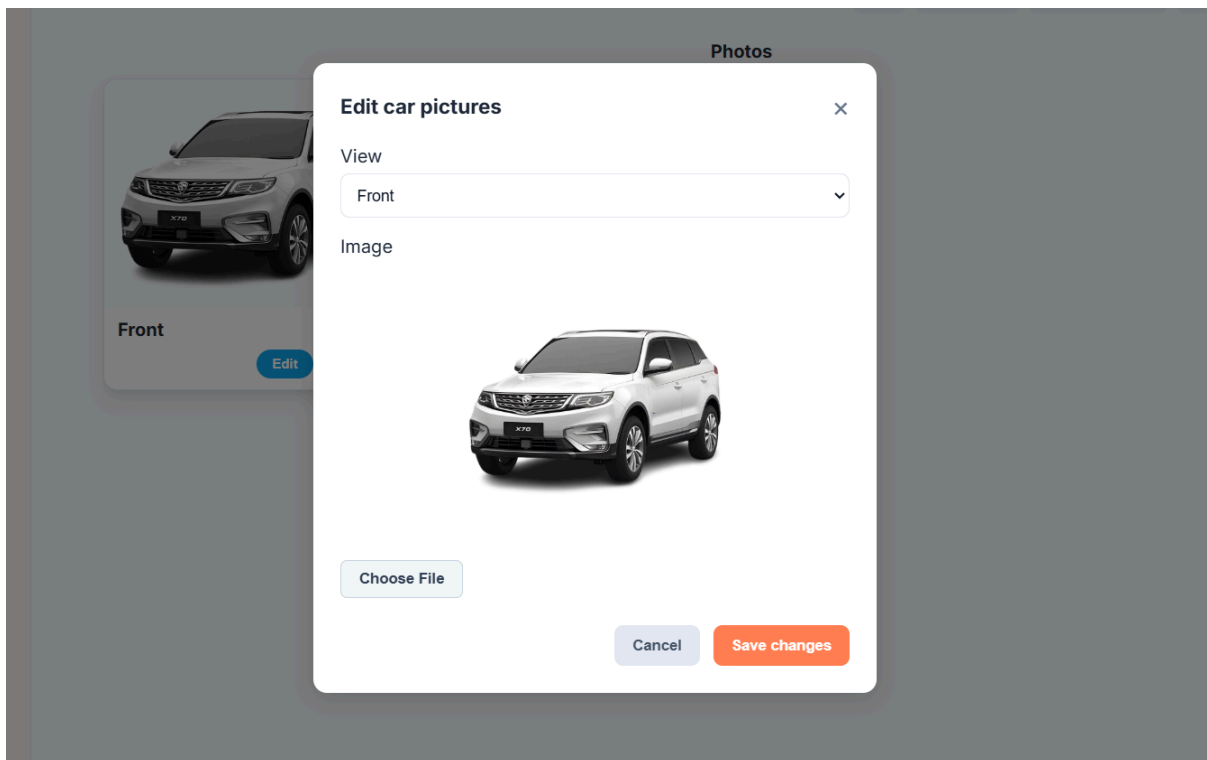


Figure 6.3.3 Update Car Photos

DML#3: Update

```
UPDATE car_photos
```

```
SET
```

```
path = 'Fleet/UTM137/UTM137_front_updated.png',
```

```
label = 'Front',
```

```
updated_at = CURRENT_TIMESTAMP
```

```
WHERE id = 7;
```

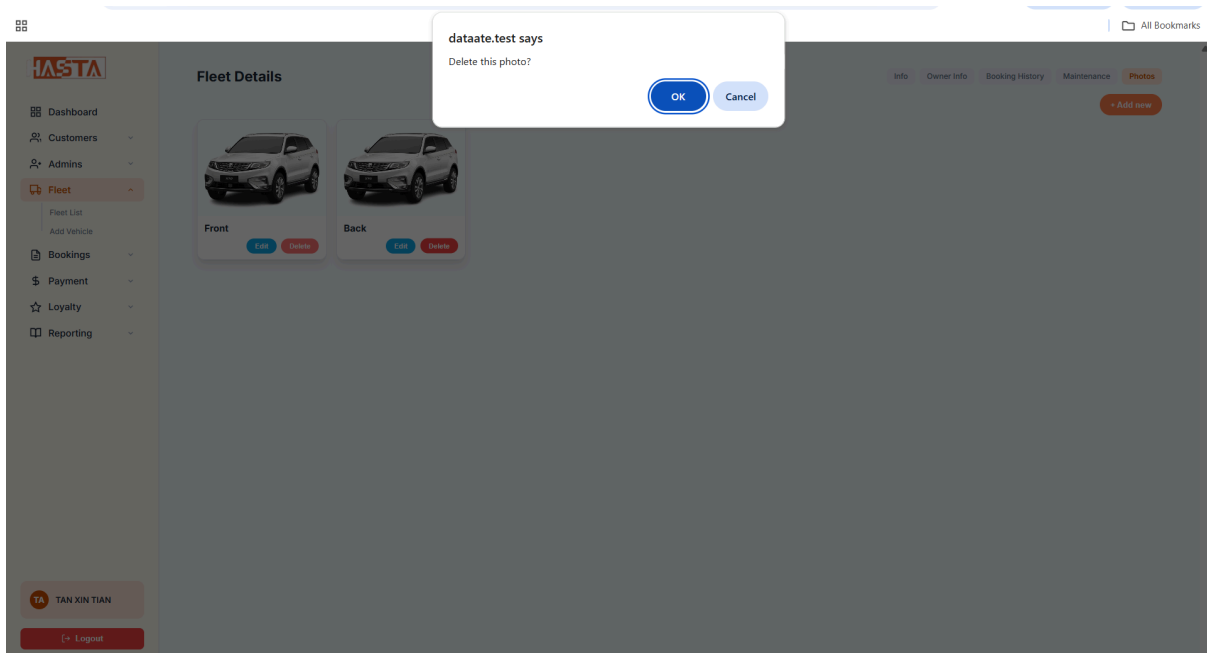


Figure 6.3.4 Delete Car Photos

DML#4: Delete

```
DELETE FROM car_photos
```

```
WHERE id = 7;
```

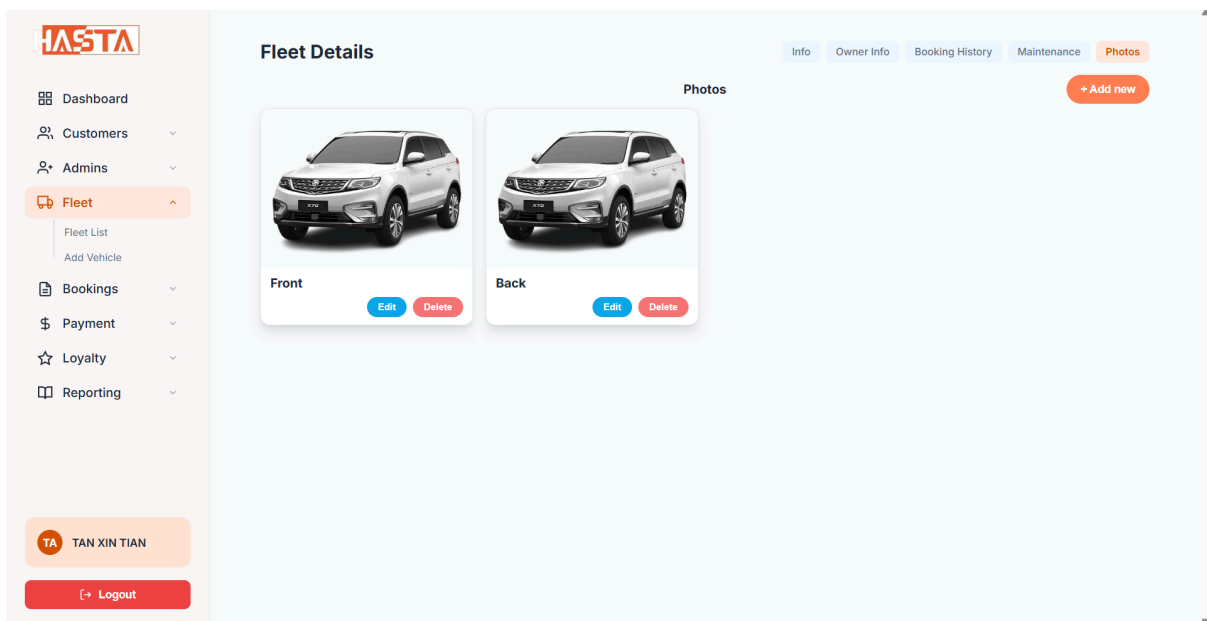


Figure 6.3.5 Dummy Data of Car Photos

DML#5: Insert Dummy Data (more in SQL file)

```
INSERT INTO `car_photos` (`id`, `plate_no`, `path`, `label`, `created_at`, `updated_at`)
VALUES
```


(9, 'VC6522', 'Fleet/VC6522/VC6522_front.png', 'Front', '2026-01-22 10:00:00', '2026-01-22 10:00:00'),
(10, 'VC6522', 'Fleet/VC6522/VC6522_back.png', 'Back', '2026-01-22 10:00:00', '2026-01-22 10:00:00'),
(11, 'VC6522', 'Fleet/VC6522/VC6522_left.png', 'Left', '2026-01-22 10:00:00', '2026-01-22 10:00:00'),
(12, 'VC6522', 'Fleet/VC6522/VC6522_right.png', 'Right', '2026-01-22 10:00:00', '2026-01-22 10:00:00'),
(13, 'VC6522', 'Fleet/VC6522/VC6522_interior.png', 'Interior', '2026-01-22 10:00:00', '2026-01-22 10:00:00'),
(14, 'NDD7803', 'Fleet/NDD7803/NDD7803_front.png', 'Front', '2026-01-22 10:01:00', '2026-01-22 10:01:00'),
(15, 'NDD7803', 'Fleet/NDD7803/NDD7803_back.png', 'Back', '2026-01-22 10:01:00', '2026-01-22 10:01:00'),
(16, 'NDD7803', 'Fleet/NDD7803/NDD7803_left.png', 'Left', '2026-01-22 10:01:00', '2026-01-22 10:01:00'),
(17, 'NDD7803', 'Fleet/NDD7803/NDD7803_right.png', 'Right', '2026-01-22 10:01:00', '2026-01-22 10:01:00'),
(18, 'NDD7803', 'Fleet/NDD7803/NDD7803_interior.png', 'Interior', '2026-01-22 10:01:00', '2026-01-22 10:01:00'),
(19, 'JPN1416', 'Fleet/JPN1416/JPN1416_front.png', 'Front', '2026-01-22 10:02:00', '2026-01-22 10:02:00'),
(20, 'JPN1416', 'Fleet/JPN1416/JPN1416_back.png', 'Back', '2026-01-22 10:02:00', '2026-01-22 10:02:00'),
(21, 'JPN1416', 'Fleet/JPN1416/JPN1416_left.png', 'Left', '2026-01-22 10:02:00', '2026-01-22 10:02:00'),
(22, 'JPN1416', 'Fleet/JPN1416/JPN1416_right.png', 'Right', '2026-01-22 10:02:00', '2026-01-22 10:02:00'),
(23, 'JPN1416', 'Fleet/JPN1416/JPN1416_interior.png', 'Interior', '2026-01-22 10:02:00', '2026-01-22 10:02:00'),

6.4 customers Table

DDL:

```
CREATE TABLE customers (  
  customer_id VARCHAR(50) NOT NULL,  
  status VARCHAR(20) NOT NULL DEFAULT 'active',  
  user_id BIGINT UNSIGNED NOT NULL,  
  fullname VARCHAR(100),  
  nationality VARCHAR(100),  
  matric_staff_no VARCHAR(20),  
  ic_passport VARCHAR(15),  
  gender VARCHAR(50),  
  faculty VARCHAR(100),  
  residential_college VARCHAR(100),  
  address TEXT,  
  phone VARCHAR(20),  
  emergency_contact_name VARCHAR(100),  
  relationship VARCHAR(50),  
  emergency_phone VARCHAR(20),  
  license_expiry DATE,  
  license_image VARCHAR(255),  
  identity_card_image VARCHAR(255),  
  matric_staff_image VARCHAR(255),  
  documents_status ENUM('pending','approved','rejected'),  
  documents_rejection_reason TEXT,  
  documents_submitted_at TIMESTAMP NULL,  
  documents_approved_at TIMESTAMP NULL,  
  documents_verified_by VARCHAR(255),  
  bank_name VARCHAR(100),  
  bank_account VARCHAR(30),  
  created_at TIMESTAMP NULL,  
  updated_at TIMESTAMP NULL,
```

```

CONSTRAINT pk_customers PRIMARY KEY (customer_id),
CONSTRAINT uq_customers_matric_staff_no UNIQUE (matric_staff_no),
CONSTRAINT uq_customers_phone UNIQUE (phone),
CONSTRAINT fk_customers_user
    FOREIGN KEY (user_id)
    REFERENCES users(id)
    ON DELETE CASCADE
);

```

NAME	CUSTOMER ID	EMAIL	PHONE	IC/PASSPORT	MATRIC/STAFF NO	GENDER	NATIONALITY	FACULTY	COLLEGE	ADDRESS	EMERGENCY CONTACT	LICENSE NO	LICENSE EXPIRY
Ambiga Vasudevan a/l Prem Raju	C000057	haani42@example.org	+6010-319 8711	970306132572	A24CS9982	Female	Malaysia	Faculty of Computing	KDSE	No. 78, Jalan BH, Bukit St...	Nur Hjh Wadhia Rais binti Haaf (Parent) +6014-387 3435	N/A	18 Apr 2028
B. C. Murugesan	C000043	nazira20@example.net	+6019-915 2678	880708107875	A21CS2602	Male	Malaysia	Faculty of Computing	KDSE	Lot 106, Jalan Tun Dr Ism...	Muhammed Huzair bin Zulayri (Sibling) +6017-374 6385	N/A	04 May 2029
Betty Siu Cher Pek	C000010	umasunder123@example.com	+6012-587 1912	680327015356	A24JH5818	Female	Malaysia	Faculty of Electrical Engineering	KDO	Lot 88, Jalan Keliling 9/6...	Lew Lar Hua (Parent) +6019-990 7784	N/A	09 Oct 2029
Blacklist1	C000003	blacklist@example.com	+60112-3456780	123456-12-3456	A24CS1234	Male	Malaysia	Azman Hashim International Business School (AHBS)	KTR	KRP	Mother (Parent) +6010-3456789	N/A	21 Jan 2030
Chen Choo Shok	C000044	arifuddin.thayaparam@example.com	+6012-342 5555	980112084855	A24JH1546	Male	Malaysia	Faculty of Chemical & Energy Engineering	KDSE	942, Jalan 900, Laman M...	Nur Emlyia Mazlan binti Kholis (Parent) +6011-3016 3015	N/A	03 May 2026
Hajjah Norsyafiqah Yuzelen	C000028	henry83@example.net	+6011-1931 4157	030816128477	A23JH7746	Male	Malaysia	Faculty of Electrical Engineering	KRP	3, Jin 5/3H, Kota Setia Ja...	Helmy Zain (Spouse) +6012-588 4105	N/A	26 Aug 2029
Haran Thevdran	C000040	einyadi@example.org	+6016-032 6099	070607023383	A25KE3372	Male	Malaysia	Faculty of Electrical Engineering	KTDI	1, Jalan Baru 275, Taman ...	Muhammed Abdulloh Asyran bin Zamrol Syazwan (Parent) +6019-240 278	N/A	27 May 2029
Hj Alzam Zukri bin Saiful Shamsul	C000049	salam.quikang@example.net	+6013-155 7794	720619063853	A20JH3001	Male	Malaysia	Faculty of Computing	KDSE	8, Jalan Aniffin BGV, Batu ...	Charanru Sivanesan a/l Rohan Sambanthan (Spouse) +6019-624 9102	N/A	06 Oct 2029
Hj Wan Sabri Shafel bin Che Luthfi Sanorhizam	C000012	bchandran@example.com	+6012-380 6413	780427018621	A22KM8878	Male	Malaysia	Faculty of Chemical & Energy Engineering	KTDI	8, Jalan 7, Taman Amang...	Ko Da Xiao (Sibling) +6012-022 3222	N/A	24 Aug 2026
Hjh Arwina Erfan binti Huzafy	C000047	jimmy83@example.net	+6010-730 3957	990802748634	A25JH5638	Female	Malaysia	Faculty of Computing	KRP	No. 96, Jin 1/BL, RUSTH...	Muhamad Wan Sham bin Wan Rauzyan (Parent) +6019-084 9307	N/A	16 Apr 2030
Huzir a/l Shan Rishyakaran	C000046	lmalayalam@example.com	+6018-285 9585	071004109342	A22KE6790	Female	Malaysia	Faculty of Electrical Engineering	KTDI	8-7, Lorong 6/86, Seky...	R. S. Analka (Spouse) +6013-483 0142	N/A	23 Jan 2030
Kamalathathan Manikavasagam a/l Ramesh Perumal	C000056	genapathy.elaine@example.org	+6014-407 3525	780915011458	A20KE4313	Female	Malaysia	Faculty of Electrical Engineering	KTDI	72, Jin 7/2, Bandar Maluri...	Liew Ren Kuong (Spouse) +6011-1647 7046	N/A	28 Apr 2030

Figure 6.4.1 Customer List Page

DML#1: Select, Join

SELECT

```

c.customer_id,
c.fullname,
u.email,
c.phone,
c.ic_passport,
c.matric_staff_no,
c.gender,

```

```
c.nationality,  
c.faculty,  
c.residential_college,  
c.address,  
c.emergency_contact_name,  
c.emergency_phone,  
c.license_expiry  
FROM customers c  
JOIN users u  
  ON c.user_id = u.id  
ORDER BY c.created_at DESC;
```

[Back to Profile](#)

Personal Data

Personal Information

Emergency Contact

Documents

TY

Username

TAN XIN YEE

Full Name

Enter full name

Gender

Select Gender

Nationality

Malaysia

Matric / Staff Number

e.g. A24CS1234

Faculty

Select Faculty

Kolej

Select Kolej

Residential Address

Enter your address

Phone Number

e.g. +(60)12-3456789

Bank Name

e.g. Maybank, CIMB

Bank Number

Please enter your bank account number

Next: Emergency Contact →

Figure 6.4.2 Customer Profile Page (Insert Details)

DML#2: Insert

INSERT INTO customers (

customer_id,
user_id,
status,
fullname,
gender,
nationality,
matric_staff_no,

```
faculty,  
residential_college,  
address,  
phone,  
bank_name,  
bank_account,  
created_at,  
updated_at  
) VALUES (  
    'C000058',  
    66,  
    'active',  
    'Crystal Tan Xin Yee',  
    'Female',  
    'Malaysia',  
    'A24CS0198',  
    'Faculty of Electrical Engineering',  
    'Kolej Tun Hussein Onn',  
    '17 Jalan Sasa 16',  
    '+601121609266',  
    'RHB',  
    '1234567891011',  
    NOW(),  
    NOW()  
);
```

[Back to Profile](#)

Personal Data

Personal Information

Emergency Contact

Documents

TY

Username

TAN XIN YEE

Full Name

Crystal Tan Xin Yee

Gender

Female

▼

Nationality

Malaysia

Matric / Staff Number

A24CS0198

Faculty

Faculty of Electrical Engineering

▼

Kolej

Kolej Tun Hussein Onn

▼

Residential Address

17 Jalan Sasa 16

Phone Number

+6012-8889999

Bank Name

RHB

Bank Number

1234567891011

Next: Emergency Contact →

Figure 6.4.3 Customer Profile Page (Update Details)

DML#3: Update

UPDATE customers

SET

phone = '+6012-888 9999',

updated_at = CURRENT_TIMESTAMP

WHERE customer_id = 'C000058';

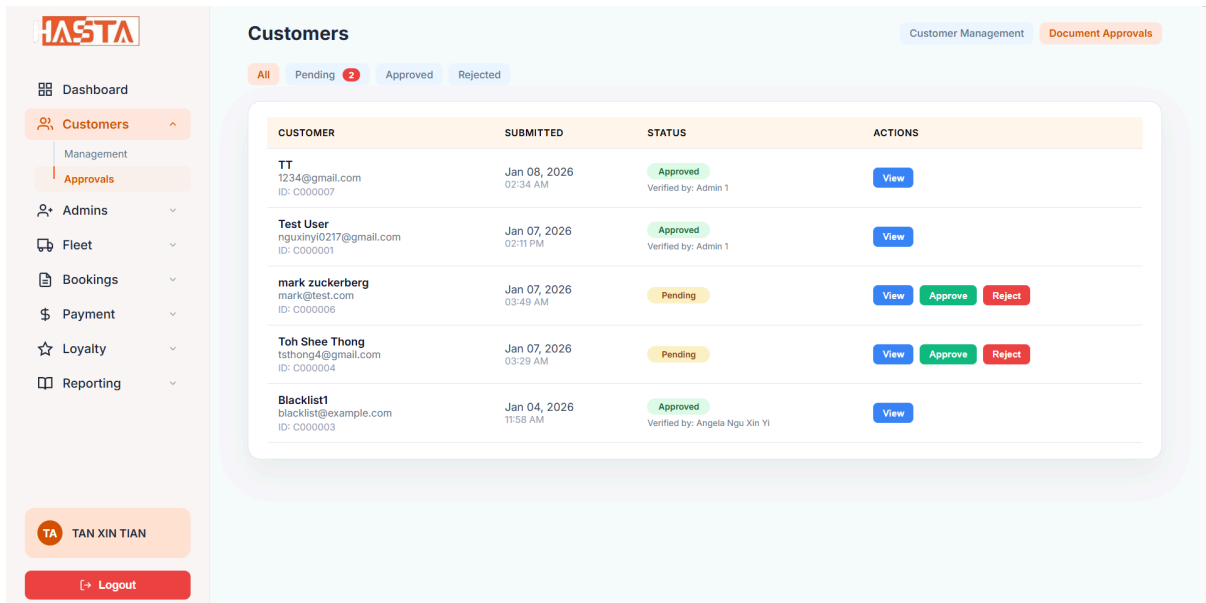


Figure 6.4.4 Customer Approval Page

DML#4: Update (Approve Customer Documents)

UPDATE customers

SET

documents_status = 'approved',

documents_approved_at = CURRENT_TIMESTAMP,

documents_verified_by = 'Admin A000001',

updated_at = CURRENT_TIMESTAMP

WHERE customer_id = 'C000004';

Customers													
Customer Management Document Approvals													
Search by customer name...													
NAME	CUSTOMER ID	EMAIL	PHONE	IC/PASSPORT	MATRIC/STAFF NO	GENDER	NATIONALITY	FACULTY	COLLEGE	ADDRESS	EMERGENCY CONTACT	LICENSE NO	LICENSE EXPIRY
Ambiga Vasudevan a/l Premia Raju	C000057	haanli42@example.org	+6010-319 8771	970306132572	A24CS9882	Female	Malaysia	Faculty of Computing	KDSE	No. 78, Jalan 8H, Bukit S...	Nuar Hjh Wadha Rals binti Hasif (Parent) +6014-287 3432	N/A	18 Apr 2028
B. C. Murugesan	C000043	nacira20@example.net	+6019-915 2678	880708107875	A21CS2602	Male	Malaysia	Faculty of Computing	KDSE	Lot 106, Jalan Tun Dr Ism...	Muhammed Hadzir bin Zulayri (Sibling) +6017-374 6385	N/A	04 May 2029
Betty Siu Chen Pek	C000070	umasundar123@example.com	+6012-587 1812	680327015356	A24JH5818	Female	Malaysia	Faculty of Electrical Engineering	KDO	Lot 89, Jalan Katilling 9/S...	Lew Lai Hua (Parent) +6019-930 7754	N/A	09 Oct 2029
Blacklist1	C000003	blacklist1@example.com	+6012-3456780	123456-12-3456	A24CS1234	Male	Malaysia	Azman Hashim International Business School (AHBS)	KTR	KRP	Mother (Parent) +6012-3456789	N/A	21 Jan 2028
Chen Choo Shok	C000044	arifuddin.thayaparan@example.com	+6012-342 6555	980112084855	A24JH1548	Male	Malaysia	Faculty of Chemical & Energy Engineering	KDSE	942, Jalan 900, Laman M...	Nur Emlyla Mazlan binti Khols (Parent) +6017-2058 585	N/A	03 May 2026
Hajjah Norsyafiqah Yuselan	C000028	henry83@example.net	+6011-1831 4157	030816108477	A23JH7746	Male	Malaysia	Faculty of Electrical Engineering	KRP	3, Jln 5/3H, Kota Setia Ja...	Helmy Zakri (Spouse) +6012-384 4785	N/A	26 Aug 2029
Haran Thevandran	C000040	eirayad@example.org	+6016-032 8099	070607023383	A25KE3372	Male	Malaysia	Faculty of Electrical Engineering	KTDI	1, Jalan Baru 275, Taman...	Muhammed Abdulloh Asyran bin Zamrud Syakwan (Parent) +6019-245 2718	N/A	27 May 2029
Hj Aizam Zukri bin Saiful Shamsul	C000049	salam.quikang@example.net	+6013-155 7794	720619063853	A20JH3001	Male	Malaysia	Faculty of Computing	KDSE	8, Jalan Ariffin 86V, Batu...	Chantany Sivanesan a/l Rohan Sambandan (Spouse) +6013-624 9102	N/A	06 Oct 2029
Hj Wan Sabri Shafel bin Che Lutfi Sanorhizam	C000012	bchandran@example.com	+6012-380 6413	780427018621	A22KM878	Male	Malaysia	Faculty of Chemical & Energy Engineering	KTDI	8, Jalan 7, Taman Amanja...	Ko Da Xia (Sibling) +6013-222 3222	N/A	24 Aug 2026
Hjh Arwina Erfan binti Huzzafy	C000047	jimmy93@example.net	+6010-730 3857	990607146834	A25JH5638	Female	Malaysia	Faculty of Computing	KRP	No. 96, Jln 1/8L, RUSW, ...	Muhammed Wan Sham bin Wan Razuan (Parent) +6015-4631942	N/A	16 Apr 2030
Huzir a/l Shan Risnykan	C000046	lmalayalam@example.com	+6018-285 6085	071004109342	A22KE6790	Female	Malaysia	Faculty of Electrical Engineering	KTDI	8-7, Lorong 6/86, Sekoye...	R. S. Anaka (Spouse) +6011-1647 7046	N/A	23 Jan 2030
Kamalanathan Manikavasagam a/l Ramesh Perumal	C000056	ganapathy.elaine@example.org	+6014-407 3523	790915011458	A20KE4313	Female	Malaysia	Faculty of Electrical Engineering	KTDI	72, Jln 7/2, Bandar Matul...	Liew Fen Kuong (Spouse) +6011-1647 7046	N/A	28 Apr 2030

Figure 6.4.5 Dummy Data of Customers

DML#5: Insert Dummy Data (more in SQL file)

INSERT INTO customers (

customer_id,

status,

user_id,

fullname,

nationality,

matric_staff_no,

ic_passport,

gender,

faculty,

residential_college,

address,

phone,

emergency_contact_name,

relationship,

emergency_phone,

license_expiry,

documents_status,

```

documents_submitted_at,
created_at,
updated_at,
bank_name,
bank_account
) VALUES
(
'C000001', 'active', 1, 'Angela Ngu Xin Yi', 'Malaysia',
'A24CS0124', '12345678902', 'Female', 'Faculty of Computing', 'KTDI',
'KDOJ', '0123456789', 'Ngu Siu Hook', 'Parent', '+6012-3456789',
'2026-01-16', 'approved', '2026-01-07 06:11:12',
'2026-01-02 01:12:05', '2026-01-07 06:27:42',
'Maybank', '1234567890'
),
(
'C000007', 'active', 10, 'TOH SHEE THONG', 'Malaysia',
'A24CS1324', '12345678902', 'Female',
'Faculty of Mechanical Engineering', 'KTDI',
'No. 2 Jalan Kristal 6', '+6012-3456789',
'Toh SSS', 'Parent', '+6012-345678',
'2026-01-13', 'approved', '2026-01-07 18:34:17',
'2026-01-07 18:32:11', '2026-01-07 18:35:04',
'Maybank', '13245789'
),
(
'C000008', 'active', 12, 'Phang Loi Chu', 'Malaysia',
'A22JH0254', '711122101867', 'Male',
'Faculty of Electrical Engineering', 'KTR',
'6, Jalan 9/4, Kondominium Hartamas', '+6010-5223001',
'Lee Qin Ruo', 'Parent', '+6017-9267413',
'2027-08-12', 'approved', '2026-01-07 19:27:22',

```

'2026-01-07 19:27:22', '2026-01-07 19:27:22',
 'RHBbank', '33224455'
),
 (
 'C000009', 'active', 13, 'Mohammed Hj Syed Sudirman bin A"zizul Suzaimi', 'Malaysia',
 'A25KE7399', '910610123209', 'Male',
 'Faculty of Electrical Engineering', 'KTR',
 'Bandar Tun Hussein Onn, Kedah', '+6011-17249252',
 'Vijandren Veeran', 'Parent', '+6014-3983641',
 '2030-01-04', 'approved', '2026-01-07 19:27:22',
 '2026-01-07 19:27:22', '2026-01-07 19:27:22',
 'BankRakyat', '55668877'
),
 (
 'C000010', 'active', 14, 'Betty Siu Cher Pek', 'Malaysia',
 'A24JH5818', '680327015356', 'Female',
 'Faculty of Electrical Engineering', 'KDOJ',
 'Lot 89, Jalan Keliling', '+6012-5971912',
 'Lew Lar Hua', 'Parent', '+6019-9907794',
 '2029-10-09', 'approved', '2026-01-07 19:27:22',
 '2026-01-07 19:27:22', '2026-01-07 19:27:22',
 'RHBbank', '33224455'
),
 (
 'C000011', 'active', 15, 'Nuur Hasmalinda Ubaidillah binti Arifin', 'Malaysia',
 'A24KE3739', '811218043718', 'Female',
 'Faculty of Electrical Engineering', 'KDOJ',
 'Taman Desa Flora, Negeri Sembilan', '+6011-23639639',
 'Umasundari a/l Diljit Devandran', 'Sibling', '+6019-0246756',
 '2030-08-09', 'approved', '2026-01-07 19:27:23',
 '2026-01-07 19:27:23', '2026-01-07 19:27:23',

'OCBC', '91888888'

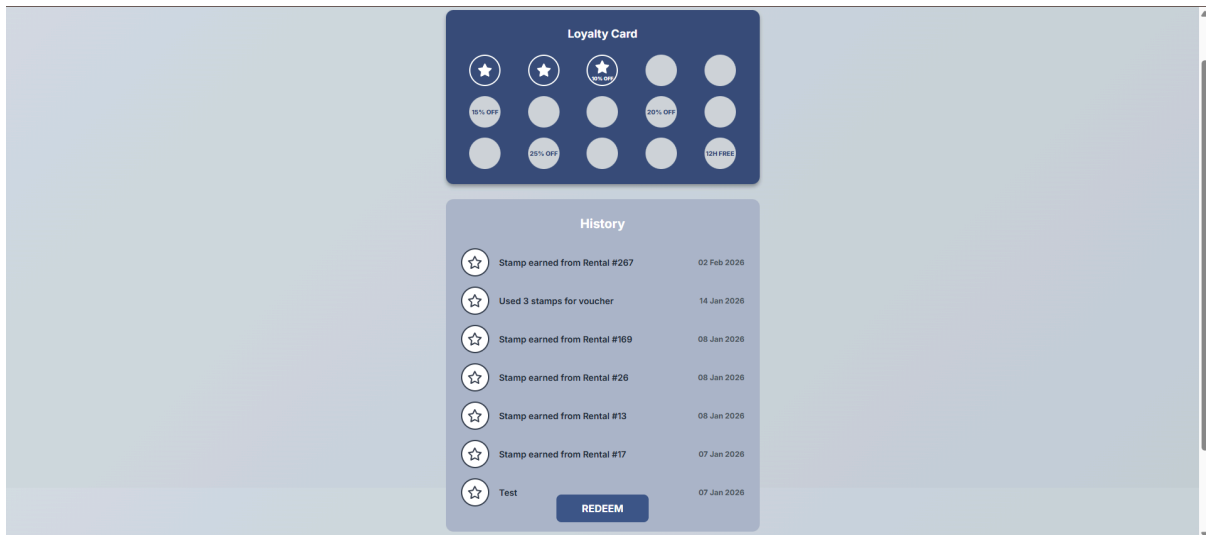
);

6.5 loyalties Table

DDL:

```
CREATE TABLE `loyalties` (  
  `id` bigint UNSIGNED NOT NULL,  
  `customer_id` varchar(50) COLLATE utf8mb4_unicode_ci NOT NULL,  
  `type` varchar(50) COLLATE utf8mb4_unicode_ci NOT NULL DEFAULT 'earned',  
  `description` varchar(255) COLLATE utf8mb4_unicode_ci NOT NULL,  
  `rental_counter` int NOT NULL,  
  `no_of_stamp` int NOT NULL,  
  `created_at` timestamp NULL DEFAULT NULL,  
  `updated_at` timestamp NULL DEFAULT NULL  
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_unicode_ci;
```

Customer Loyalty					
CUSTOMER		STAMPS	ACTIVE STATUS	CURRENT INVENTORY	PAST HISTORY
Dashboard Customers Admins Fleet Bookings Payment Loyalty Vouchers Customer Loyalties Rules	M. Rishyakaran rishyad@exampl.org	★ 3	None	-	REWARD-D4V13H (18% Off) [Used]
	Lam Kek Yang emantang@example.net	★ 0	None	-	REWARD-PGQ4QF (58% Off) [Expired]
	Hj Khalirul Nawawi bin Syed Hafizh shoba.chu@example.com	★ 0	None	-	-
	Palanivel a/ Vinod Shammugaratnam rasiah.anand@example.com	★ 1	None	-	REWARD-NC5QBC (2h Free) [Expired]
System Manager	Sybil a/ Manickavasagam Puaneswaran basianpila.kavindra@example.com	★ 0	None	-	REWARD-GU2NCK (18h Free) [Used]
	Loo Yan Wi saankar@example.net	★ 0	None	-	-
	Saninatass Jayaseelan a/ Sivarasa othargit@example.com	★ 0	None	-	REWARD-2U68BH (18% Off) [Expired], REWARD-8F388F (18h Free) [Expired]
	Puvaneswaran Xavier clew@example.org	★ 1	1 Active	REWARD-VPHU3T (5h Free)	-
Log Out	Nur Zatalini Basri binti Nik Salim muhammad57@example.net	★ 2	None	-	-
	Loganathan a/ Weeratunge devan.thang@example.com	★ 1	1 Active	REWARD-FC2SDJ (2h Free)	-
	K. Rajannaidu devsivar.gulide@example.net	★ 2	None	-	REWARD-LTT0GV (20% Off) [Expired]
	Nur Hjh Siti binti Wan Syazwan jojoee.arishah@example.com	★ 2	None	-	-
Sangeeta Ramadass a/ Selvaraju Sinnadavar batrisya00@example.com		★ 1	None	-	-



DML#1: Insert (Insert Sample Loyalties)

```
INSERT INTO `loyalties` (`id`, `customer_id`, `type`, `description`, `rental_counter`,  
`no of stamp`, `created at`, `updated at`) VALUES
```

(1, 'C000043', 'earned', 'Earned from rental #31', 1, 1, '2025-10-05 13:31:19', '2025-10-05 13:31:19'),

(2, 'C000056', 'earned', 'Earned from rental #39', 1, 1, '2025-10-08 13:31:19', '2025-10-08 13:31:19'),

```
(3, 'C000021', 'earned', 'Earned from rental #1', 1, 1, '2025-10-07 13:31:19', '2025-10-07 13:31:19'),
```

The screenshot shows the IASTA Customer Loyalty management interface. On the left is a sidebar with navigation links: Dashboard, Customers, Admins, Fleet, Bookings, Payment, Loyalty (selected), and Reporting. The Loyalty section has sub-links for Vouchers, Customer Loyalties, and Rules. At the bottom of the sidebar are buttons for System Manager and Logout. The main content area is titled 'Customer Loyalty' and includes a search bar with the email 'rasiah.anand@example.com'. Below the search bar is a table with the following data:

CUSTOMER	STAMPS	ACTIVE STATUS	CURRENT INVENTORY	PAST HISTORY
Palanivel a/ Vinod Shanmugaratnam rasiah.anand@example.com	★ 1	None	-	REWARD-NC5QBC (2h Free) [Expired]

Figure 6.5.2 Loyalty Stamps Filtered by Customer Email

DML#2: Select (Retrieve Loyalty Records)

```
SELECT id, customer_id, type, description, rental_counter, no_of_stamp
FROM loyalties
WHERE email= 'rasiah.anand@example.com';
```

6.6 notifications Table

DDL:

```
CREATE TABLE `notifications` (  
  `id` char(36) COLLATE utf8mb4_unicode_ci NOT NULL,  
  `type` varchar(100) COLLATE utf8mb4_unicode_ci NOT NULL,  
  `notifiable_type` varchar(100) COLLATE utf8mb4_unicode_ci NOT NULL,  
  `notifiable_id` varchar(50) COLLATE utf8mb4_unicode_ci NOT NULL,  
  `data` text COLLATE utf8mb4_unicode_ci NOT NULL,  
  `read_at` timestamp NULL DEFAULT NULL,  
  `created_at` timestamp NULL DEFAULT NULL,  
  `updated_at` timestamp NULL DEFAULT NULL  
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_unicode_ci;
```

DML#1: Insert Notification (More in SQL file)

```
INSERT INTO `notifications` (`id`, `type`, `notifiable_type`, `notifiable_id`, `data`, `read_at`,  
`created_at`, `updated_at`) VALUES
```

```
('a1111111-0000-1111-2222-abcdef000001', 'App\\Notifications\\CarPickupSuccess',  
'App\\Models\\User', '2', '{\"message\": \"Pickup for Honda Dash 125 successful! Safe  
driving.\", \"action_url\": \"http:\\\\example.com\\booking\\countdown?rental_id=7\", \"statu  
s\": \"success\", \"rental_id\": 7}', '2026-01-31 02:01:35', '2026-01-31 01:57:46', '2026-01-31  
02:01:35'),
```

```
('a1111111-0000-1111-2222-abcdef000002', 'App\\Notifications\\DepositDeductionNotification', 'App\\Models\\User', '2',  
'{\"booking_id\": 10, \"message\": \"Your security deposit has been deducted by RM30.00 due  
to missing fuel (3 levels less than pickup).\", \"type\": \"deposit_deduction\", \"amount\": 30}',  
'2026-01-31 02:01:35', '2026-01-31 01:57:46', '2026-01-31 02:01:35'),
```

```
('a1111111-0000-1111-2222-abcdef000003', 'App\\Notifications\\RentalReadyForPickup',  
'App\\Models\\User', '2', '{\"message\": \"Your rental for Perodua Bezza 2013 is ready for  
pickup!\", \"action_url\": \"http:\\\\example.com\\booking\\pickup?rental_id=8\", \"status\": \"  
success\", \"rental_id\": 8}', '2026-01-31 02:01:35', '2026-01-31 01:57:46', '2026-01-31  
02:01:35');
```

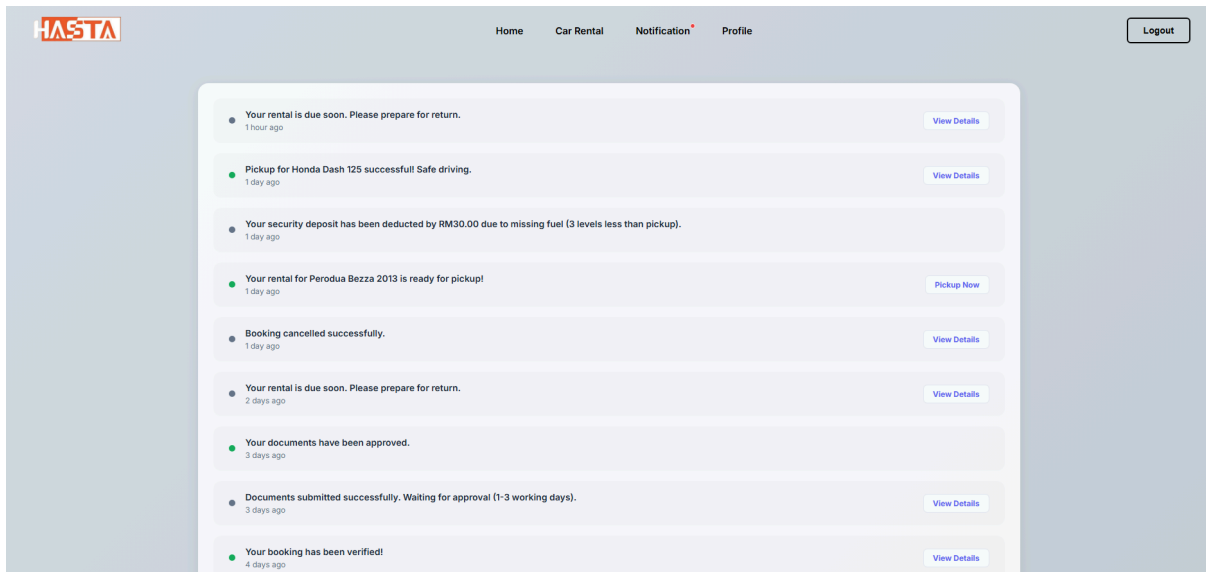



Figure 6.6.1 Notification Records

DML#2: Select (Retrieve Notification)

```
SELECT id, type, data, read_at, created_at
FROM notifications
WHERE notifiable_id = '2'
ORDER BY created_at DESC;
```

DML#3: Update (Mark a Notification as Read)

```
UPDATE notifications
SET read_at = NOW(),
    updated_at = NOW()
WHERE id = '184c4ab6-c1dd-445f-abe2-5ef20ca8ad3f';
```

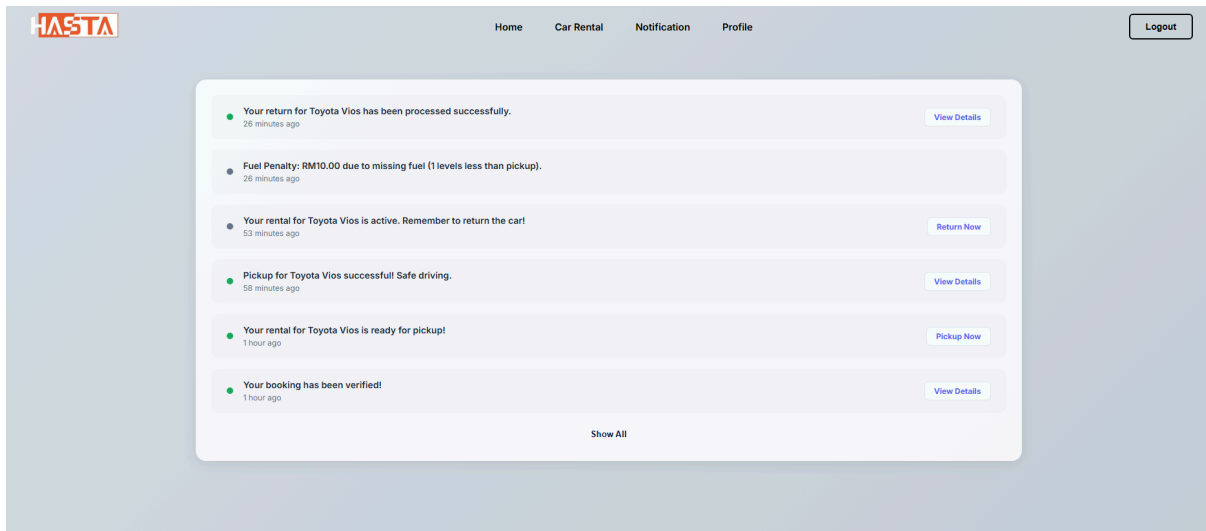


Figure 6.6.2 Notification Red Button Disappear when User Read the Notification

DML#4: Delete (Remove Old Read Notifications)

DELETE FROM notifications

WHERE read_at IS NOT NULL

AND created_at < DATE_SUB(NOW(), INTERVAL 30 DAY);

DML#5: Select (Retrieve Unread Notifications Only)

SELECT id, data, created_at

FROM notifications

WHERE notifiable_id = '2'

AND read_at IS NULL;

6.7 payments Table

DDL:

```
CREATE TABLE `payments` (  
  `payment_id` varchar(20) COLLATE utf8mb4_unicode_ci NOT NULL,  
  `staff_id` varchar(50) COLLATE utf8mb4_unicode_ci DEFAULT NULL,  
  `amount` decimal(10,2) NOT NULL,  
  `receipt_path` varchar(255) COLLATE utf8mb4_unicode_ci DEFAULT NULL,  
  `verification_status` varchar(50) COLLATE utf8mb4_unicode_ci NOT NULL DEFAULT  
'pending',  
  `reject_reason` text COLLATE utf8mb4_unicode_ci,  
  `penalty` decimal(10,2) NOT NULL DEFAULT '0.00',  
  `deposit` decimal(10,2) NOT NULL DEFAULT '0.00',  
  `payment` varchar(20) COLLATE utf8mb4_unicode_ci DEFAULT NULL COMMENT  
'Payment method or reference',  
  `payment_type` varchar(20) COLLATE utf8mb4_unicode_ci NOT NULL DEFAULT  
'booking',  
  `refund_amount` decimal(10,2) NOT NULL DEFAULT '0.00',  
  `created_at` timestamp NULL DEFAULT NULL,  
  `updated_at` timestamp NULL DEFAULT NULL  
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_unicode_ci;
```

BOOKING ID	DATE	CUSTOMER	TYPE	AMOUNT	DEPOSIT	RECEIPT	STATUS	VERIFIED BY	DETAILS
#71	Feb 01, 2026 10:05 PM	Toh Shee Thong tsthongt@gmail.com	Rental	MYR 415.00	MYR 50.00	View	Verified	Custom Admin	View
#46	Jan 07, 2026 02:31 PM	Ding Kak Woo kakoo@example.net	Rental	MYR 960.41	MYR 50.00	View	Verified	Mohd Hj Nur bin Amil Azraai	View
#57	Jan 07, 2026 12:02 PM	Mok Chor Dong hokkiee.munee@example.org	Rental	MYR 321.87	MYR 50.00	View	Verified	Welson Chee Yit Bow	View
#47	Jan 07, 2026 04:07 AM	Shanthi Sathasivam a/l Magendran Nagappan saravanan.adityana@example.org	Rental	MYR 312.55	MYR 50.00	View	Verified	Mohd Hj Nur bin Amil Azraai	View
#48	Jan 06, 2026 09:26 PM	Haani a/l Munshi Rayer rasamnah.vost@example.org	Rental	MYR 948.25	MYR 50.00	View	Verified	Mohd Haji Radzi Noorizman bin Rezai Bahari	View
#62	Jan 06, 2026 02:13 PM	Shamini Thirunavukarasu a/l Sanisvara shamini@example.com	Rental	MYR 640.06	MYR 50.00	View	Verified	Mohd Hj Nur bin Amil Azraai	View
#52	Jan 06, 2026 03:19 PM	Bishan a/l Govindasamy khar@example.net	Rental	MYR 547.83	MYR 50.00	View	Verified	Welson Chee Yit Bow	View
#58	Jan 06, 2026 04:13 AM	S. A. Singh sine@example.net	Rental	MYR 718.66	MYR 50.00	View	Verified	Mohd Hj Nur bin Amil Azraai	View
#53	Jan 05, 2026 09:20 AM	Kuan Khian Ong michele.ashtal@example.net	Rental	MYR 187.02	MYR 50.00	View	Verified	Custom Admin	View
#42	Jan 05, 2026 04:20 AM	Shi Ee Lex mipoosathal@example.net	Rental	MYR 988.49	MYR 50.00	View	Verified	Nur Serene binti Syed Samsuddin	View
#59	Jan 05, 2026 02:44 AM	B. C. Palanisamy yase.sarathi@example.org	Rental	MYR 890.94	MYR 50.00	View	Verified	Nur Serene binti Syed Samsuddin	View
#51	Jan 04, 2026 11:27 PM	S. A. Singh sine@example.net	Rental	MYR 643.09	MYR 50.00	View	Verified	Mohd Hj Nur bin Amil Azraai	View

Figure 6.7.1 Admin Site of Payment

DML#1: Insert Sample Payments

```
INSERT INTO `payments` (`payment_id`, `staff_id`, `amount`, `receipt_path`,
`verification_status`, `reject_reason`, `penalty`, `deposit`, `payment`, `payment_type`,
`refund_amount`, `created_at`, `updated_at`) VALUES
```

```
('PAY-1769954628-1137-', '1', 817.24, 'seed_assets/avatar.png', 'failed', 'Amount transferred
does not match booking price.', 0.00, 50.00, 'QR Payment', 'booking', 0.00, '2025-07-02
00:11:52', '2026-02-01 18:21:09'),
```

```
('PAY-1769954628-1310-', '2', 197.17, 'seed_assets/avatar.png', 'verified', NULL, 0.00, 50.00,
'QR Payment', 'booking', 0.00, '2024-10-13 18:08:19', '2026-02-01 18:21:09'),
```

```
('PAY-1769954628-2357-', '7', 677.80, 'seed_assets/avatar.png', 'verified', NULL, 0.00, 50.00,
'QR Payment', 'booking', 0.00, '2025-09-10 08:47:20', '2026-02-01 18:21:09'),
```

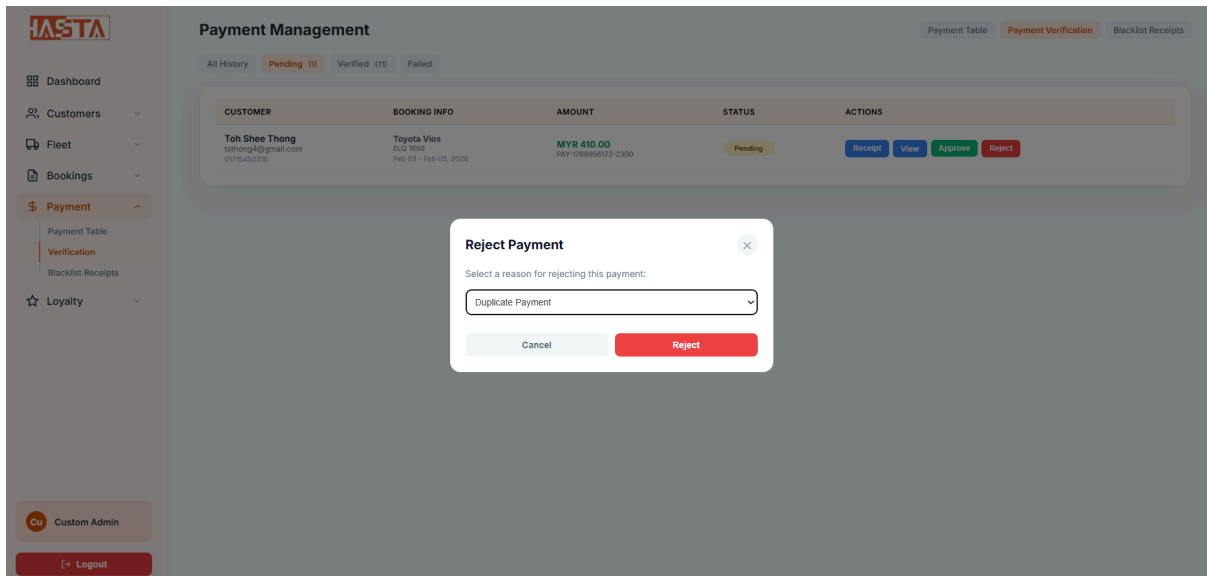


Figure 6.7.2 Admin Site for Reject Transaction

DML#2: Update (Reject a Transaction)

UPDATE payments

SET verification_status = 'failed',

reject_reason = 'Invalid Account/Transaction',

staff_id = A000001,

updated_at = NOW()

WHERE payment_id = 'PAY-1769956173-2300';

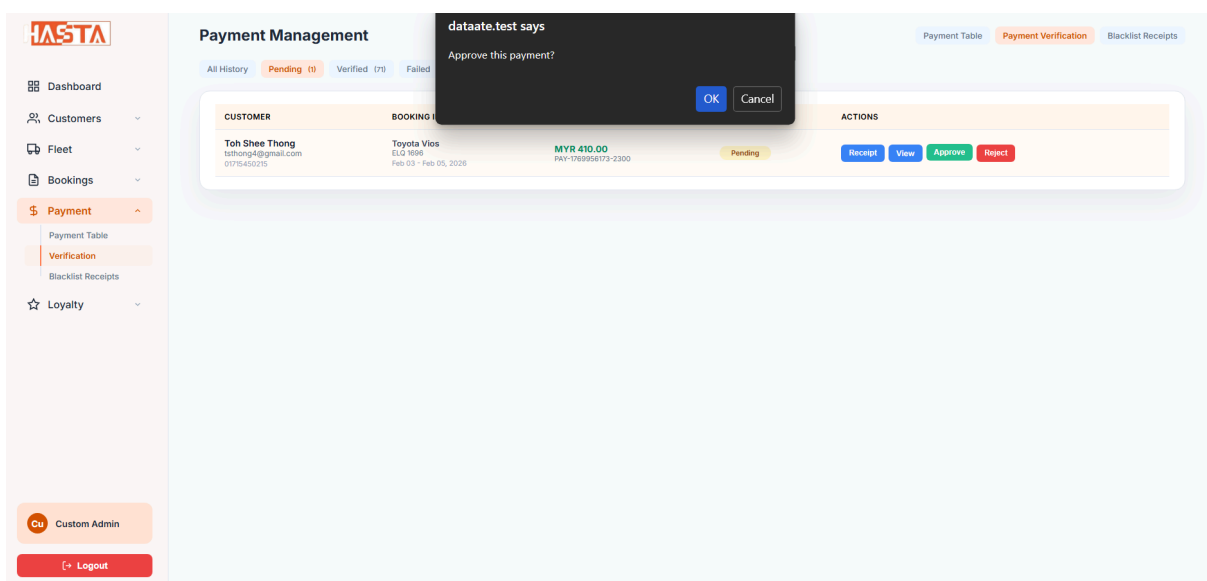


Figure 6.7.3 Admin Site for Approving Transaction

DML#3: Update (Approve a Transaction)

```

UPDATE payments
SET verification_status = 'approved',
    staff_id = A000001,
    updated_at = NOW()
WHERE payment_id = 'PAY-1769956173-2300';

```

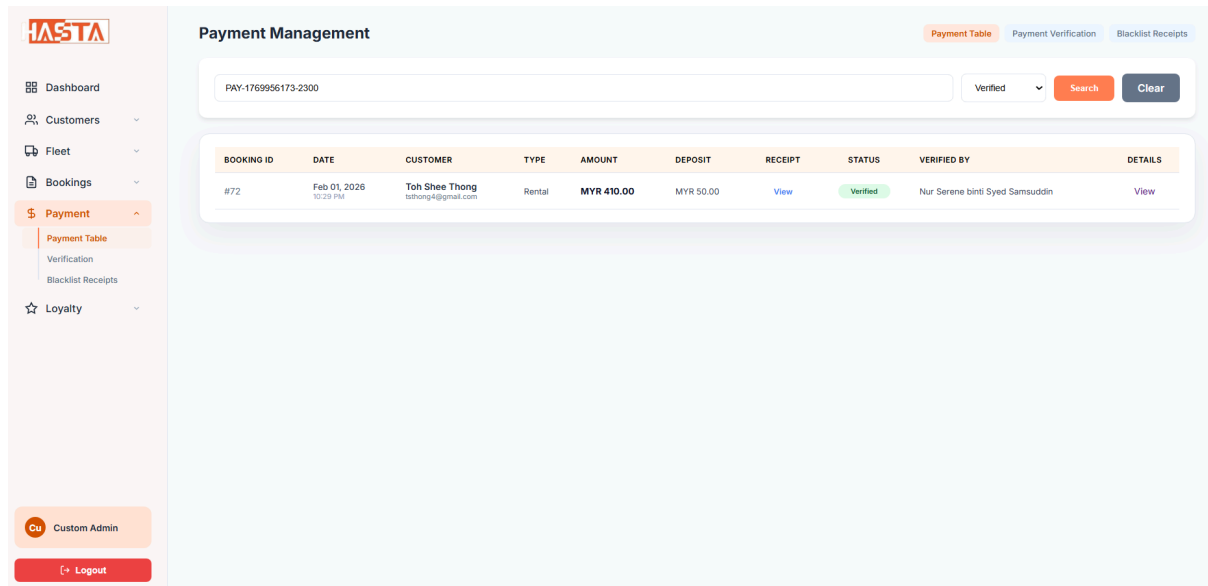


Figure 6.7.4 Admin Site for Filter Customer using customer_id and Status

DML#4: Select (Filter by customer_id and status)

```

SELECT payment_id, amount, verification_status, created_at
FROM payments
WHERE payment_id = 'PAY-1769956173-2300'
AND verification_status = 'verified';

```

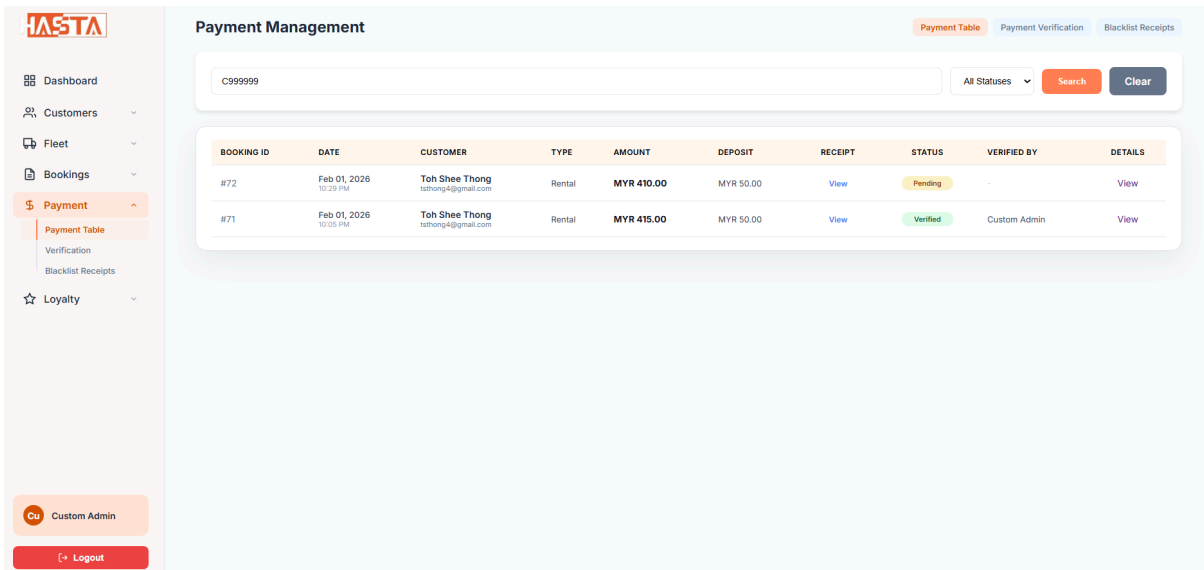


Figure 6.7.5 Admin Site for Finding Payment History for One Customer

DML#5: Select (View Payments Made by Customer)

```
SELECT payment_id, amount, verification_status, created_at
FROM payments
WHERE customer_id = 'C999999'
ORDER BY created_at DESC;
```

6.8 renewal_reminders Table

DDL:

```
CREATE TABLE `renewal_reminders` (  
  `id` BIGINT UNSIGNED NOT NULL AUTO_INCREMENT,  
  `plate_no` VARCHAR(20) NOT NULL,  
  `service_type` ENUM(  
    'Insurance',  
    'Roadtax',  
    'Investor Commission',  
    'Monthly Installment',  
    'Others'  
  ) NOT NULL,  
  `due_date` DATE NOT NULL,  
  `note` text COLLATE utf8mb4_unicode_ci,  
  `status` ENUM(  
    'upcoming',  
    'notified',  
    'completed'  
  ) NOT NULL DEFAULT 'upcoming',  
  `created_at` TIMESTAMP NULL DEFAULT CURRENT_TIMESTAMP,  
  `updated_at` TIMESTAMP NULL DEFAULT CURRENT_TIMESTAMP  
    ON UPDATE CURRENT_TIMESTAMP,  
  PRIMARY KEY (`id`),  
  KEY `idx_plate_no` (`plate_no`),  
  CONSTRAINT `fk_renewal_plate_no`  
    FOREIGN KEY (`plate_no`)  
    REFERENCES `car` (`plate_no`)  
    ON DELETE CASCADE  
)
```

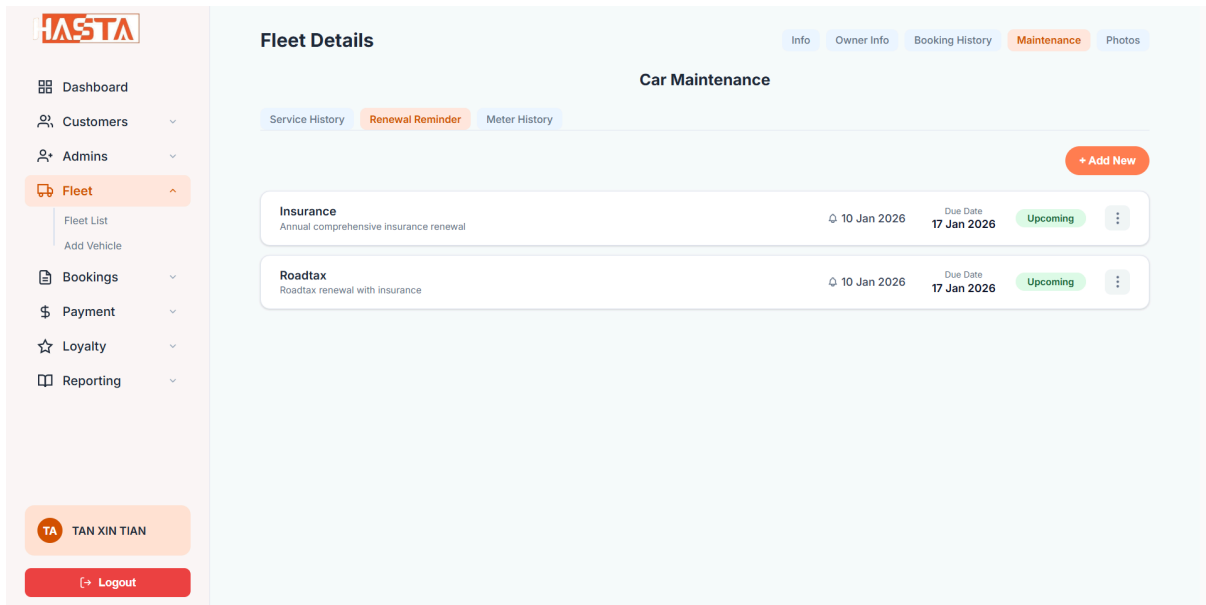



Figure 6.8.1 Renewal Reminder List

DML#1: Select

SELECT

id,
plate_no,
service_type,
note,
due_date,
status,
created_at

FROM renewal_reminders

WHERE plate_no = 'UTM137'

ORDER BY due_date ASC;

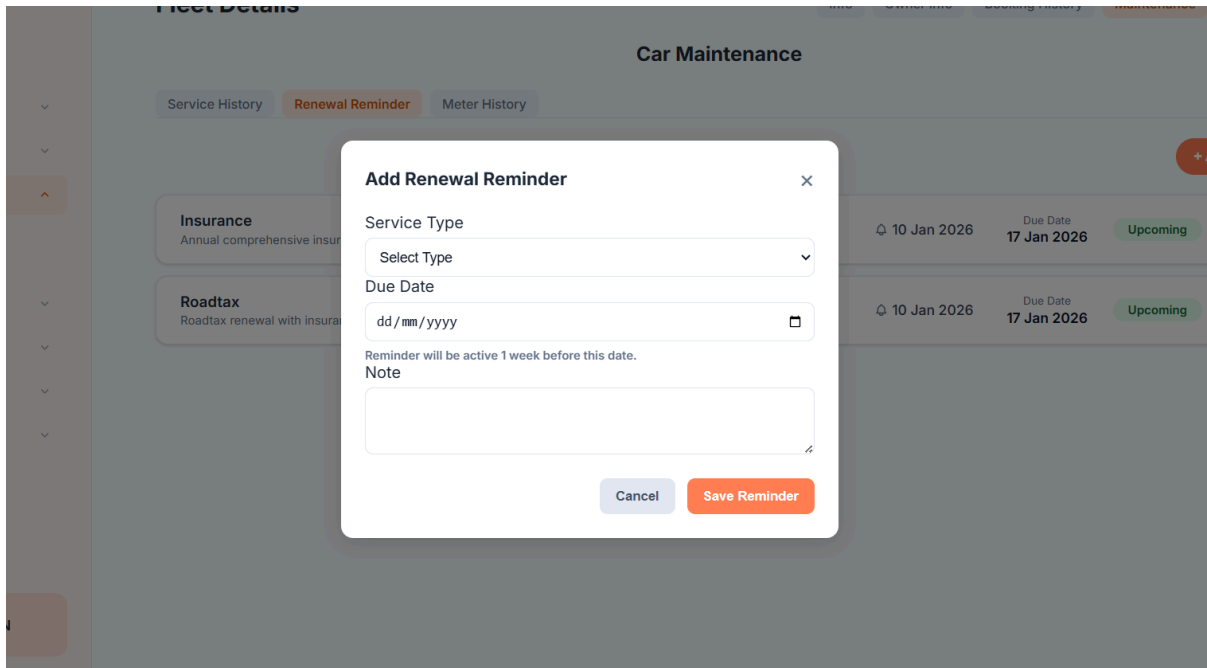


Figure 6.8.2 Add Renewal Reminder

DML#2: Insert

```
INSERT INTO renewal_reminders (
    plate_no,
    service_type,
    due_date,
    note,
    status,
    created_at,
    updated_at
)
VALUES (
    'UTM137',
    'Roadtax',
    '2026-02-15',
    'Roadtax renewal with insurance',
    'upcoming',
    CURRENT_TIMESTAMP,
    CURRENT_TIMESTAMP
);
```

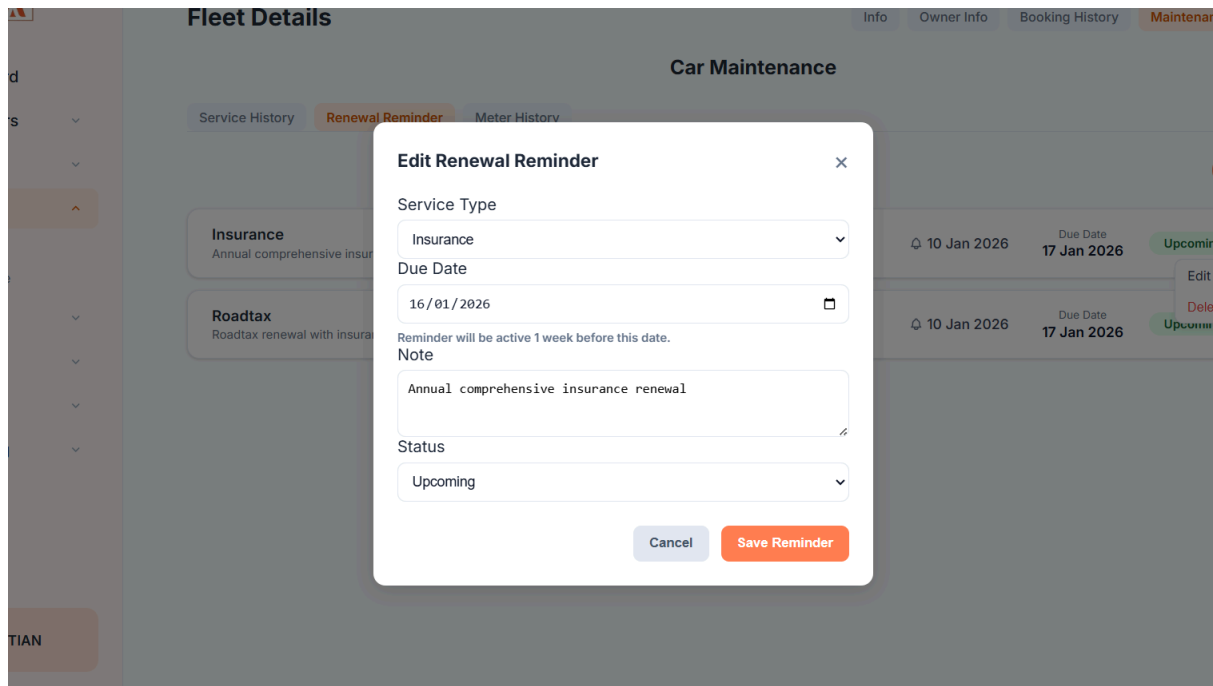


Figure 6.8.3 Update Renewal Reminder

DML#3: Update

```
UPDATE renewal_reminders
SET
    service_type = 'Insurance',
    due_date = '2026-01-16',
    note = 'Annual comprehensive insurance renewal',
    status = 'upcoming',
    updated_at = CURRENT_TIMESTAMP
WHERE id = 1;
```

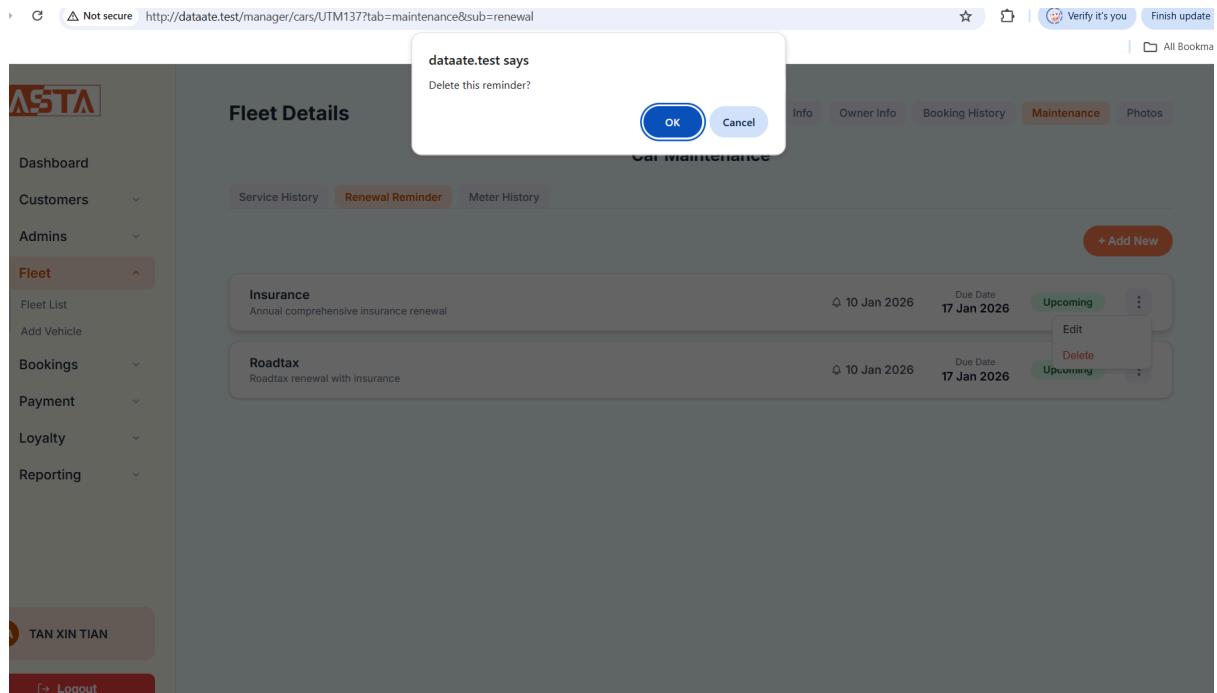


Figure 6.8.4 Delete Renewal Reminder

DML#4: Delete

DELETE FROM renewal_reminders

WHERE id = 1;

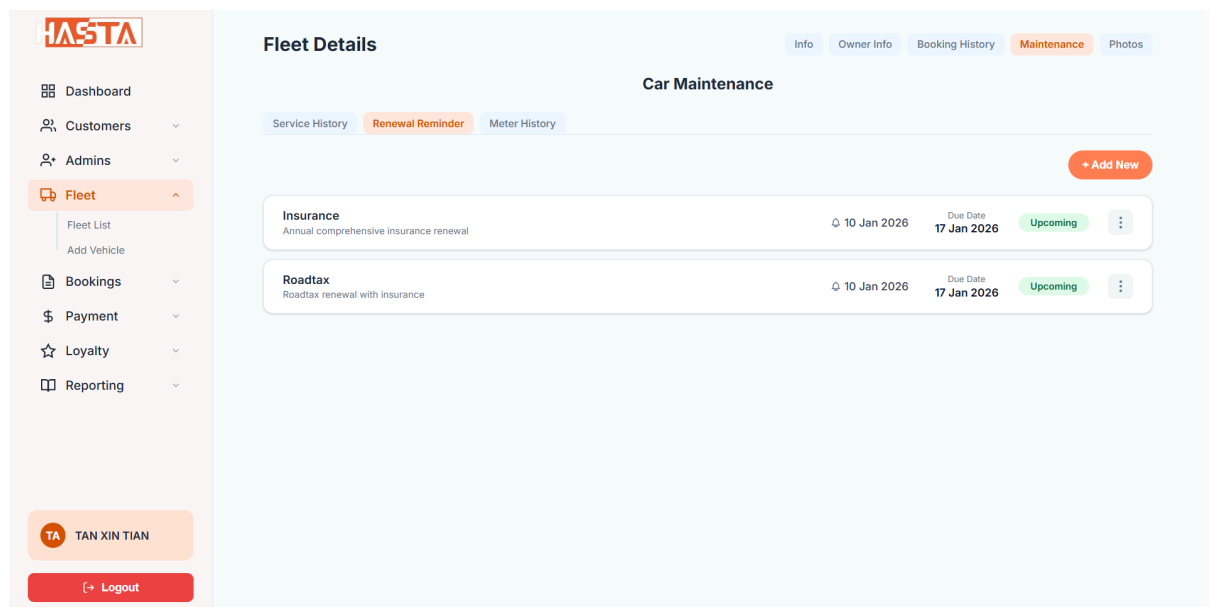


Figure 6.8.5 Insert Dummy Data of Renewal Reminder

DML#5: Insert Dummy Data (more in SQL file)

INSERT INTO renewal_reminders (
plate_no,

```

service_type,
due_date,
note,
status,
created_at,
updated_at
)
VALUES
('UTM137', 'Insurance', '2026-01-17',
'Annual comprehensive insurance renewal',
'upcoming', '2026-01-07 15:07:56', '2026-01-07 15:07:56'),

('UTM137', 'Roadtax', '2026-01-17',
'Roadtax renewal with insurance',
'upcoming', '2026-01-07 15:08:20', '2026-01-07 15:08:20');

```

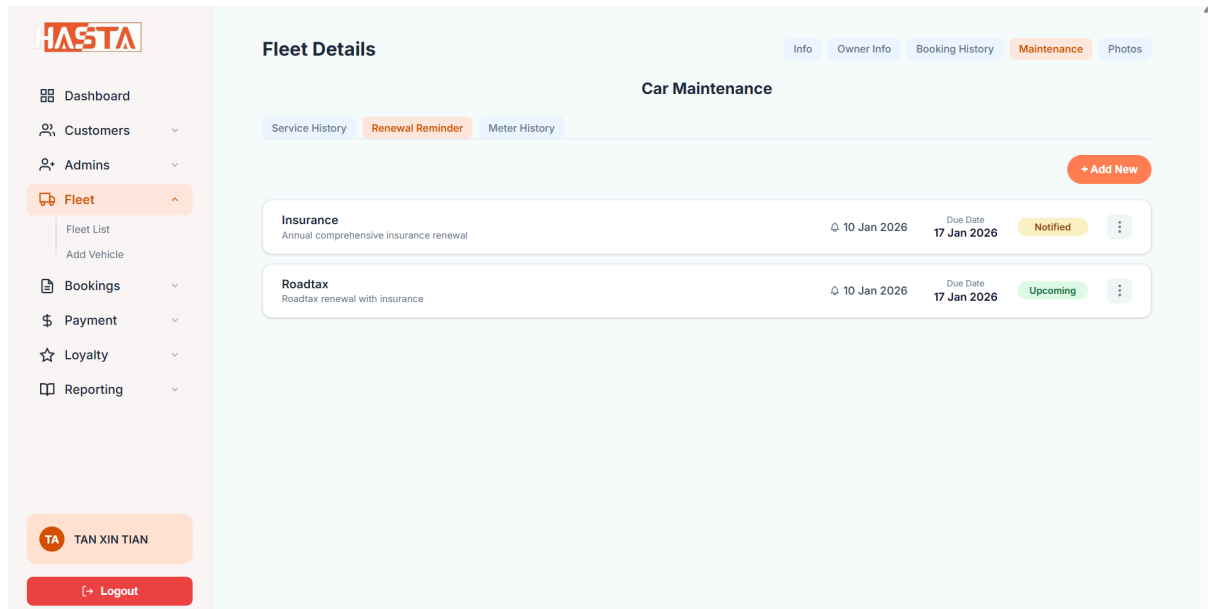


Figure 6.8.6 Update Dummy Data of Renewal Reminder

DML#6: Update Dummy Data

UPDATE renewal_reminders

SET

status = 'notified',

```
    updated_at = CURRENT_TIMESTAMP  
WHERE plate_no = 'UTM137'  
AND service_type = 'Insurance';
```

6.9 rental Table

DDL:

```
CREATE TABLE `rental` (  
  `rental_id` bigint UNSIGNED NOT NULL,  
  `customer_id` varchar(50) COLLATE utf8mb4_unicode_ci NOT NULL,  
  `plate_no` varchar(10) COLLATE utf8mb4_unicode_ci NOT NULL,  
  `payment_id` varchar(50) COLLATE utf8mb4_unicode_ci NOT NULL,  
  `verification_status` varchar(50) COLLATE utf8mb4_unicode_ci DEFAULT NULL,  
  `withdraw_status` varchar(20) COLLATE utf8mb4_unicode_ci NOT NULL DEFAULT  
'pending',  
  `withdraw_receipt_path` varchar(255) COLLATE utf8mb4_unicode_ci DEFAULT NULL,  
  `destination` varchar(255) COLLATE utf8mb4_unicode_ci DEFAULT NULL,  
  `payment_status` varchar(50) COLLATE utf8mb4_unicode_ci DEFAULT NULL,  
  `invoice_path` varchar(255) COLLATE utf8mb4_unicode_ci DEFAULT NULL,  
  `created_at` timestamp NULL DEFAULT NULL,  
  `updated_at` timestamp NULL DEFAULT NULL,  
  `destination_lat` decimal(10,7) DEFAULT NULL,  
  `destination_lng` decimal(10,7) DEFAULT NULL,  
  `voucher_code` varchar(50) COLLATE utf8mb4_unicode_ci DEFAULT NULL,  
  `fuel_penalty` decimal(8,2) NOT NULL DEFAULT '0.00',  
  `late_penalty` decimal(10,2) NOT NULL DEFAULT '0.00',  
  `due_notification_sent` timestamp NULL DEFAULT NULL,  
  `penalty_notification_sent` timestamp NULL DEFAULT NULL  
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_unicode_ci;
```

BOOKING ID	CUSTOMER	CAR MODEL	PICK UP TIME	TOTAL PRICE	DEPOSIT RETURN	PAYMENT	STATUS	WITHDRAWAL	ACTIONS
#72	Toh Shee Thong C095999	Toyota Vios ELG 1696	03 Feb 2026 08:00 AM	RM 410.00	RM 50.00	Pending	Pending	Pending Deposit Refund Rental not complete	View
#71	Toh Shee Thong C095999	Toyota Vios ELG 1696	01 Feb 2026 08:00 AM	RM 415.00	RM 40.00 (-RM 10.00) Reason: Missing 1 fuel tank	Verified	Done	Done Deposit Refund View Receipt	View
#46	Ding Kak Woo C000004	Perodua Bezza 2013 Z08 8046	07 Jan 2026 02:31 PM	RM 960.41	RM 50.00	Verified	Done	Approved Deposit Refund	View
#57	Mok Chor Dong C000038	Honda City GPS 1603	07 Jan 2026 12:02 PM	RM 321.87	RM 50.00	Verified	Done	Approved Deposit Refund	View
#47	Shanthi Sathasivam a/i Magendran Nagappan C000017	Honda Dash 125 NHL 0406	07 Jan 2026 04:07 AM	RM 312.55	RM 50.00	Verified	Done	Approved Deposit Refund	View
#48	Haanli a/i Munshi Rayer C000028	Perodua Myvi LEF 6732	06 Jan 2026 09:28 PM	RM 948.25	RM 50.00	Verified	Done	Completed Deposit Refund	View
#62	Shamini Thirunavukarasu a/i Sanisvara C000039	Toyota Vios YAT 1827	06 Jan 2026 05:13 PM	RM 640.06	RM 50.00	Verified	Done	Pending Deposit Refund Upload Receipt	View
#52	Bishan a/i Govindasamy C000029	Toyota Vios HKK 4874	06 Jan 2026 03:18 PM	RM 547.63	RM 50.00	Verified	Done	Completed Deposit Refund	View
#58	S. A. Singh C000043	Honda Dash 125 NHL 0406	06 Jan 2026 04:13 AM	RM 718.66	RM 50.00	Verified	Done	Completed Deposit Refund	View

Figure 6.9.1 Rental Records

DML#1: Insert

```
INSERT INTO `rental` (`rental_id`, `customer_id`, `plate_no`, `payment_id`,
`verification_status`, `withdraw_status`, `withdraw_receipt_path`, `destination`,
`payment_status`, `invoice_path`, `created_at`, `updated_at`, `destination_lat`,
`destination_lng`, `voucher_code`, `fuel_penalty`, `late_penalty`, `due_notification_sent`,
`penalty_notification_sent`) VALUES
```

```
(1, 'C000025', 'WZL 9187', 'PAY-1769954628-9553-0-6', 'done', 'completed', NULL, 'Johor
Bahru', 'paid', 'rentals/invoices/inv_PAY-1769954628-9553-0-6.pdf', '2025-05-31 10:09:34',
'2025-06-02 21:39:34', 1.4927000, 103.7414000, NULL, 0.00, 0.00, '2025-06-02 21:39:34',
NULL),
```

```
(2, 'C000062', 'TVR 6628', 'PAY-1769954628-2491-1-39', 'done', 'pending', NULL, 'Johor
Bahru', 'paid', 'rentals/invoices/inv_PAY-1769954628-2491-1-39.pdf', '2024-10-08 20:45:55',
'2024-10-13 00:15:55', 1.4927000, 103.7414000, NULL, 0.00, 0.00, '2024-10-13 00:15:55',
NULL),
```

```
(3, 'C000062', 'LOT 3933', 'PAY-1769954628-2675-2-29', 'done', 'approved', NULL, 'Johor
Bahru', 'paid', 'rentals/invoices/inv_PAY-1769954628-2675-2-29.pdf', '2025-12-10 04:33:46',
'2025-12-14 23:03:46', 1.4927000, 103.7414000, NULL, 0.00, 0.00, '2025-12-14 23:03:46',
NULL),
```

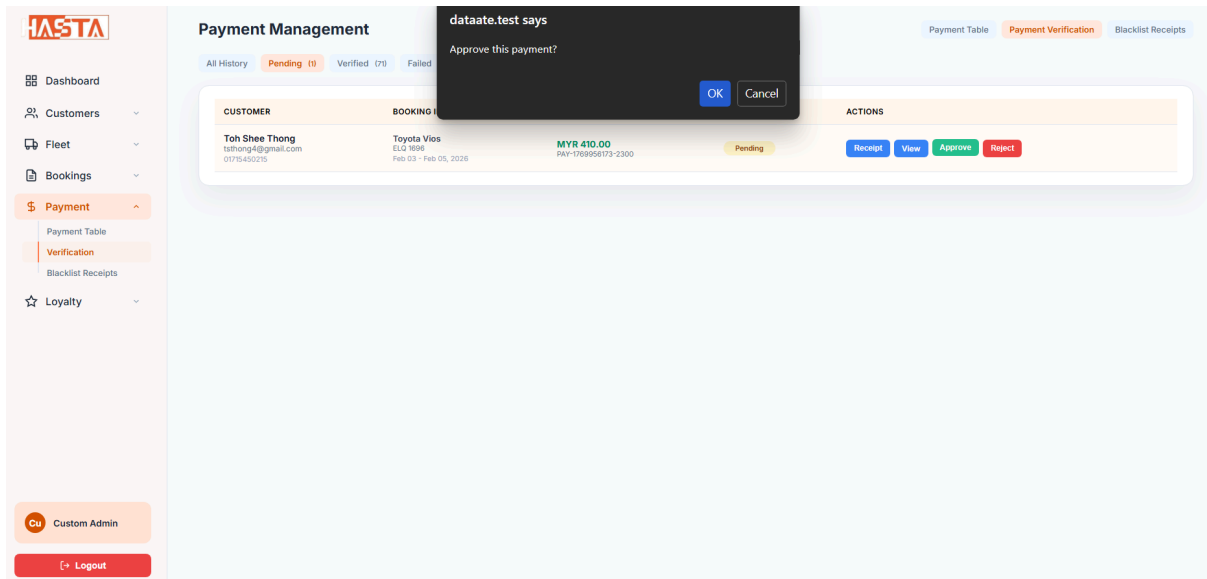



Figure 6.9.2 Approve Pending Booking

DML#2: Update (Approve Pending Booking)

UPDATE rental

SET withdraw_status = 'approved',

updated_at = CURRENT_TIMESTAMP

WHERE rental_id = 72;

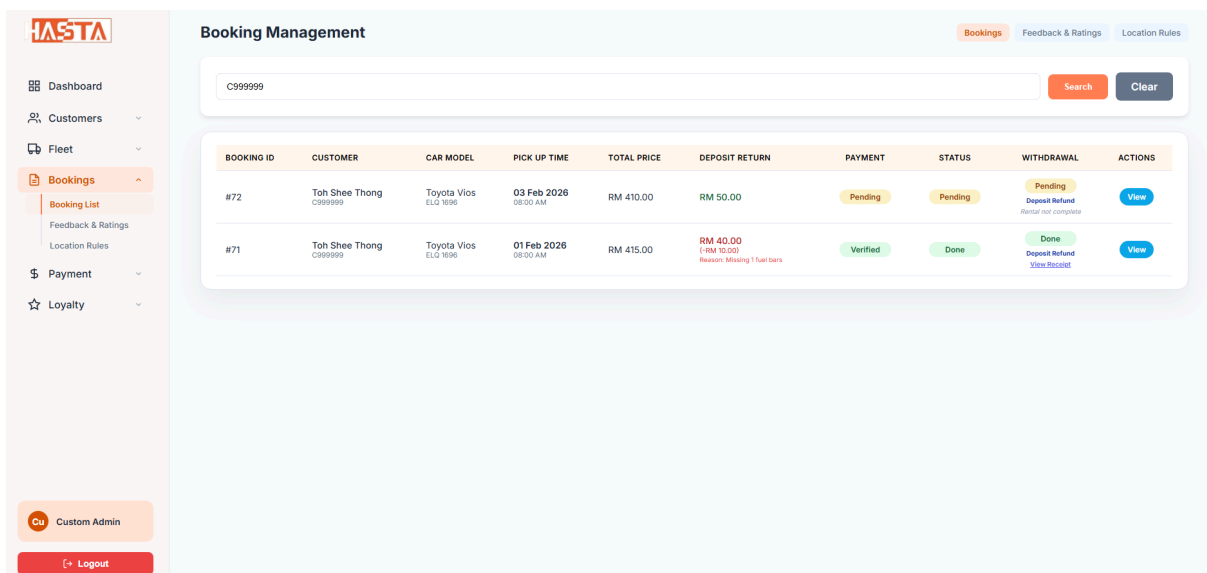


Figure 6.9.3 Search Customer ID

DML#3: Select (Find Customer using Customer ID)

SELECT rental_id, plate_no, payment_id, withdraw_status, payment_status

FROM rental

WHERE customer_id = 'C999999'

BOOKING ID	RETURN DATE	CUSTOMER	VEHICLE	VEHICLE RATING	SERVICE RATING	FEEDBACK
#71	Feb 01, 2026 10:13 PM	Toh Shee Thong tsthong4@gmail.com	ELQ 1696	5 ★	5 ★	Good for everything Condition: Good and no damage
#41	Jan 08, 2026 12:02 PM	Kam An Mee kam.kam@example.com	OMJ 3744	2 ★	4 ★	Incidunt cumque dolor fugit. Condition: Est impedit eae et
#42	Jan 10, 2026 12:39 PM	Shi Ee Lex mopusanatham@example.net	LOT 3933	1 ★	2 ★	Tempora cupiditate animi ab exercitationem. Condition: Asperiores consequatur eos consequatur beatae et
#43	Jan 07, 2026 10:50 PM	Shanthi a/i Kamal seetho.amy@example.net	VEA 6459	2 ★	1 ★	Expedita excepturi dolores et veritatis ea reprehenderit id et. Condition: Veniam aperiam distinctio velit autem voluptatibus...
#45	Jan 03, 2026 08:16 AM	Mohamad Aniq bin Che Hazmie nuraveena.ar@example.org	NQI 7484	5 ★	2 ★	Oditi aut est consequuntur necessitatibus. Condition: Neque et maiores nam totam nostrum eligendi repell...
#46	Jan 12, 2026 09:03 PM	Ding Kak Woo kasturi.vethamuthu@example.net	ZXB 8045	5 ★	3 ★	lure quia repudiandae sapiente neque. Condition: Sit totam similique ipsam non.
#47	Jan 08, 2026 08:37 AM	Shanthi Sathasivam a/i Magendran Nagappan saravanan.adityana@example.org	NHL 0406	5 ★	2 ★	Vitae assumenda laboriosam doloremque facilis quiquam commodi iusto. Condition: Et non earum architecto beatae error blanditiis.
#48	Jan 09, 2026 02:56 PM	Haani a/i Munshi Rayer rasammah.ros@example.org	UEF 6732	2 ★	1 ★	Et consequatur et non voluptas. Condition: Dectus totam sint voluptates nihil aspernatur.
#49	Jan 04, 2026 02:39 AM	Mohammed Nik Fendi Fairoz bin Hamzah Amiruddin pyakot@example.com	MZS 0786	3 ★	4 ★	Enim pariatur sint aut et voluptatem cum. Condition: Incidunt et maiores tenetur.
#44	Jan 07, 2026 09:02 AM	Mohamed Hj Nik Ramlan Saad jett.ew@example.com	FKN 1231	4 ★	3 ★	Placemat impedit et quis repellat exercitationem qui fugit ea. Condition: Maxime curque assumenda est voluptate rerum aperia...
#44	Jan 07, 2026	Nur Najlan binti Syed Miza Shazani	...	...	...	Sint quis voluptatem dolores voluptate doloribus voluptate...

Figure 6.9.4 Update for Feedback

DML#4: Update (Record Customer Rating & Feedback)

UPDATE rental

SET rating = 5,

return_feedback = 'Good for everything',

updated_at = CURRENT_TIMESTAMP

WHERE id = 71;

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Booking Management

Bookings Feedback & Ratings Location Rules

Withdrawal receipt uploaded successfully and marked as done.

Search name, Booking ID, or Customer ID...

BOOKING ID	CUSTOMER	CAR MODEL	PICK UP TIME	TOTAL PRICE	DEPOSIT RETURN	PAYMENT	STATUS	WITHDRAWAL	ACTIONS
#72	Toh Shee Thong C999999	Toyota Vios ELQ 1696	03 Feb 2026 08:00 AM	RM 410.00	RM 50.00	Pending	Pending	Pending Deposit Refund Rental not complete	View
#71	Toh Shee Thong C999999	Toyota Vios ELQ 1696	01 Feb 2026 08:00 AM	RM 415.00	RM 40.00 (-RM 10.00) Reason: Missing 1 fuel tank	Verified	Done	Done Deposit Refund View Receipt	View
#46	Ding Kak Woo C000004	Perodua Bezza 2013 ZX8 8048	07 Jan 2026 02:33 PM	RM 960.41	RM 50.00	Verified	Done	Approved Deposit Refund	View
#57	Mok Chor Dong C000038	Honda City GPS 1003	07 Jan 2026 12:02 PM	RM 321.87	RM 50.00	Verified	Done	Approved Deposit Refund	View
#47	Shanthi Sathasivam a/l Magendran Nagappan C000017	Honda Dash 125 NHL 0408	07 Jan 2026 04:07 AM	RM 312.55	RM 50.00	Verified	Done	Approved Deposit Refund	View
#48	Haanli a/l Munshi Rayer C000028	Perodua Myvi UEF 6732	06 Jan 2026 09:28 PM	RM 948.25	RM 50.00	Verified	Done	Completed Deposit Refund	View
#62	Shamini Thirunavukarasu a/l Sanisvara C000030	Toyota Vios YXY 2827	06 Jan 2026 05:13 PM	RM 640.06	RM 50.00	Verified	Done	Done Deposit Refund View Receipt	View
#52	Bishan a/l Govindasamy C000029	Toyota Vios KWK 4874	06 Jan 2026 03:18 PM	RM 547.63	RM 50.00	Verified	Done	Completed Deposit Refund	View

Dashboard Customers Fleet Bookings Booking List Feedback & Ratings Location Rules Payment Loyalty Custom Admin Logout

Figure 6.9.5 Update Withdraw Status to Done when Receipt Uploaded for Rental #71

DML#5: Update (Update withdraw_status to done)

UPDATE rental

SET withdraw_status = 'done',

withdraw_receipt_path = 'Booking_Info/payment/receipt_1769957481.png',

updated_at = CURRENT_TIMESTAMP

WHERE rental_id = 71;

6.10 pickups Table

DDL:

```
CREATE TABLE `pickups` (  
  `id` bigint UNSIGNED NOT NULL,  
  `rental_id` bigint UNSIGNED NOT NULL,  
  `start_time` datetime NOT NULL,  
  `pick_up_location` varchar(255) COLLATE utf8mb4_unicode_ci NOT NULL,  
  `pickup_lat` decimal(10,7) DEFAULT NULL,  
  `pickup_lng` decimal(10,7) DEFAULT NULL,  
  `car_condition_pickup` text COLLATE utf8mb4_unicode_ci,  
  `dashboard_image` varchar(255) COLLATE utf8mb4_unicode_ci DEFAULT NULL,  
  `damage_image` varchar(255) COLLATE utf8mb4_unicode_ci DEFAULT NULL,  
  `car_description_pickup` text COLLATE utf8mb4_unicode_ci,  
  `pickup_fuel_level` int DEFAULT NULL,  
  `agreement_form` varchar(255) COLLATE utf8mb4_unicode_ci DEFAULT NULL,  
  `acknowledgement` tinyint(1) NOT NULL DEFAULT '0',  
  `pickup_report_path` varchar(255) COLLATE utf8mb4_unicode_ci DEFAULT NULL,  
  `created_at` timestamp NULL DEFAULT NULL,  
  `updated_at` timestamp NULL DEFAULT NULL  
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_unicode_ci;
```

Figure 6.10.1 Pickup Forms

DML#1: Insert Pickup Record

INSERT INTO pickups (

rental_id,

start_time,

pick_up_location,

pickup_lat,

pickup_lng,

car_condition_pickup,

dashboard_image,

damage_image,

car_description_pickup,

pickup_fuel_level,

agreement_form,

acknowledgement,

pickup_report_path,

created_at,

updated_at

)

VALUES (

71,

```

'2026-02-01 08:00:00',
'Student Mall',
1.5585570,
103.6366470,
'Excellent',
'Booking_Info/71/pickup/dashboard.jpg',
'Booking_Info/71/pickup/damage.jpg',
'No visible damage',
5,
'Booking_Info/71/pickup/agreement.pdf',
1,
'Booking_Info/71/pickup/report.pdf',
NOW(),
NOW()
);

```

Important Reminder

Each one-level fuel reduction incurs a RM10 deposit deduction.
Return the vehicle on time to avoid additional charges.
Charges for any vehicle damage will be decided by company.

Final Confirmation

All booking details provided are accurate and complete.
Rental date, time, and location have been reviewed and confirmed.
Vehicle condition has been acknowledged based on provided photos.
Additional charges (if any) are understood and accepted.
Rental terms and conditions have been read and agreed to.
Booking confirmation is final upon submission.

☒ I agree to the above terms

Submit and Pick Up the car

Figure 6.10.2 Update Acknowledgement for Pickup

DML#2: Update (Confirm Acknowledgement)

UPDATE pickups

SET acknowledgement = 1,

updated_at = CURRENT_TIMESTAMP

WHERE rental_id = '72';

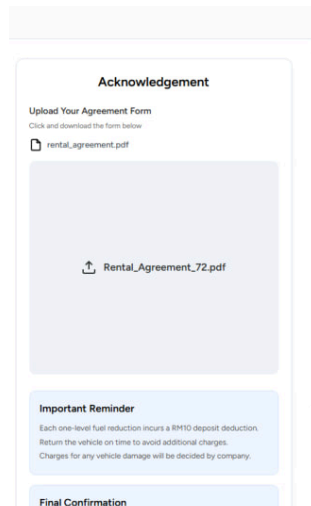


Figure 6.10.3 Update Report

DML#3: Update (Attach Generated Pickup Reports)

```
UPDATE pickups
SET pickup_report_path = 'reports/rental_agreement_72.pdf',
    updated_at = CURRENT_TIMESTAMP
WHERE rental_id = '72';
```

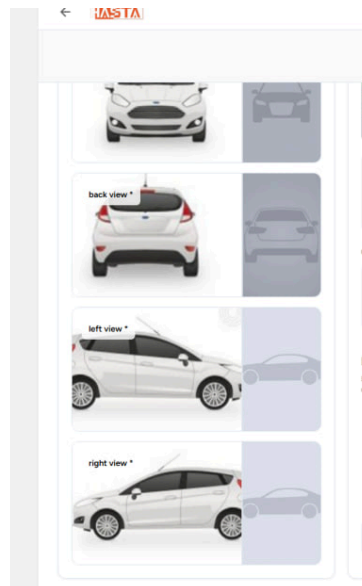


Figure 6.10.3 Update for Car Condition for Different View

DML#4: Update (Update Pickup Info after Uploading Files)

```
UPDATE pickups
SET
    front_view = 'images/front_C0001_R2026001.jpg',
```

```

back_view = 'images/back_C0001_R2026001.jpg',
left_view = 'images/left_C0001_R2026001.jpg',
right_view = 'images/right_C0001_R2026001.jpg',
updated_at = CURRENT_TIMESTAMP
WHERE rental_id = '72';

```

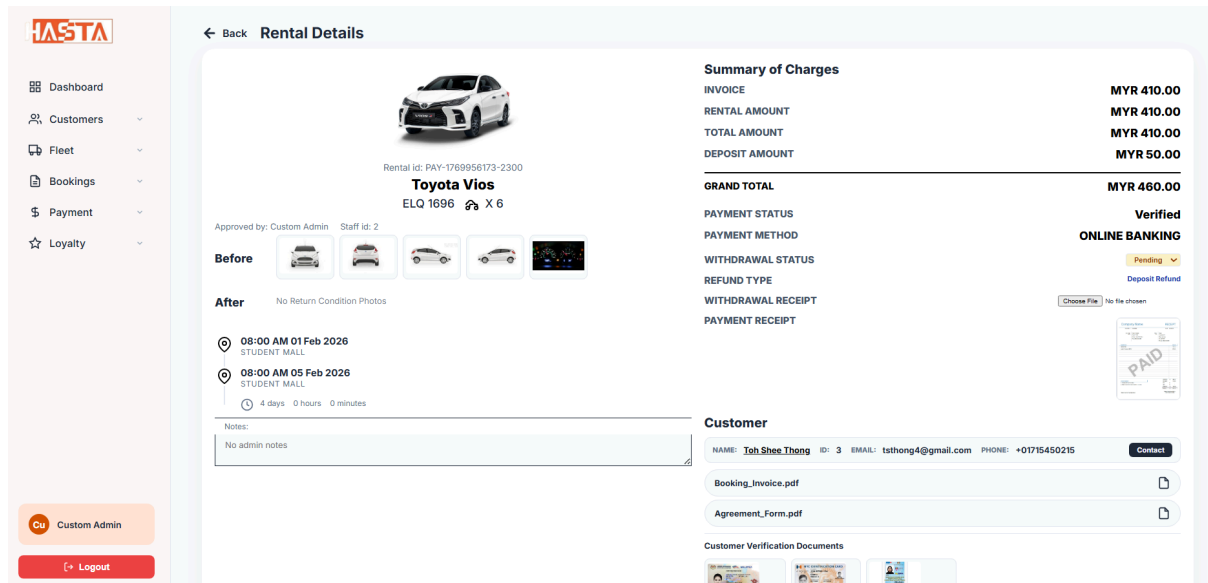


Figure 6.10.5 Find Pickup Records using rental_id

DML#5: Select (Show Pickup Details for Specific Booking)

```

SELECT *
FROM pickups
WHERE rental_id = 72;

```


6.11 returns Table

DDL:

```
CREATE TABLE `returns` (  
  `id` bigint UNSIGNED NOT NULL,  
  `rental_id` bigint UNSIGNED NOT NULL,  
  `end_time` datetime NOT NULL,  
  `return_location` varchar(255) COLLATE utf8mb4_unicode_ci NOT NULL,  
  `return_lat` decimal(10,7) DEFAULT NULL,  
  `return_lng` decimal(10,7) DEFAULT NULL,  
  `car_condition_return` text COLLATE utf8mb4_unicode_ci,  
  `dashboard_image_return` varchar(255) COLLATE utf8mb4_unicode_ci DEFAULT NULL,  
  `damage_image_return` varchar(255) COLLATE utf8mb4_unicode_ci DEFAULT NULL,  
  `car_description_return` text COLLATE utf8mb4_unicode_ci,  
  `return_fuel_level` int DEFAULT NULL,  
  `service_rating` int DEFAULT NULL,  
  `vehicle_rating` int DEFAULT NULL,  
  `return_feedback` text COLLATE utf8mb4_unicode_ci,  
  `return_report_path` varchar(255) COLLATE utf8mb4_unicode_ci DEFAULT NULL,  
  `frontview_confirm` tinyint(1) NOT NULL DEFAULT '0',  
  `backview_confirm` tinyint(1) NOT NULL DEFAULT '0',  
  `leftview_confirm` tinyint(1) NOT NULL DEFAULT '0',  
  `rightview_confirm` tinyint(1) NOT NULL DEFAULT '0',  
  `created_at` timestamp NULL DEFAULT NULL,  
  `updated_at` timestamp NULL DEFAULT NULL  
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_unicode_ci;
```

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Upload car images

Photography Tips:

- Capture the entire vehicle clearly in each image
- Ensure photos are sharp and in focus.
- Do not use filters or heavy editing

front view *

☐ The front of the car doesn't have damage

back view *

☐ The back of the car doesn't have damage

Car Issue Related

Photography Tips:

- Capture the entire vehicle clearly in each image
- Ensure photos are sharp and in focus.
- Do not use filters or heavy editing
- Clearly photograph any existing damage (scratches, dents)

dashboard *

Capture the current odometer reading

Car Damaged area (Optional)

Take close-up photo for any new damage

Return & Feedback

Rate Our Service

How was your experience with our service?

★★★★★★

Rate the Vehicle

How was the vehicle condition?

★★★★★★

Additional Feedback (Optional)

Anything else you'd like to share?

Your comments about the service or vehicle...

Figure 6.11.1 Return Forms

Upload car images

Photography Tips:

Capture the entire vehicle clearly in each image

Ensure photos are sharp and in focus.

Do not use filters or heavy editing

front view *



☒ The front of the car doesn't have damage

back view *



☒ The back of the car doesn't have damage

left view *



☒ The left side of the car doesn't have damage

right view *



☒ The right side of the car doesn't have damage

Figure 6.11.2 Update Car Condition for Return

DML#1: Update (Car Condition)

UPDATE returns

SET

```
frontview_confirm = 1,  
backview_confirm = 1,  
leftview_confirm = 1,  
rightview_confirm = 1,  
updated_at = NOW()
```

WHERE rental_id = 72;

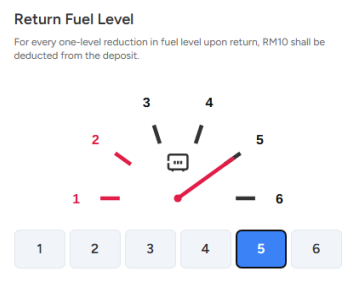


Figure 6.11.3 Update for return fuel level

DML#2: Update (Return Fuel Level)

UPDATE returns

SET

```
return_fuel_level = 5,  
updated_at = CURRENT_TIMESTAMP
```

WHERE rental_id = 72;

Figure 6.11.4 Update for Car Condition for Different View

UPDATE returns

SET

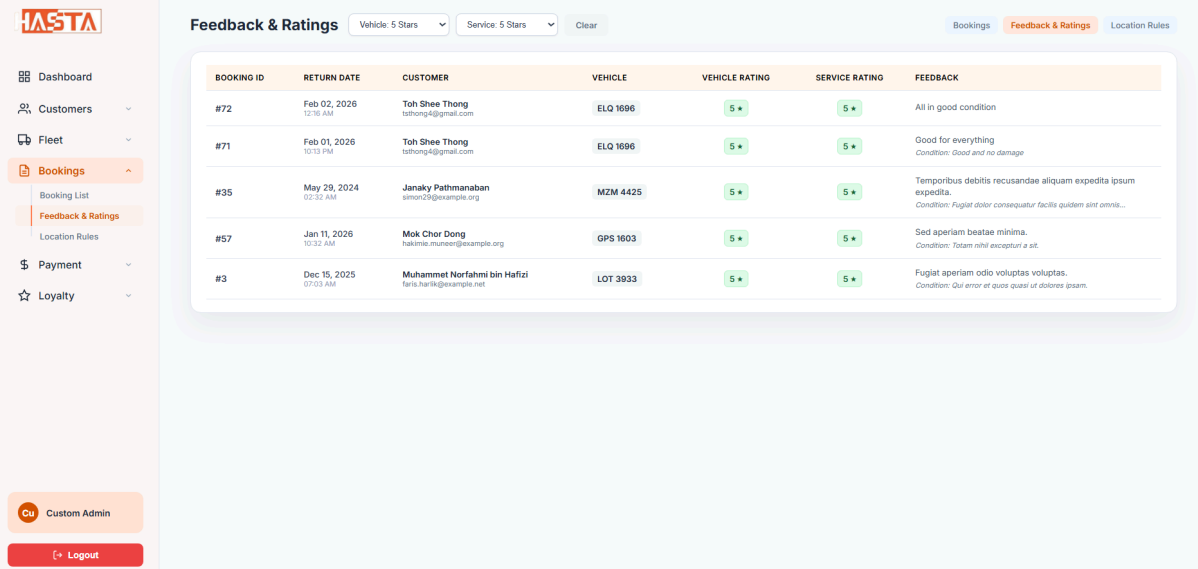
```
service_rating = 5,
vehicle_rating = 5,
return_feedback = 'All in good condition. ',
updated_at = CURRENT_TIMESTAMP
```

WHERE rental_id = 72;

Figure 6.11.5 Find Return Records using rental_id

DML#4: Select (Find Specific Return Details using rental_id)

```
SELECT *  
FROM pickups  
WHERE rental_id = 72;
```



The screenshot shows a web application interface for JASTA. On the left is a sidebar with navigation links: Dashboard, Customers, Fleet, Bookings (selected), Feedback & Ratings, Location Rules, Payment, and Loyalty. Below the sidebar is a 'Custom Admin' button and a 'Logout' button. The main content area is titled 'Feedback & Ratings' and includes filters for 'Vehicle: 5 Stars' and 'Service: 5 Stars'. A table displays the following data:

BOOKING ID	RETURN DATE	CUSTOMER	VEHICLE	VEHICLE RATING	SERVICE RATING	FEEDBACK
#72	Feb 02, 2026 12:16 AM	Toh Shee Thong tshong4@gmail.com	ELQ 1696	5 ★	5 ★	All in good condition
#71	Feb 01, 2026 10:13 PM	Toh Shee Thong tshong4@gmail.com	ELQ 1696	5 ★	5 ★	Good for everything Condition: Good and no damage
#35	May 29, 2024 02:32 AM	Janaky Pathmanaban janon2@gmail.com	MZM 4425	5 ★	5 ★	Temporibus debitis recusandae aliquam expedita ipsum expedita. Condition: Fugiat dolor consequat facilis quidem sint omnis...
#57	Jan 11, 2026 10:32 AM	Mok Chor Dong hakimio.munoor@example.org	GPS 1603	5 ★	5 ★	Sed aperiam beatae minima. Condition: Totam nihil eucepturi a sit.
#3	Dec 15, 2025 07:03 AM	Muhammet Norfahmi bin Hafizi faris.hafizi@example.net	LOT 3933	5 ★	5 ★	Fugiat aperiam odio voluptas voluptas. Condition: Qui error et quos quasi ut dolores ipsum.

Figure 6.11.6 Find Feedback According to Ratings

DML#5: Select (Find Feedback According to Ratings)

```
SELECT rental_id, service_rating, vehicle_rating, return_feedback  
FROM returns  
WHERE service_rating >= 5  
AND vehicle_rating >= 5  
ORDER BY service_rating DESC, vehicle_rating DESC;
```

6.12 service_histories Table

DDL:

```
CREATE TABLE `service_histories` (  
  `id` BIGINT UNSIGNED NOT NULL AUTO_INCREMENT,  
  `plate_no` VARCHAR(20) NOT NULL,  
  `vehicle_name` VARCHAR(100) NOT NULL,  
  `completed_date` DATE NOT NULL,  
  `odometer` INT DEFAULT NULL,  
  `previous_odometer` INT DEFAULT NULL,  
  `previous_odometer_date` DATE DEFAULT NULL,  
  `vendor` VARCHAR(100) DEFAULT NULL,  
  `service_type` VARCHAR(50) DEFAULT NULL,  
  `description` TEXT DEFAULT NULL,  
  `amount` DECIMAL(10,2) NOT NULL DEFAULT 0.00,  
  `created_at` TIMESTAMP NULL DEFAULT CURRENT_TIMESTAMP,  
  `updated_at` TIMESTAMP NULL DEFAULT CURRENT_TIMESTAMP ON UPDATE  
  CURRENT_TIMESTAMP,  
  PRIMARY KEY (`id`),  
  KEY `service_histories_plate_no_index` (`plate_no`),  
  CONSTRAINT `service_histories_plate_no_fk`  
    FOREIGN KEY (`plate_no`)  
    REFERENCES `car` (`plate_no`)  
    ON DELETE CASCADE  
)
```

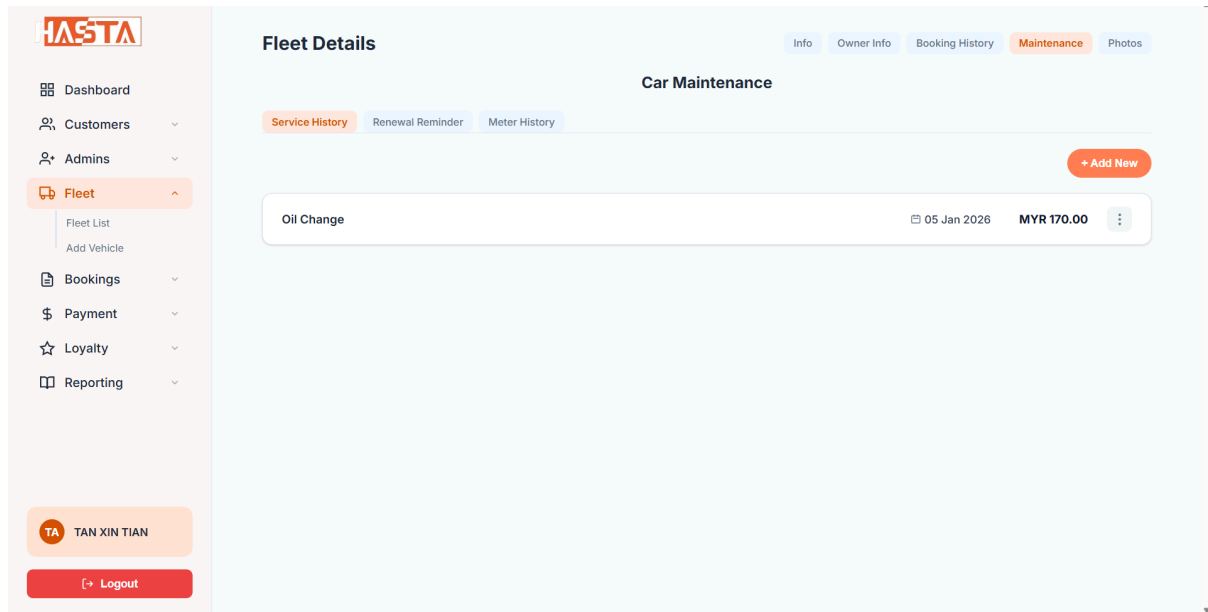


Figure 6.12.1 Service History Page

DML#1:Select (Service History)

SELECT

id,

service_type,

completed_date,

amount

FROM service_histories

WHERE plate_no = 'UTM137'

ORDER BY completed_date DESC;

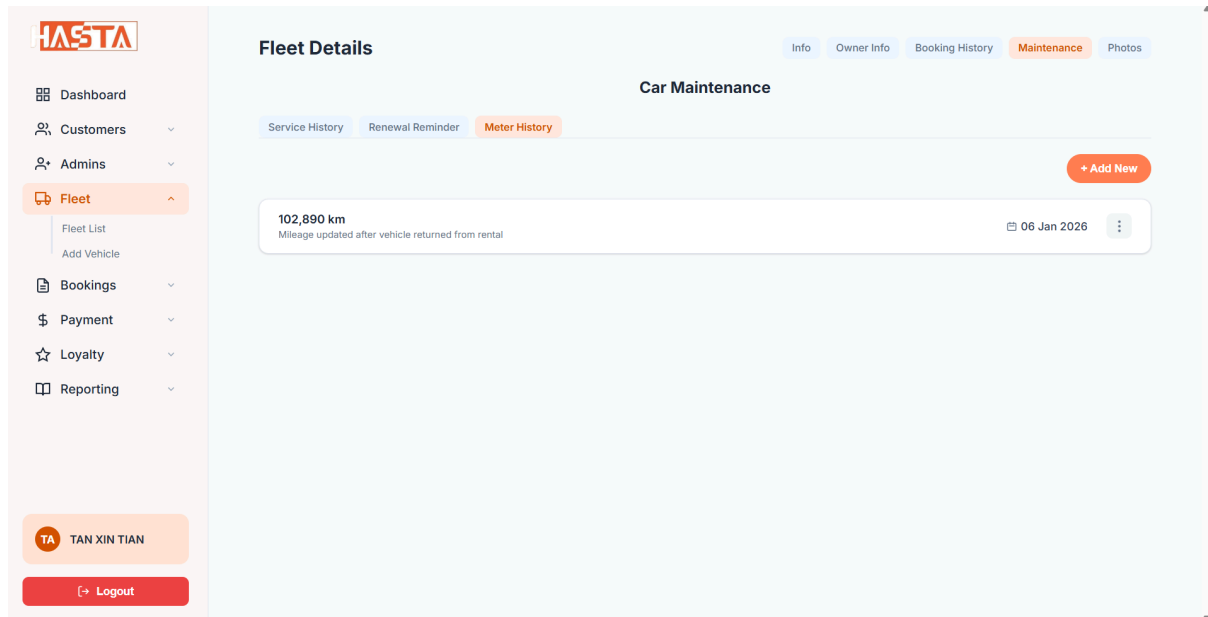


Figure 6.12.2 Meter History Page

DML#2: Select (Meter History)

SELECT

id,

odometer,

description,

completed_date

FROM service_histories

WHERE plate_no = 'UTM137'

AND service_type = 'Meter Update'

ORDER BY completed_date DESC;

Figure 6.12.3 Add Service History

DML#3: Insert (Service History)

```
INSERT INTO service_histories (
    plate_no,
    vehicle_name,
    completed_date,
    odometer,
    vendor,
    service_type,
    amount
) VALUES (
    'UTM137',
    'Proton X70',
    '2026-01-05',
    102850,
    'Proton Authorized Service Centre',
    'Oil Change',
    170.00
);
```

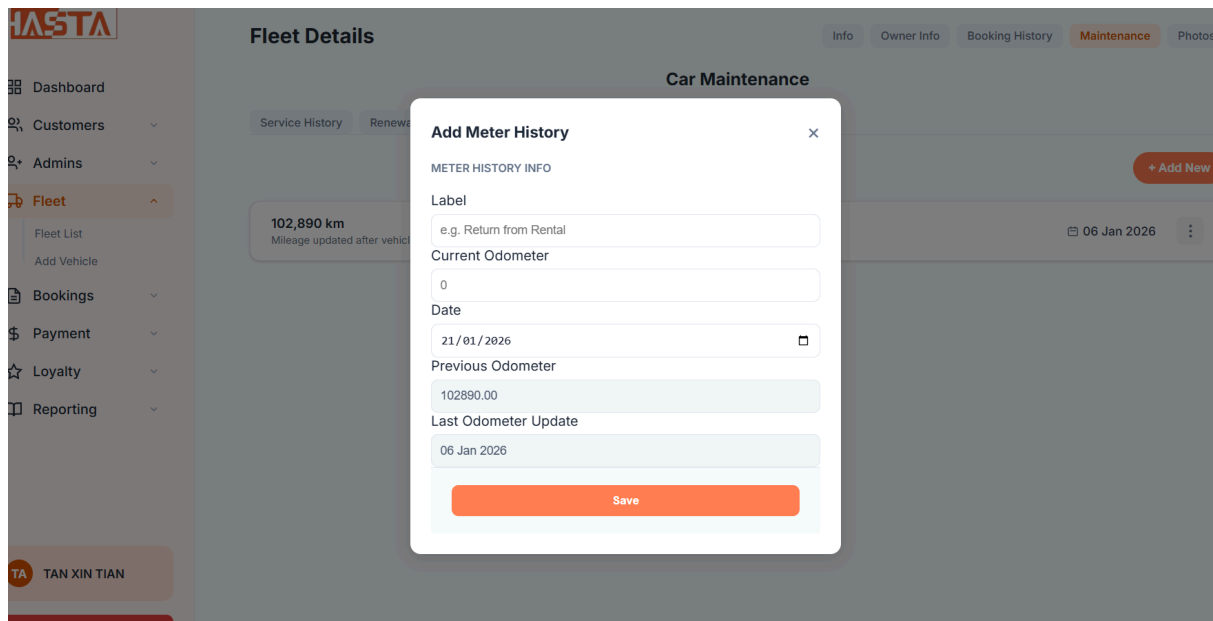


Figure 6.12.4 Add Meter History

DML#4: Insert (Meter History)

```
INSERT INTO service_histories (
    plate_no,
    vehicle_name,
    odometer,
    previous_odometer,
    previous_odometer_date,
    service_type,
    description,
) VALUES (
    'UTM137',
    'Proton X70',
    102890,
    102850,
    '2026-01-05',
    'Meter Update',
    'Mileage updated after vehicle returned from rental',
);
```

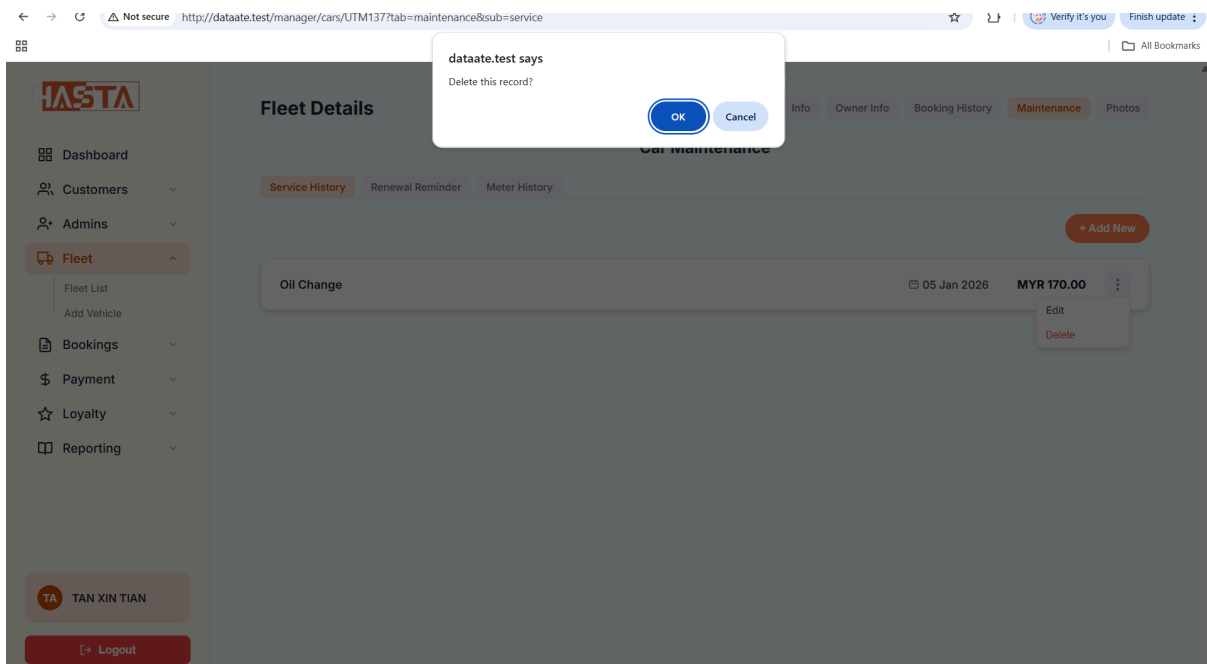


Figure 6.12.5 Delete Service History

DML#5: Delete (Service History)

DELETE FROM service_histories

WHERE id = 2

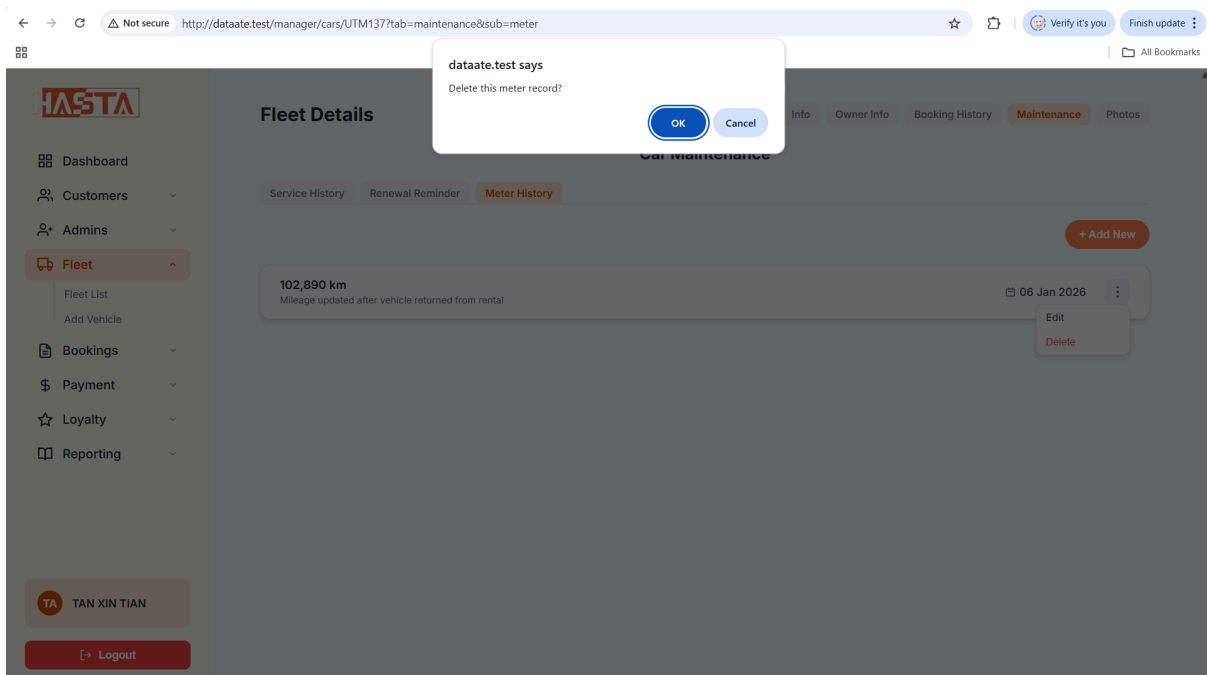


Figure 6.12.6 Delete Meter History

DML#6: Delete (Meter History)

```
DELETE FROM service_histories  
WHERE id = 1
```

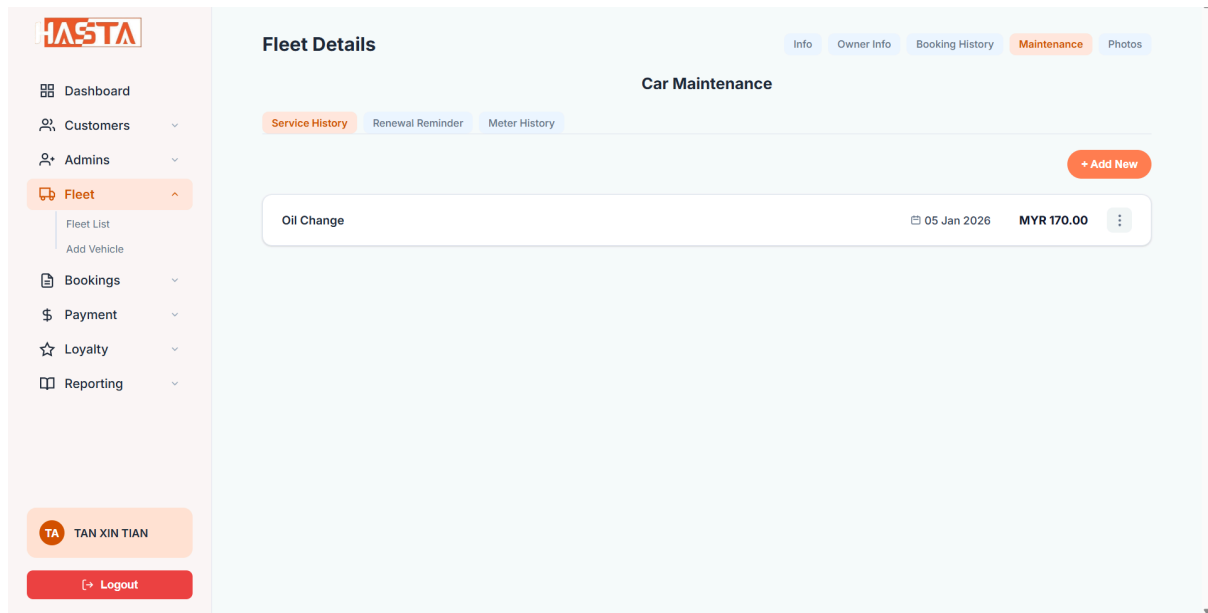


Figure 6.12.7 Insert Dummy Data of Service History

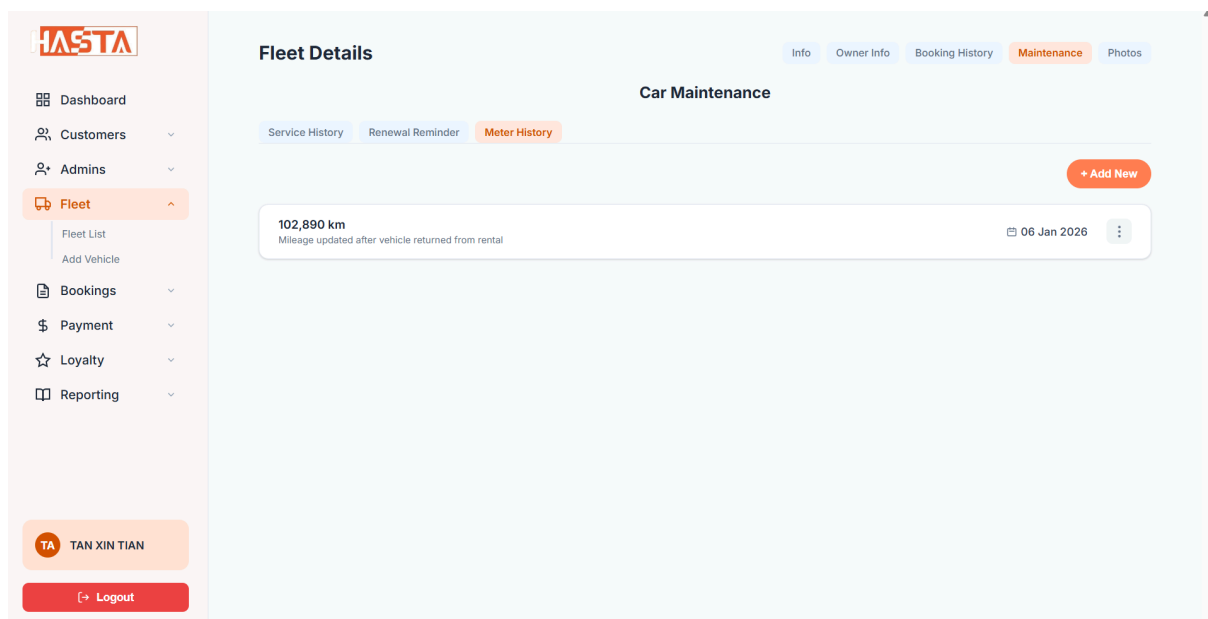


Figure 6.12.8 Insert Dummy Data of Meter History

DML#7: Insert Dummy Data (More in SQL file)

```
INSERT INTO `service_histories`  
(`plate_no`, `vehicle_name`, `completed_date`, `odometer`, `previous_odometer`,  
`previous_odometer_date`, `vendor`, `service_type`, `description`, `amount`,
```

```

    `created_at`, `updated_at`)
VALUES
(
    'UTM137',
    'Proton X70',
    '2026-01-05',
    102850,
    101920,
    '2025-12-15',
    'Proton Authorized Service Centre',
    'Oil Change',
    'Engine oil and oil filter replacement',
    170.00,
    NOW(),
    NOW()
),
(
    'UTM137',
    'Proton X70',
    '2026-01-06',
    102890,
    102850,
    '2026-01-05',
    NULL,
    'Meter Update',
    'Mileage updated after vehicle returned from rental',
    0.00,
    NOW(),
    NOW()
);

```

6.13 users Table

DDL:

```
CREATE TABLE users (  
    id BIGINT UNSIGNED NOT NULL AUTO_INCREMENT,  
    name VARCHAR(50) NOT NULL,  
    email VARCHAR(100) NOT NULL,  
    role VARCHAR(50) NOT NULL DEFAULT 'customer',  
    status VARCHAR(20) NOT NULL DEFAULT 'active',  
    password VARCHAR(255) NOT NULL,  
    avatar VARCHAR(255),  
    remember_token VARCHAR(100),  
    created_at TIMESTAMP NULL,  
    updated_at TIMESTAMP NULL,  
  
    CONSTRAINT pk_users PRIMARY KEY (id),  
    CONSTRAINT uq_users_email UNIQUE (email)  
);
```



Figure 6.13.1 Sign Up Page

DML#1: Insert New User

```
INSERT INTO `users` (  

```

```
`id`,
`name`,
`email`,
`role`,
`status`,
`created_at`,
`updated_at`
) VALUES (
    6,
    'TEOH XIN YEE',
    'xinyeeteoh0610@gmail.com',
    'customer',
    'active',
    '2026-01-06 19:31:39',
    '2026-01-06 19:31:39'
);
```

Update Password
Ensure your account is using a long, random password to stay secure.

Current Password

New Password

Confirm Password

Save

Delete Account
Once your account is deleted, all of its resources and data will be permanently deleted. Before deleting your account, please download any data or information that you wish to retain.

Delete Account

[\[→ Sign Out\]](#)

Figure 6.13.2 Update User Page

DML#2: Update

UPDATE users

SET password = '\$2y\$12\$NEW_HASHED_PASSWORD_VALUE',

updated_at = CURRENT_TIMESTAMP

WHERE id = 3;

Update Password
Ensure your account is using a long, random password to stay secure.

Current Password

New Password

Confirm Password

SAVE

Delete Account
Once your account is deleted, all of its resources and data will be permanently deleted. Before deleting your account, please download any data or information that you wish to retain.

DELETE ACCOUNT

Are you sure you want to delete your account?
Once your account is deleted, all of its resources and data will be permanently deleted. Please enter your password to confirm you would like to permanently delete your account.

CANCEL **DELETE ACCOUNT**

Figure 6.13.3 Update User Page (Deactivate)

DML#3: Update (Deactivate)

```
UPDATE users
SET status = 'deactivated',
    updated_at = CURRENT_TIMESTAMP
WHERE id = 8;
```

DML#4: Insert Dummy Data (more in SQL file)

```
INSERT INTO users (
    id,
    name,
    email,
    role,
    status,
    password,
    created_at,
    updated_at
) VALUES
(3, 'Daniel Wong', 'daniel.wong@hasta.my', 'admin', 'active', '$2y$12$hash', '2026-01-02 09:17:07', '2026-01-22 02:34:39'),
(8, 'Melissa Tan', 'melissa.tan@hasta.my', 'admin', 'active', '$2y$12$hash', '2026-01-07 06:12:50', '2026-01-22 02:34:39'),
(9, 'Ahmad Firdaus', 'ahmad.firdaus@hasta.my', 'admin', 'active', '$2y$12$hash', '2026-01-07 17:00:22', '2026-01-22 02:34:39'),
(11, 'Nur Aisyah', 'nur.aisyah@hasta.my', 'admin', 'active', '$2y$12$hash', '2026-01-08 03:06:55', '2026-01-22 02:34:39'),
(63, 'Jason Lim', 'jason.lim@hasta.my', 'admin', 'active', '$2y$12$hash', '2026-01-12 09:25:20', '2026-01-22 02:34:39');
```

DML#5: Update Dummy Data

```
UPDATE users
SET
    status = CASE id
        WHEN 8 THEN 'deactivated'
```

```
        WHEN 9 THEN 'deactivated'
    END,
    updated_at = CURRENT_TIMESTAMP
WHERE id IN (8, 9);
```

6.14 Vouchers Table

DDL:

```
CREATE TABLE `vouchers` (
  `id` bigint UNSIGNED NOT NULL,
  `customer_id` varchar(50) COLLATE utf8mb4_unicode_ci NOT NULL,
  `voucher_code` varchar(50) COLLATE utf8mb4_unicode_ci NOT NULL,
  `description` varchar(255) COLLATE utf8mb4_unicode_ci DEFAULT NULL,
  `discount_percent` int DEFAULT NULL,
  `free_hours` int DEFAULT NULL,
  `expiry_date` date NOT NULL,
  `status` varchar(50) COLLATE utf8mb4_unicode_ci NOT NULL DEFAULT 'active',
  `created_at` timestamp NULL DEFAULT NULL,
  `updated_at` timestamp NULL DEFAULT NULL
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_unicode_ci;
```

CODE	CUSTOMER	TYPE	STATUS	EXPIRY	ACTIONS
REWARD-KHS4FV	zabrina62@example.net	20% Off	Active	31 Mar 2026	Edit Delete
REWARD-WG9W7W	haszlan.bai@example.com	5H Free	Active	29 Mar 2026	Edit Delete
REWARD-AGBRNV	rajiv12@example.net	50% Off	Active	29 Mar 2026	Edit Delete
REWARD-TWVXIV	qshah@example.net	50% Off	Active	28 Mar 2026	Edit Delete
REWARD-ERUGGR	qshah@example.net	10H Free	Active	27 Mar 2026	Edit Delete
REWARD-6JZWVC	zainuddin.keguk@example.com	50% Off	Active	23 Mar 2026	Edit Delete
REWARD-GKME9Q	inew@example.net	10% Off	Active	22 Mar 2026	Edit Delete
REWARD-UYX8GH	hakimie.muneer@example.org	10% Off	Active	14 Mar 2026	Edit Delete
REWARD-3DS3AO	ktalib@example.net	10% Off	Active	12 Mar 2026	Edit Delete
REWARD-MEULVD	sam.kavita@example.com	50% Off	Active	09 Mar 2026	Edit Delete
REWARD-6TRATJ	faris.harlik@example.net	5H Free	Active	09 Mar 2026	Edit Delete
REWARD-C4EZ2K	thiahien.yew@example.org	10H Free	Active	06 Mar 2026	Edit Delete
REWARD-MMNDAP	qraju@example.com	20% Off	Active	05 Mar 2026	Edit Delete
REWARD-NMEX9D	nuraveena.er@example.org	50% Off	Used	31 Jan 2026	Edit Delete
REWARD-7WKI3B	inew@example.net	20% Off	Used	30 Jan 2026	Edit Delete

Figure 6.14.1 Admin Site for All Voucher Records

DML#1: Insert New Vouchers

```
INSERT INTO `vouchers` (`id`, `customer_id`, `voucher_code`, `description`,
`discount_percent`, `free_hours`, `expiry_date`, `status`, `created_at`, `updated_at`) VALUES
(1, 'C000007', 'REWARD-BJZWVC', '50% Off', 50, NULL, '2026-03-23', 'active',
'2026-01-23 14:03:47', '2026-01-23 14:03:47');
```

(2, 'C000049', 'REWARD-TWVXIV', '20% Off', 50, NULL, '2026-03-28', 'active', '2026-01-28 14:03:47', '2026-01-28 14:03:47'),

(3, 'C000041', 'REWARD-WG9W7W', '5h Free', NULL, 5, '2026-03-29', 'active', '2026-02-01 14:03:47', '2026-02-01 14:03:47'),

...

The screenshot displays the 'Create New Voucher' interface. The left sidebar contains a navigation menu with 'Dashboard', 'Customers', 'Fleet', 'Bookings', 'Payment', and 'Loyalty' (selected). Under 'Loyalty', there are links for 'Vouchers', 'Customer Loyalties', and 'Rules'. At the bottom of the sidebar are 'Custom Admin' and 'Logout' buttons. The main content area is titled 'Create New Voucher' with a subtitle 'Issue a new voucher to a customer.' and a 'Back to list' button. The form is divided into three sections: 'Recipient & Details' with a 'Send to All Customers' checkbox and a 'Description (Optional)' field containing '15% Off Loyalty Reward'; 'Voucher Configuration' with a 'Type' dropdown set to 'Discount (%)' and an 'Amount' field set to '15'; and 'Validity (Days)' with a field set to '30'. A note below the amount field says 'Enter percentage or hours based on type.' At the bottom right of the form are 'Cancel' and 'Create Voucher' buttons.

Figure 6.14.2 Add New Voucher Page

DML#2: Insert (Issue a New Voucher)

```
INSERT INTO vouchers (customer_id, voucher_code, description, discount_percent,
expiry_date, status, created_at, updated_at)
```

```
VALUES ('C000001', 'LOYALTY-15-XYZ', '15% Off Loyalty Reward', 15, '2026-06-01',
'active', NOW(), NOW());
```

IASTA

Edit Voucher
Update voucher details for GIFT-QMXEOF7S.

[Back to list](#)

Voucher Overview

Voucher Code: GIFT-QMXEOF7S

Details: 15% Discount

Configuration

Description: 15% Off Loyalty Reward

Expiry Date: 04/03/2026

Status: **Used**

Active
Used
Expired

[Cancel](#) [Update Voucher](#)

Dashboard

Customers

Fleet

Bookings

Payment

Loyalty

Vouchers

Customer Loyalties

Rules

Custom Admin

[Logout](#)

Figure 6.14.3 Mark a Voucher as Used

DML#3: Update (Mark Voucher as Used)

```
Voucher::where('voucher_code', 'LOYALTY-15-XYZ')->update([
    'status' => 'used',
    'updated_at' => now()
]);
```

IASTA

Edit Voucher
Update voucher details for GIFT-QMXEOF7S.

[Back to list](#)

Voucher Overview

Voucher Code: GIFT-QMXEOF7S

Details: 15% Discount

Configuration

Description: 15% Off Loyalty Reward

Expiry Date: 01/02/2026

Status: **Expired**

Active
Used
Expired

[Cancel](#) [Update Voucher](#)

Dashboard

Customers

Fleet

Bookings

Payment

Loyalty

Vouchers

Customer Loyalties

Rules

Custom Admin

[Logout](#)

Figure 6.14.4 Mark a Voucher as Expired

DML#4: Update (Update Old voucher as Expired)

UPDATE vouchers

SET status = 'expired'

WHERE status = 'active' AND expiry_date < CURDATE();

JASTA

Edit Voucher
Update voucher details for GIFT-QMXEOF7S.

[Back to list](#)

Voucher Overview

Voucher Code: GIFT-QMXEOF7S

Details: 15% Discount

Configuration

Description: 15% Off Loyalty Reward

Status: Expired

Expiry Date: 21/02/2026

February, 2026

S	M	T	W	T	F	S
1	2	3	4	5	6	7
8	9	10	11	12	13	14
15	16	17	18	19	20	21
22	23	24	25	26	27	28
1	2	3	4	5	6	7
8	9	10	11	12	13	14

Clear Today

Cancel Update Voucher

Custom Admin

Logout

Figure 6.14.5 Update the Expiry Date for a Voucher

DML#5: Update (Extend Expiry Date)

UPDATE vouchers

SET expiry_date = DATE_ADD(expiry_date, INTERVAL 30 DAY),

updated_at = CURRENT_TIMESTAMP

WHERE voucher_code = 'SUMMER-2026';

7.0 Summary

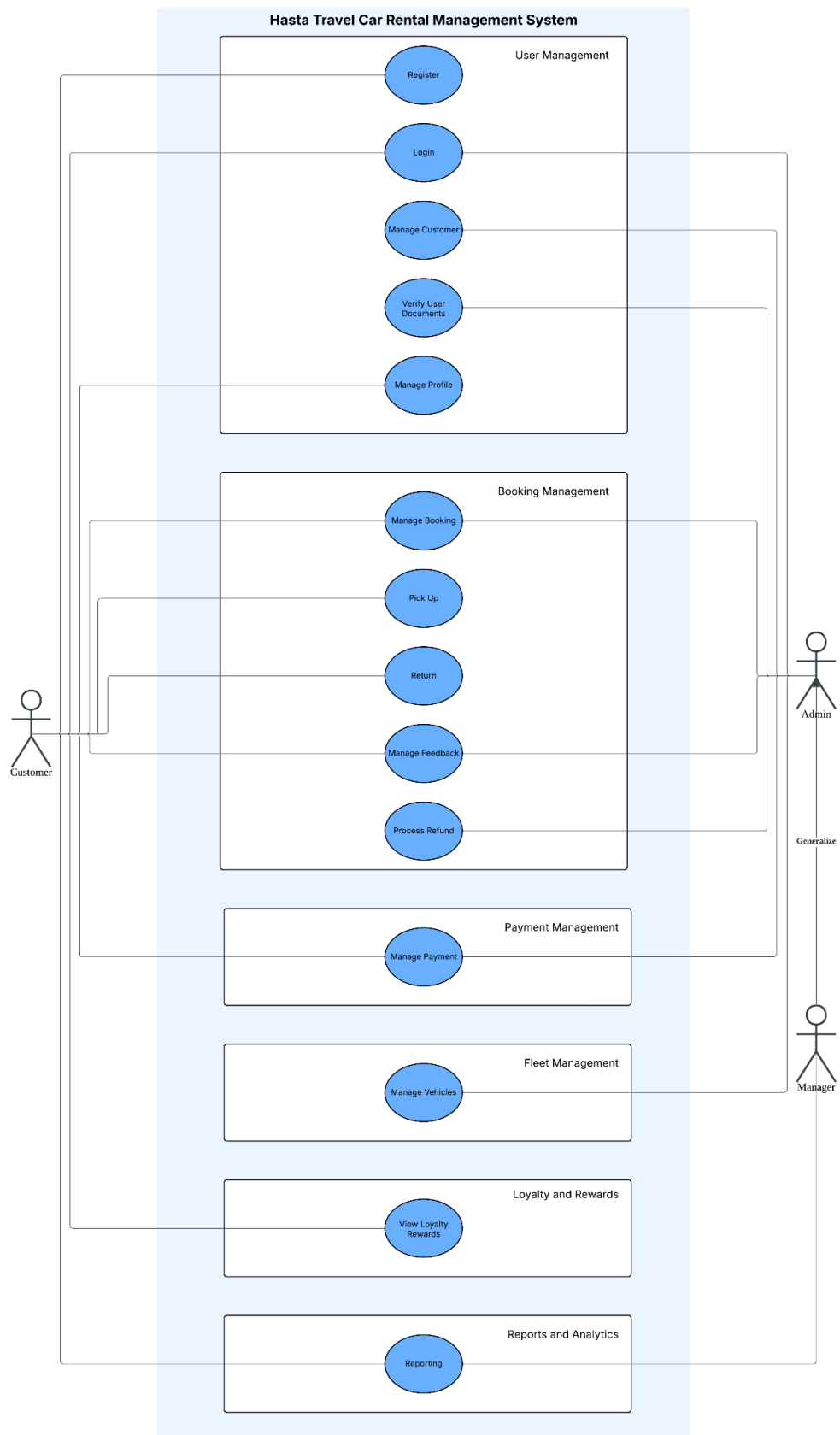
In summary, this project presents a comprehensive logical database design for the Hasta Travel vehicle rental management system. The design is derived from the system requirements and business processes identified during the analysis phase. It is implemented using a relational database model. Key system entities are clearly defined and structured to support core operations such as user management, booking, vehicle handling, payment processing, loyalty tracking and reporting.

Normalization techniques are applied to ensure data integrity, eliminate redundancy and improve maintainability. The relationships between entities are carefully designed to reflect real-world interactions. These support efficient querying and transaction processing. Other than that, supporting documentation, including ERDs, relational schemas and data dictionaries, further clarifies the database structure and ensures consistency throughout the design.

Overall, the logical design provides a robust and well-organized data foundation that meets operational needs, supports accurate reporting and enables efficient system development. This design prepares the system for reliable implementation while ensuring compliance with academic and industry database design standards.

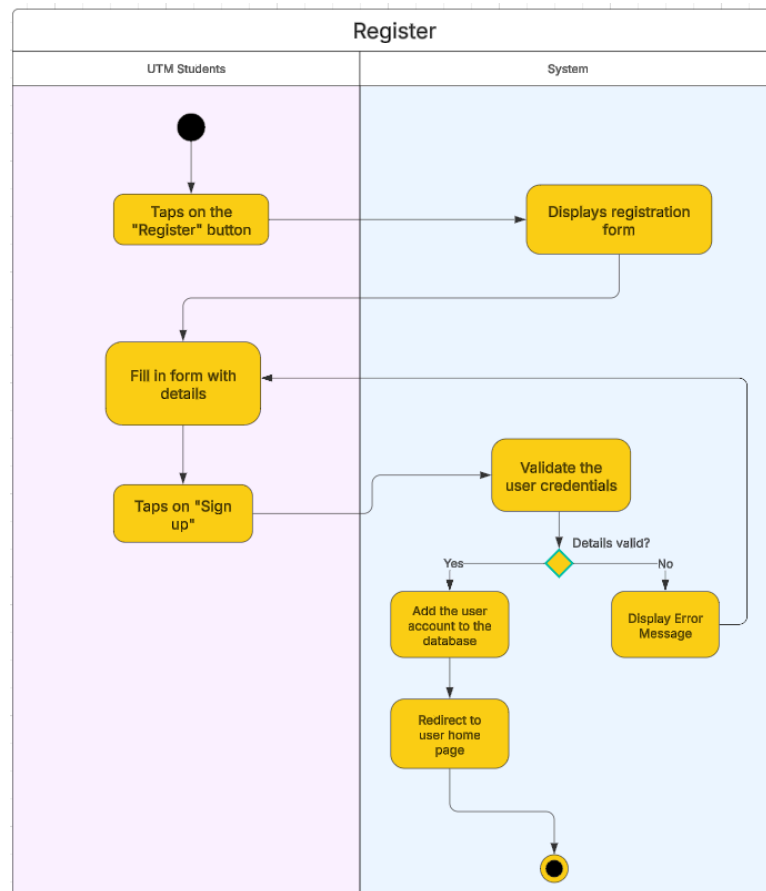
8.0 Appendices

8.1 Use Case Diagram

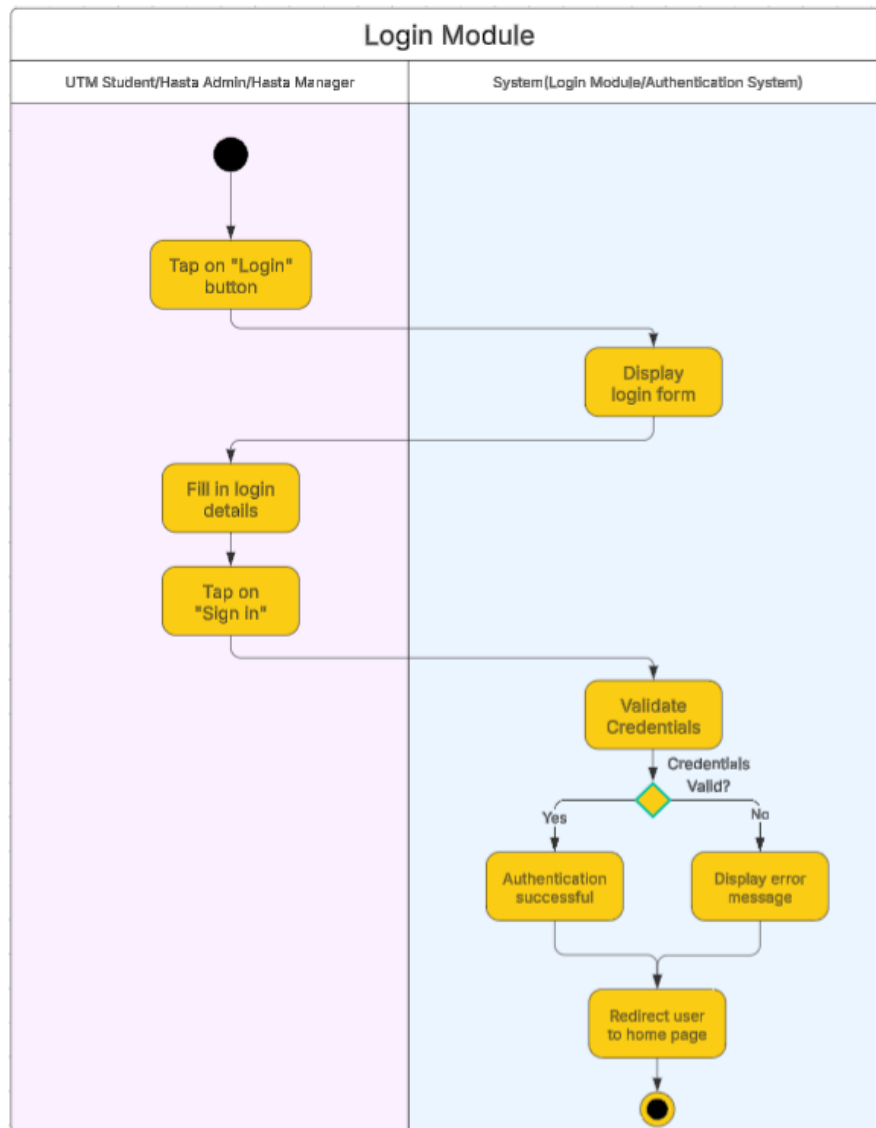


8.2 Activity Diagram

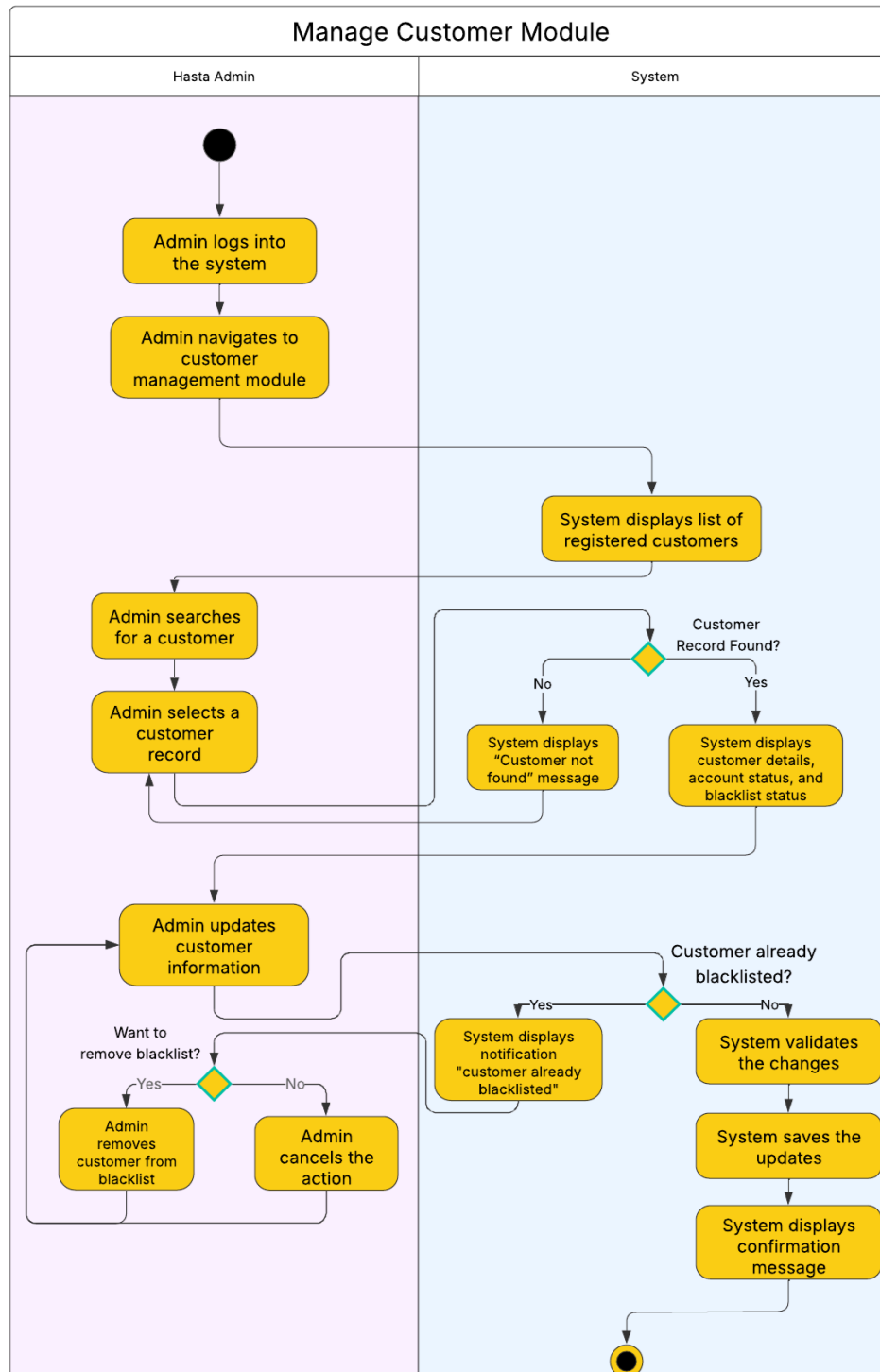
1. Register



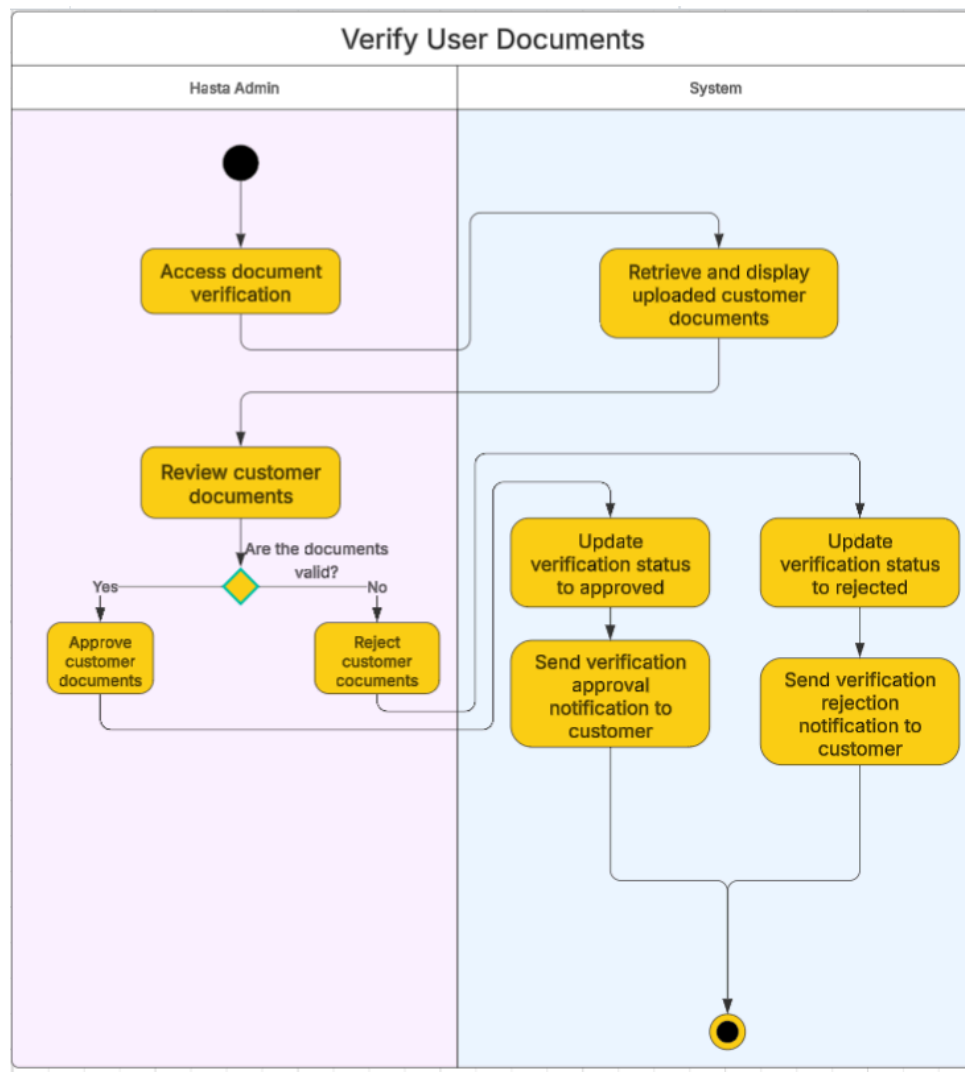
2. Login



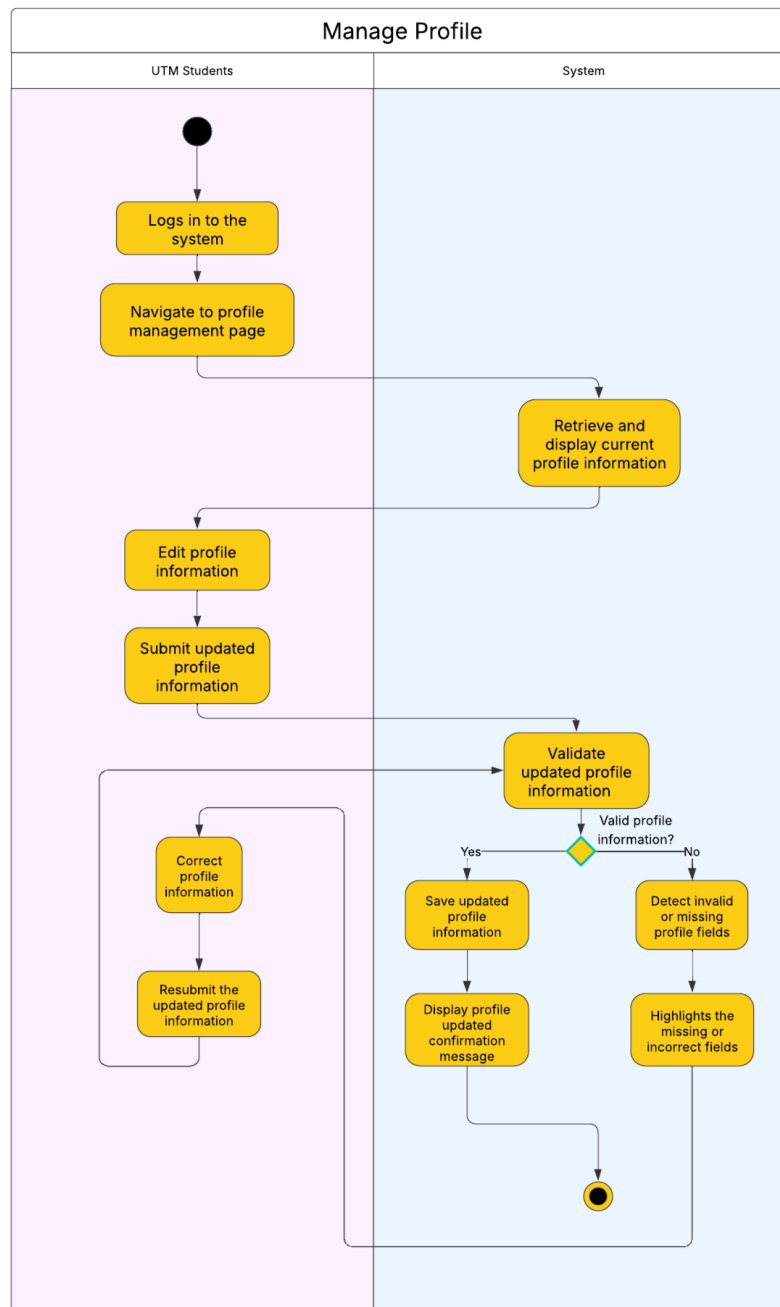
3. Manage Customer



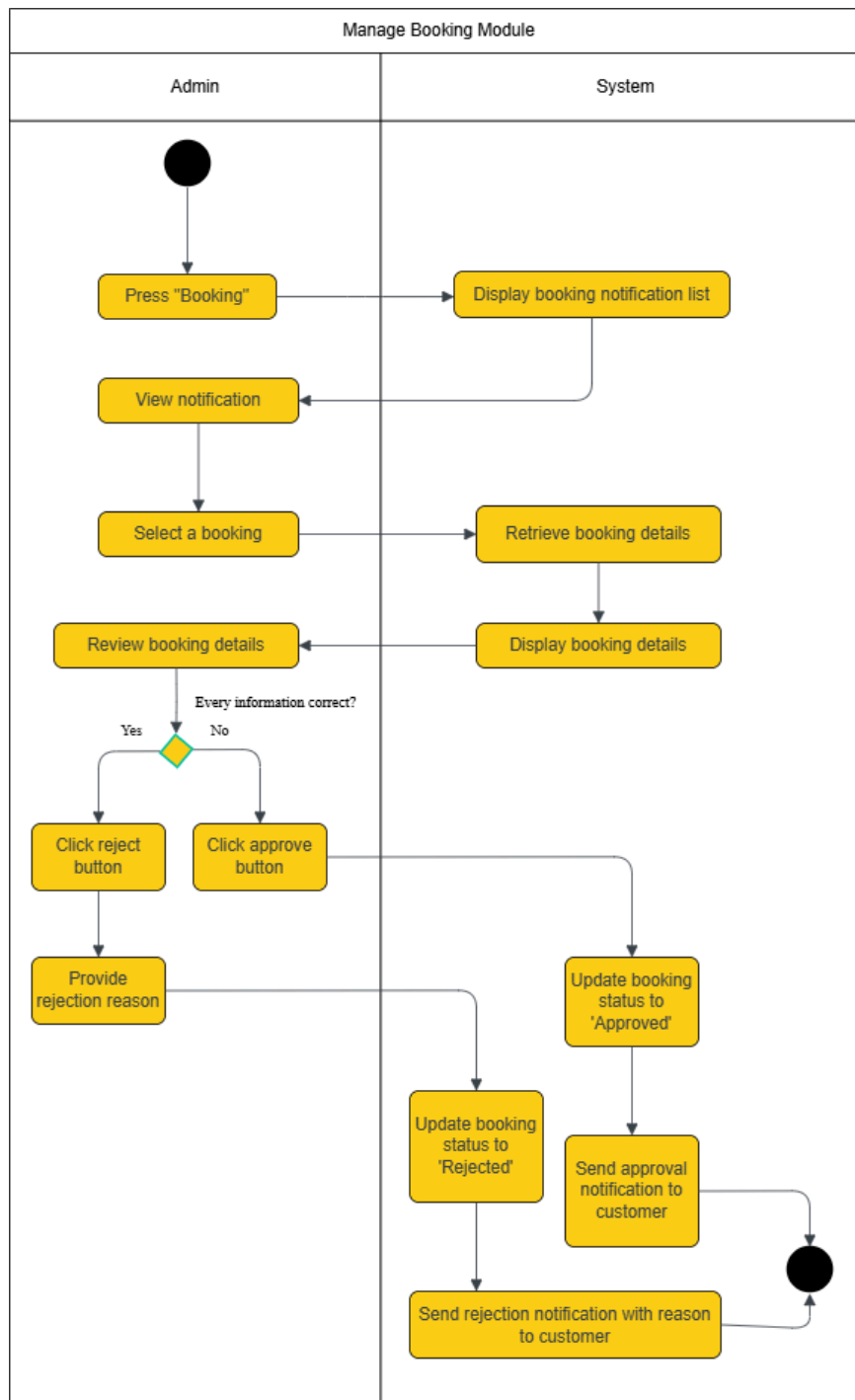
4. Verify User Documents



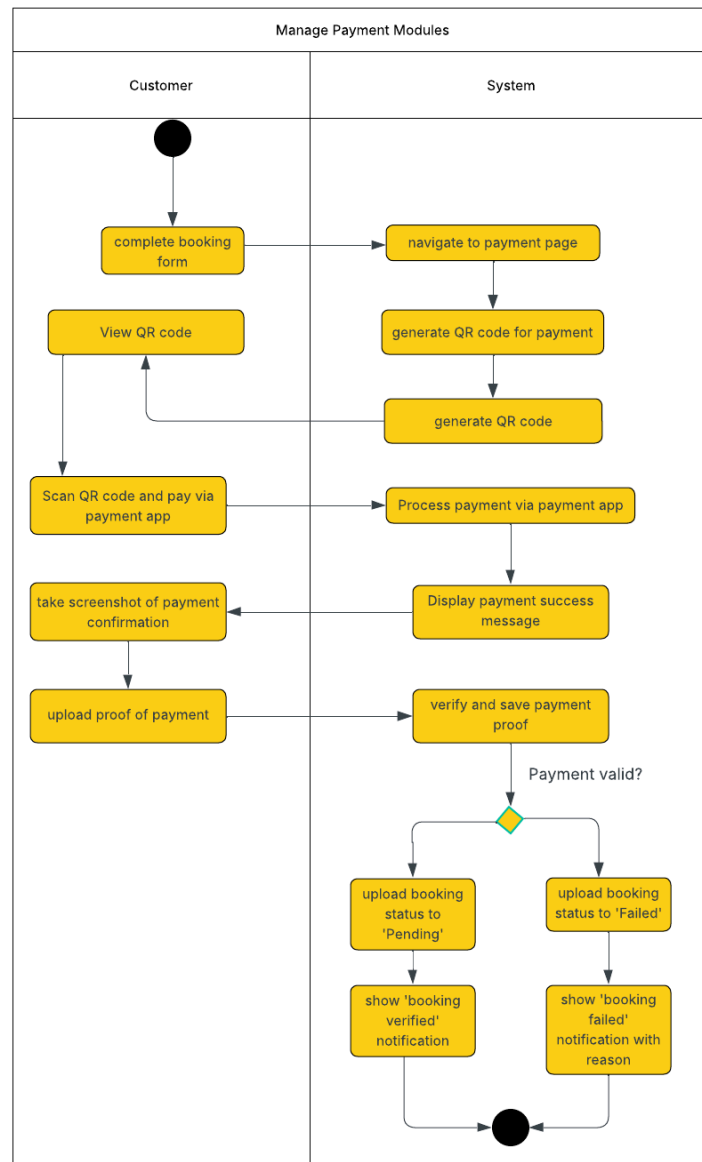
5. Manage Profile

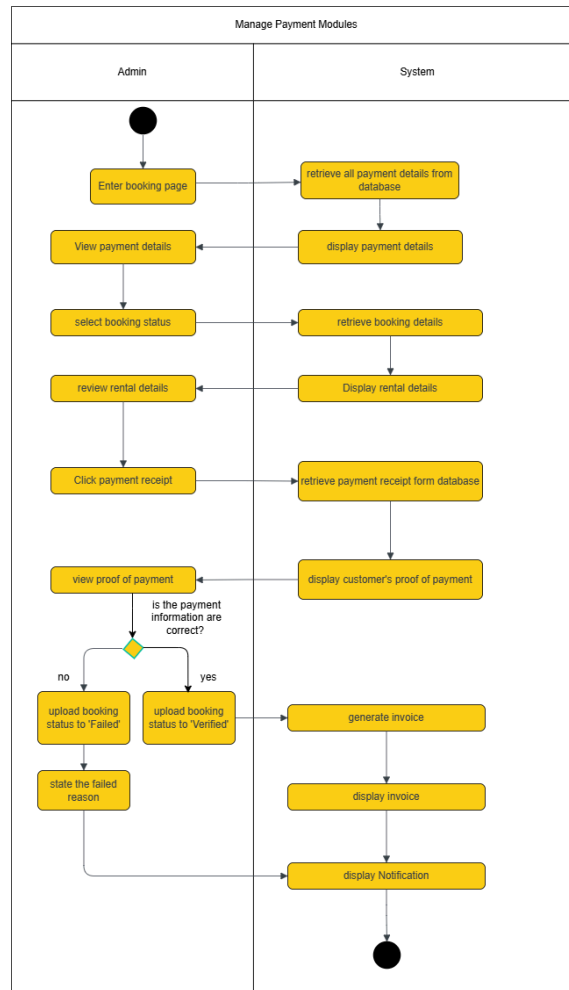


6. Manage Booking

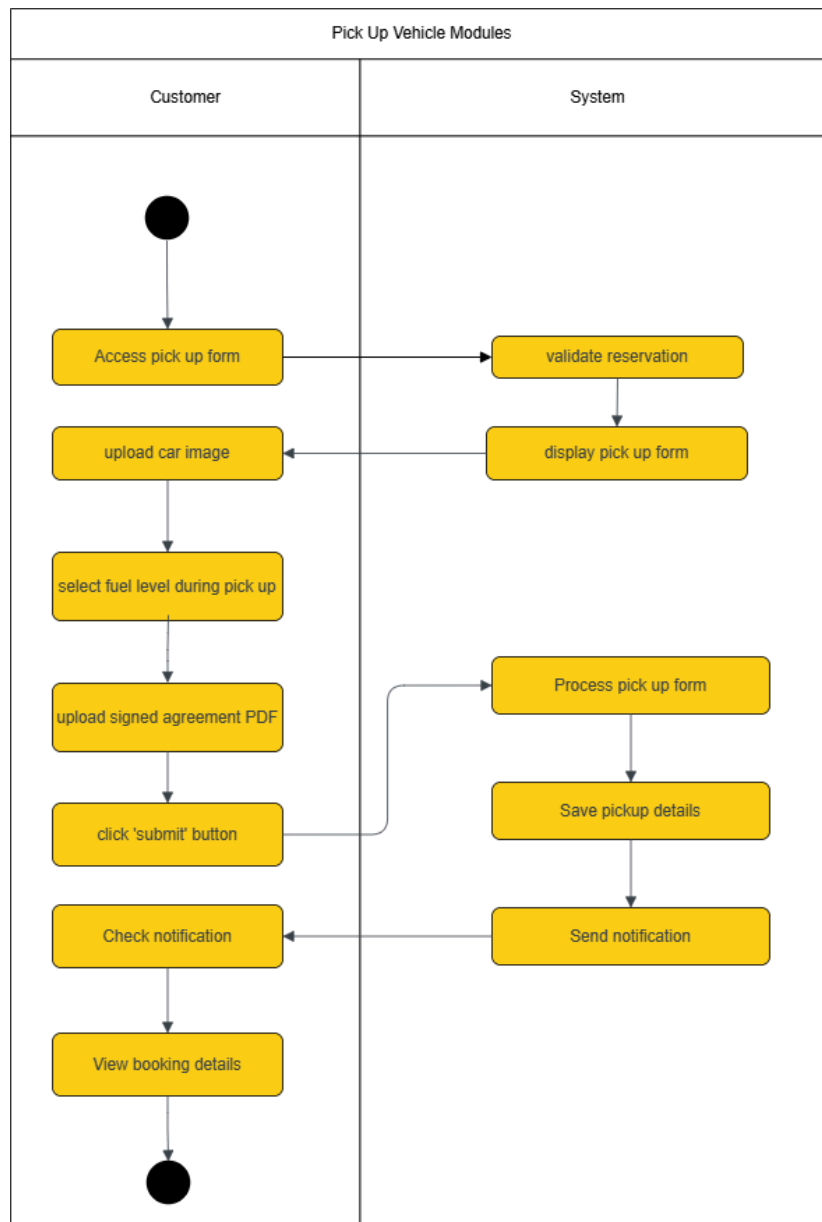


7. Manage Payment

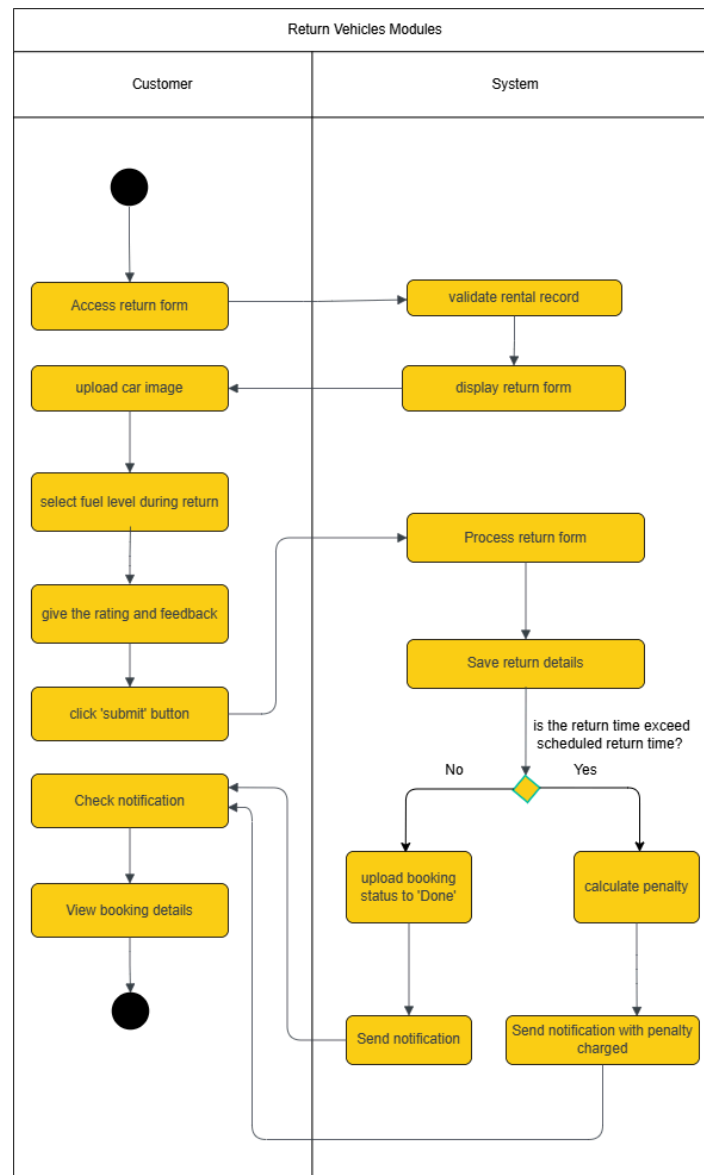




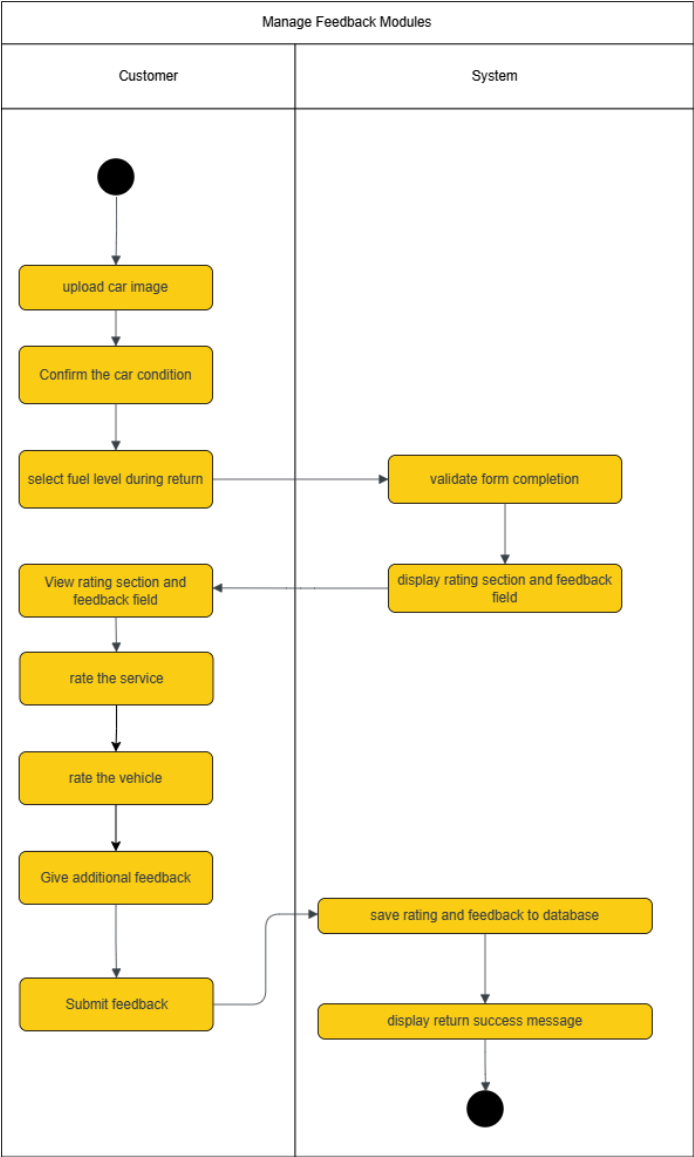
8. Pick Up Vehicle

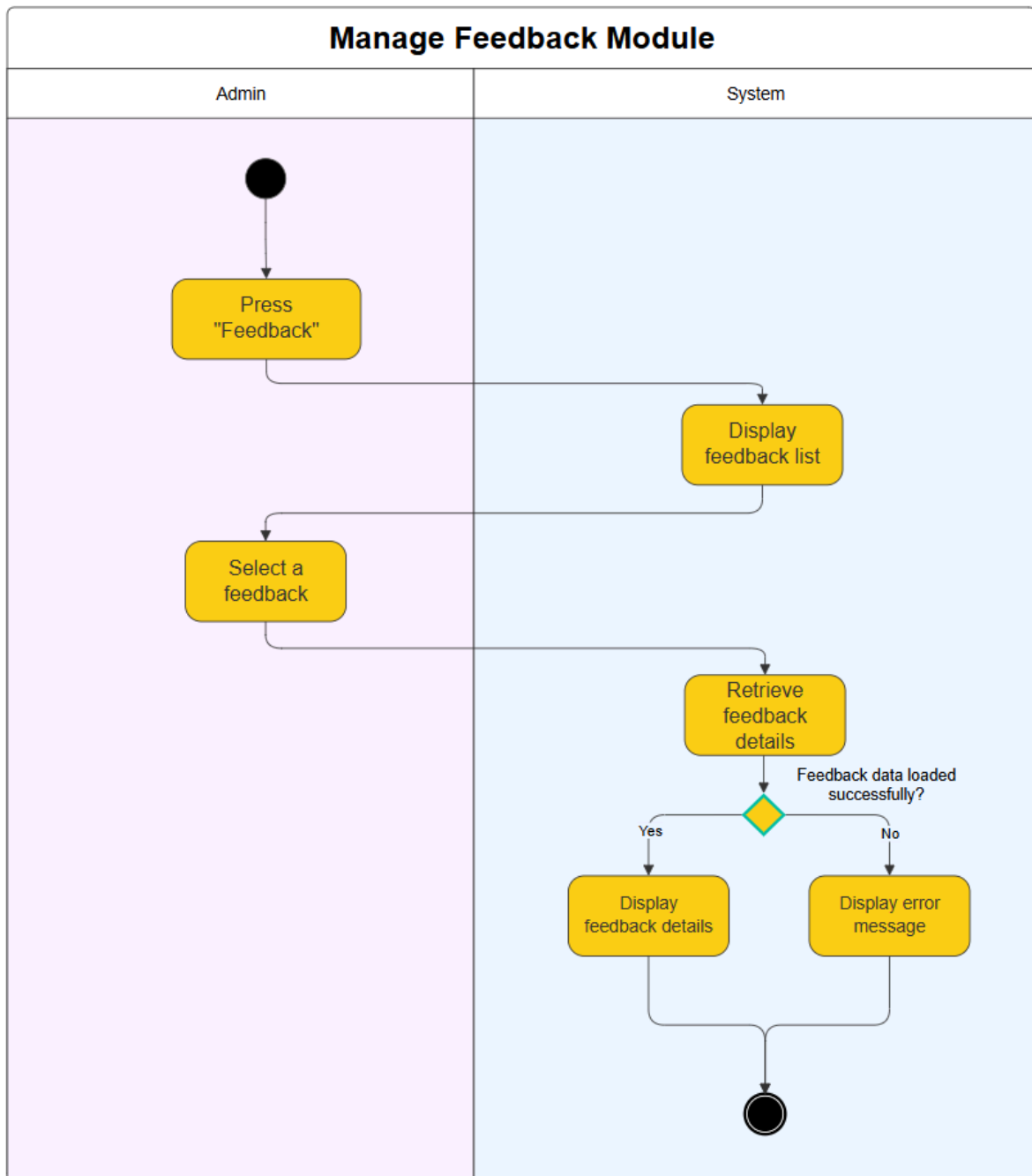


9. Return Vehicle

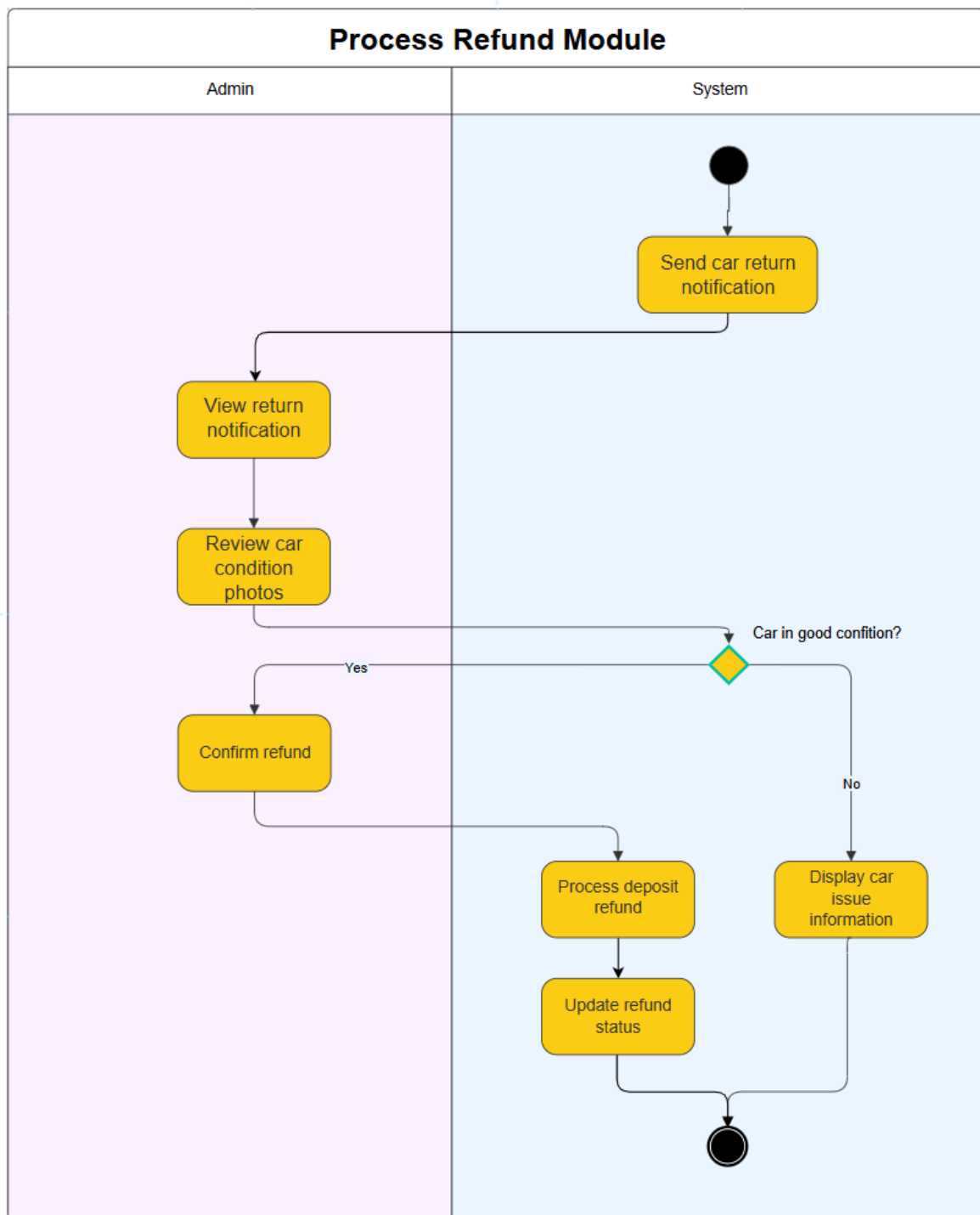


10. Manage Feedback

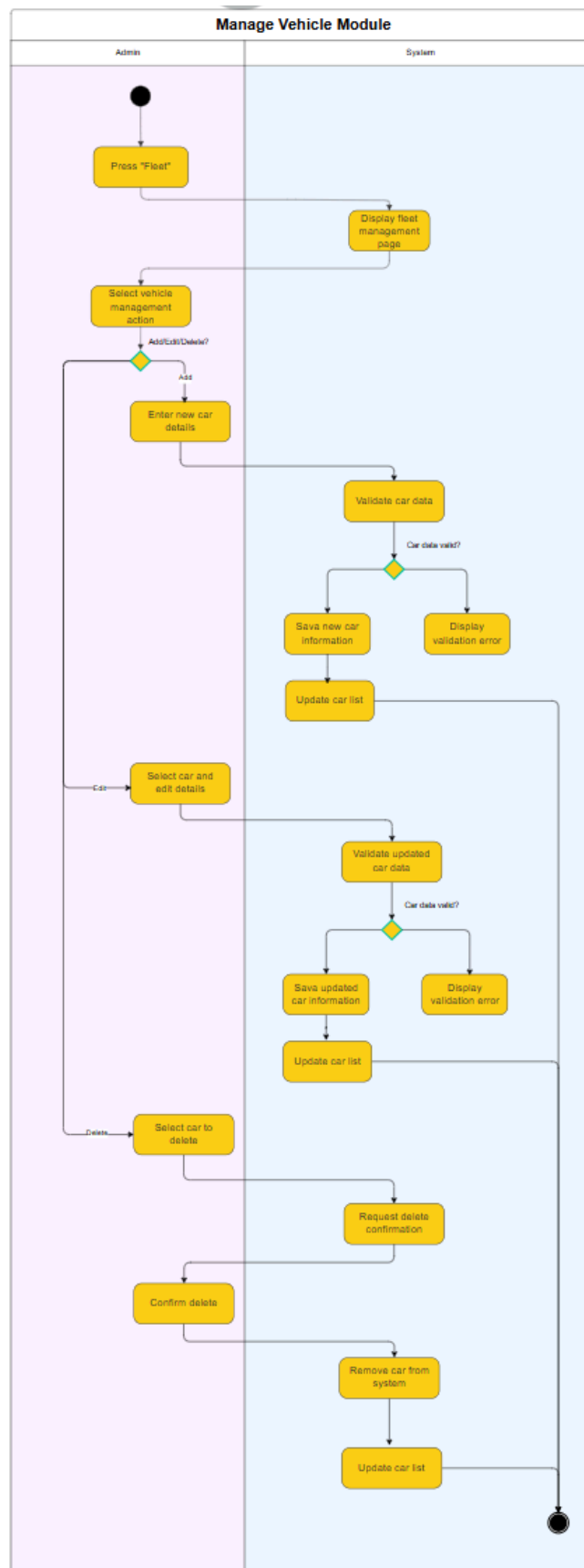




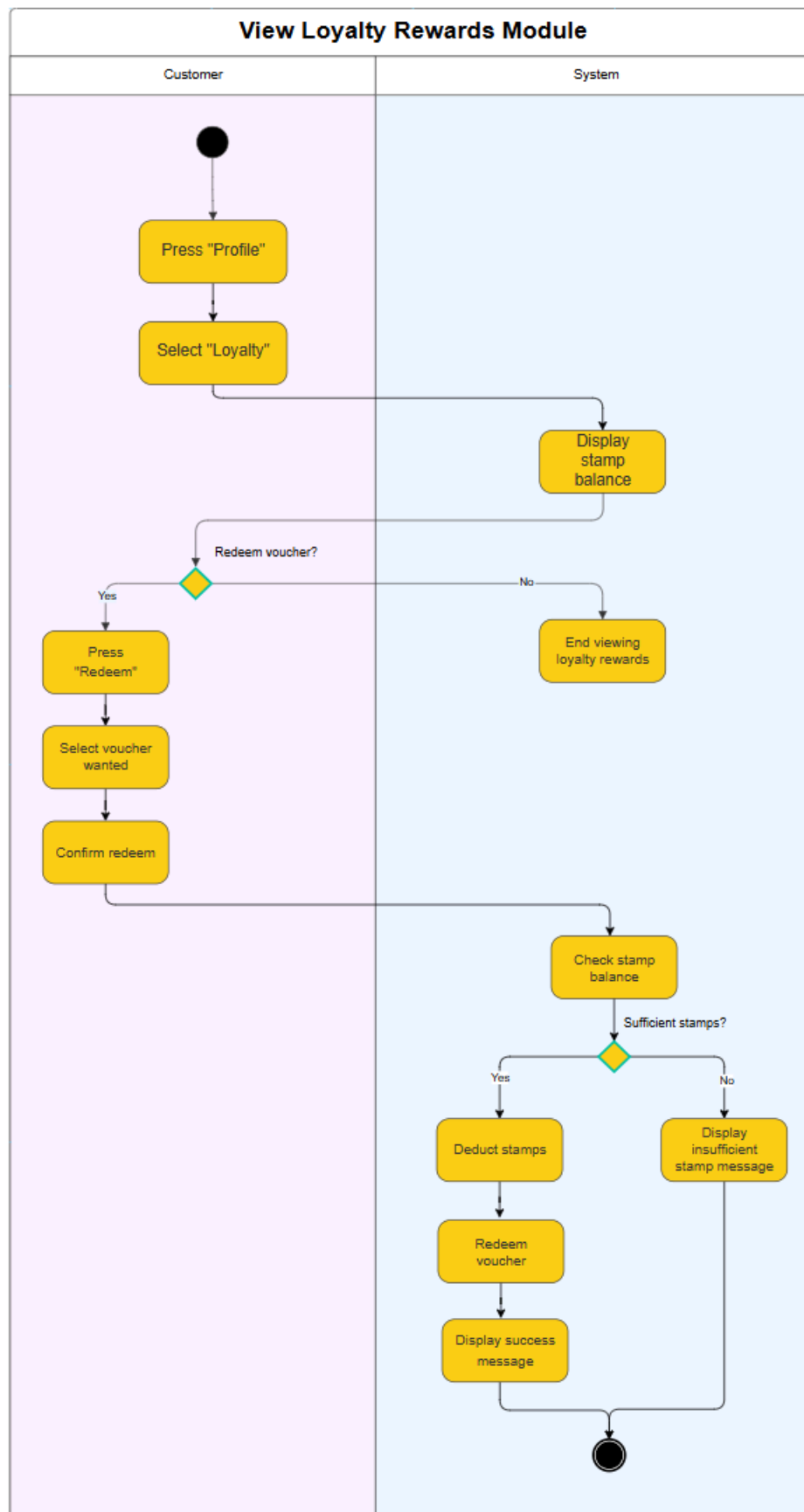
11. Process Refund



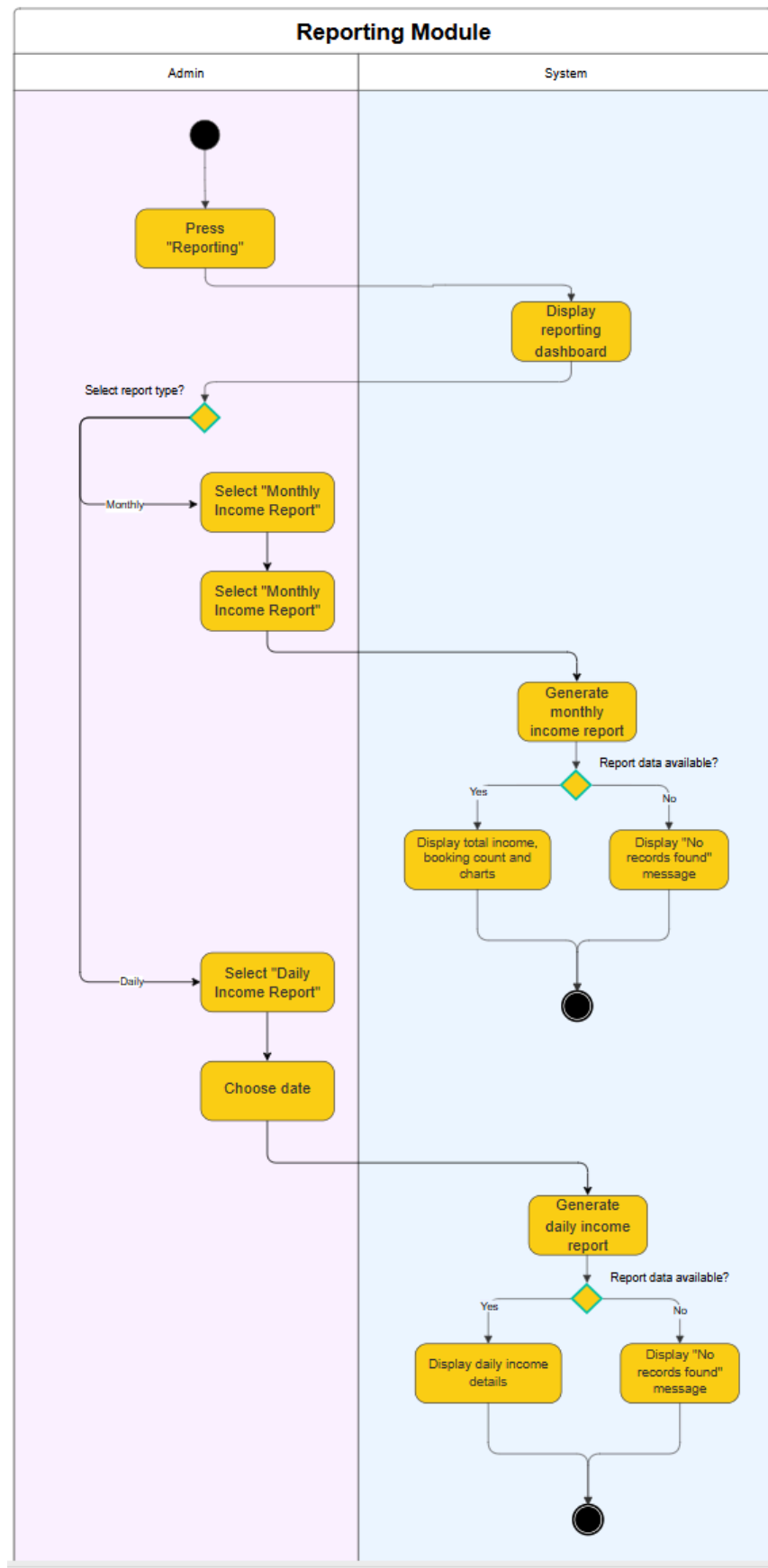
12. Manage Vehicle



13. View Loyalty Rewards



14. Reporting



8.3 Meeting Logs

MEETING MINUTES

Date/Time		21 January 2026 8:00pm	
Location		Google Meet	
Agenda		1. Review Global Conceptual ERD	
		2. Discuss about Logical ERD and Normalization (up to 3NF)	
		3. Review Relational Database Schema and SQL DDL & DML implementation	
		4. Assign Interface Mapping & Testing	
Meeting MC		Angela Ngu Xin Yi	
Attendance			
Name		Time	Reason for Absence
Angela Ngu Xin Yi		2000	-
Gavenesh A/L Batumalai		2000	-
Tan Xin Tian		2000	-
Teoh Xin Yee		2130	-
Toh Shee Thong		2000	-
Minutes			
No.	Item Discussed	Ideas/Suggestions and Person Giving It	Person in Charge
1.	Review Global Conceptual ERD	Angela: - Presented the merged Global ERD integrating all modules (User, Fleet, Booking, Payment, Loyalty, Renewal, Service History). Xin Yee: - Ensured correct relationship cardinalities (1:N, M:N resolved via associative entities).	All Members

		Group Decision: - Confirmed specialization for User → Customer & Admin - Verified foreign key consistency across modules	
2.	Discuss about Logical ERD and Normalization (up to 3NF)	Xin Tian: - Explained transformation from conceptual model to logical model. Shee Thong: - Verified normalization up to 3NF Gavenesh: - Suggested separating pricing rules into structured JSON or child table for scalability.	Xin Tian, Shee Thong and Gavenesh
3.	Review Relational Database Schema and SQL DDL & DML implementation	Angela: - Reviewed relation schemas derived from Logical ERD. Xin Yee: - Validated primary key and foreign key constraints. Xin Tian: - Implemented SQL DDL Shee Thong: - Prepared DML scripts	Angela, Xin Yee, Xin Tian and Shee Thong
4.	Assign Interface Mapping & Testing	Gavenesh: - Review Relational Database Schema and SQL DDL & DML implementation Group testing: - Shee Thong: Booking process tested end-to-end - Xin Yee: Payment verification validated - Angela: Renewal reminder notifications checked - Xin Tian: Service history logging tested	All Members

